

# popgenVCF quickstart example: chromosome 22 subset (160 samples, 8 populations)

popgenVCF

2026-08-09

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# 1 Executive summary

This PDF file is a portable, human-readable analysis report. Figures are embedded directly in the file. The HTML version can be opened in a standard web browser, while the PDF version is optimized for compact exchange, printing, and email. The accompanying `analysis_results.rds` file is a reproducibility archive for R users; it is not required to read this report.

Generated with **popgenVCF v0.10.0**.

The analysis retained **160 samples** from **8 populations**, with **1969 QC-passing SNPs** and **357 LD-pruned SNPs**.

The global Weir-Cockerham  $F_{ST}$  estimate was **0.0915**.

## 2 Quality control

| step              | variants | removed_at_step | retained_percent |
|-------------------|----------|-----------------|------------------|
| Input biallelic   | 21418    | 0               | 100.00           |
| After MAF         | 1969     | 19449           | 9.19             |
| After missingness | 1969     | 0               | 9.19             |
| After LD pruning  | 357      | 1612            | 1.67             |

## 3 Genetic diversity

| population | n_samples | n_loci | polymorphic_loci | polymorphic_fraction | observed_heterozygosity | expected_heterozygosity | inbreeding_coefficient | mean_minor_allele_frequency | mean_locus_call_rate | hwe_tested_loci | hwe_significant_loci | hwe_significant_loci_fdr | private_allele_loci |
|------------|-----------|--------|------------------|----------------------|-------------------------|-------------------------|------------------------|-----------------------------|----------------------|-----------------|----------------------|--------------------------|---------------------|
| CHB        | 20        | 1969   | 1653             | 0.8395               | 0.2683                  | 0.2855                  | 0.0603                 | 0.2133                      | 1                    | 1653            | 149                  | 55                       | 0                   |
| GBR        | 20        | 1969   | 1736             | 0.8817               | 0.2764                  | 0.2827                  | 0.0226                 | 0.2004                      | 1                    | 1736            | 99                   | 31                       | 0                   |
| ITU        | 20        | 1969   | 1829             | 0.9289               | 0.2736                  | 0.2919                  | 0.0626                 | 0.2059                      | 1                    | 1829            | 100                  | 43                       | 0                   |
| LMK        | 20        | 1969   | 1622             | 0.8238               | 0.2638                  | 0.2727                  | 0.0326                 | 0.1963                      | 1                    | 1622            | 66                   | 14                       | 0                   |
| PEL        | 20        | 1969   | 1800             | 0.9142               | 0.2330                  | 0.2579                  | 0.0968                 | 0.1851                      | 1                    | 1800            | 147                  | 45                       | 0                   |
| PUR        | 20        | 1969   | 1844             | 0.9365               | 0.2748                  | 0.2730                  | -0.0067                | 0.1871                      | 1                    | 1844            | 81                   | 21                       | 0                   |
| STU        | 20        | 1969   | 1788             | 0.9081               | 0.2561                  | 0.2863                  | 0.1056                 | 0.2013                      | 1                    | 1788            | 131                  | 33                       | 0                   |
| YRI        | 20        | 1969   | 1636             | 0.8309               | 0.2731                  | 0.2705                  | -0.0096                | 0.1944                      | 1                    | 1636            | 92                   | 18                       | 0                   |

## 4 Principal component analysis

| sample  | vcf_sample | PC1     | PC2     | PC3     | PC4     | PC5     | PC6     | PC7     | PC8     | PC9     | PC10    | population |
|---------|------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|------------|
| NA18536 | NA18536    | -0.0575 | 0.0756  | -0.1958 | -0.0176 | 0.0470  | 0.0646  | -0.0320 | -0.0509 | -0.0829 | 0.0169  | CHB        |
| NA18538 | NA18538    | -0.0665 | 0.1310  | -0.0958 | -0.0006 | 0.0158  | -0.0224 | 0.1012  | 0.0074  | -0.2047 | -0.0051 | CHB        |
| NA18541 | NA18541    | -0.0654 | 0.0991  | -0.0443 | -0.0362 | 0.0684  | 0.0917  | 0.0875  | -0.0995 | -0.0353 | 0.0002  | CHB        |
| NA18552 | NA18552    | -0.0714 | 0.1650  | -0.1811 | -0.1448 | 0.0223  | 0.0384  | 0.0671  | 0.0537  | 0.0328  | -0.0593 | CHB        |
| NA18553 | NA18553    | -0.0616 | -0.0660 | -0.1590 | 0.0237  | -0.2193 | -0.0004 | -0.1910 | -0.0064 | -0.1380 | 0.0197  | CHB        |
| NA18555 | NA18555    | -0.0312 | 0.0930  | -0.1309 | -0.1100 | 0.0308  | 0.0717  | -0.0562 | 0.0124  | 0.0901  | -0.1218 | CHB        |
| NA18557 | NA18557    | -0.0893 | 0.2057  | -0.1264 | -0.0737 | 0.0509  | 0.0121  | 0.0892  | 0.0377  | -0.0913 | 0.0984  | CHB        |
| NA18562 | NA18562    | -0.0381 | 0.0209  | -0.0813 | 0.0459  | 0.0625  | -0.0312 | 0.1107  | -0.1086 | 0.0245  | -0.1299 | CHB        |
| NA18564 | NA18564    | -0.0305 | 0.0841  | -0.1174 | -0.1036 | -0.0883 | -0.0264 | -0.1778 | 0.0857  | 0.2782  | -0.1910 | CHB        |
| NA18573 | NA18573    | -0.0685 | 0.0633  | 0.0378  | 0.0394  | -0.0765 | 0.0588  | -0.0318 | -0.1708 | 0.0201  | -0.0102 | CHB        |
| NA18574 | NA18574    | -0.0873 | 0.1260  | -0.1222 | -0.0647 | 0.0656  | 0.0098  | 0.0426  | -0.0156 | -0.0750 | -0.0113 | CHB        |
| NA18606 | NA18606    | -0.0584 | 0.0752  | -0.1100 | 0.0288  | 0.0297  | -0.0415 | 0.1097  | 0.1042  | -0.0327 | -0.0739 | CHB        |
| NA18611 | NA18611    | -0.0717 | 0.1273  | -0.0419 | -0.0468 | -0.0176 | 0.1518  | -0.0208 | 0.0271  | -0.1508 | -0.0482 | CHB        |
| NA18612 | NA18612    | -0.0504 | -0.0424 | -0.2118 | 0.1398  | -0.1597 | -0.0342 | -0.1189 | -0.0957 | -0.0276 | 0.1465  | CHB        |
| NA18613 | NA18613    | -0.0583 | 0.1108  | -0.0705 | -0.0469 | 0.0092  | 0.0932  | 0.0875  | 0.0691  | -0.1390 | 0.1136  | CHB        |
| NA18616 | NA18616    | -0.0173 | 0.0510  | -0.1403 | -0.1915 | 0.0842  | 0.1383  | -0.0259 | -0.1606 | 0.0728  | -0.0742 | CHB        |
| NA18617 | NA18617    | -0.0307 | 0.1223  | -0.1382 | -0.1085 | -0.0277 | 0.0449  | -0.0945 | 0.0113  | 0.0748  | -0.1828 | CHB        |
| NA18638 | NA18638    | -0.0722 | 0.1224  | -0.0679 | -0.0460 | -0.0106 | 0.1791  | -0.0069 | -0.1120 | -0.1554 | 0.0068  | CHB        |
| NA18643 | NA18643    | -0.0437 | 0.0467  | -0.1922 | 0.0077  | 0.0101  | -0.0009 | 0.0329  | -0.0462 | 0.0704  | -0.0114 | CHB        |
| NA18647 | NA18647    | -0.0558 | 0.1011  | -0.1989 | -0.0603 | 0.1633  | 0.1660  | 0.0094  | -0.0507 | 0.0072  | 0.0560  | CHB        |

## 5 PCA SNP loadings (top 20 by |loading|; see 31\_PCA\_loadings.tsv for the complete top-N list)

| axis | snp_id | chromosome | position | contribution | magnitude |
|------|--------|------------|----------|--------------|-----------|
| PC7  | 10937  | 22         | 20437004 | 0.2848       | 0.2848    |
| PC7  | 10502  | 22         | 20371297 | 0.2541       | 0.2541    |
| PC7  | 11374  | 22         | 20502771 | 0.2524       | 0.2524    |
| PC8  | 13593  | 22         | 20753041 | 0.2191       | 0.2191    |
| PC6  | 10937  | 22         | 20437004 | -0.2191      | 0.2191    |
| PC8  | 13990  | 22         | 20765538 | 0.2178       | 0.2178    |
| PC8  | 14246  | 22         | 20773711 | 0.2034       | 0.2034    |
| PC5  | 4933   | 22         | 20159049 | -0.2010      | 0.2010    |
| PC3  | 15906  | 22         | 20832523 | 0.1980       | 0.1980    |
| PC8  | 13078  | 22         | 20733667 | 0.1952       | 0.1952    |
| PC7  | 10415  | 22         | 20360856 | 0.1932       | 0.1932    |
| PC8  | 14080  | 22         | 20768038 | -0.1925      | 0.1925    |
| PC9  | 3724   | 22         | 20121080 | -0.1900      | 0.1900    |
| PC6  | 10502  | 22         | 20371297 | -0.1887      | 0.1887    |
| PC10 | 11570  | 22         | 20630754 | 0.1883       | 0.1883    |
| PC9  | 1516   | 22         | 20047587 | -0.1869      | 0.1869    |
| PC3  | 14848  | 22         | 20794119 | 0.1865       | 0.1865    |
| PC9  | 2006   | 22         | 20065314 | -0.1859      | 0.1859    |
| PC8  | 14377  | 22         | 20777533 | -0.1806      | 0.1806    |
| PC10 | 11593  | 22         | 20639324 | 0.1763       | 0.1763    |

## 6 Population differentiation

| population_1 | population_2 | n_1 | n_2 | fst    |
|--------------|--------------|-----|-----|--------|
| CHB          | GBR          | 20  | 20  | 0.1210 |
| CHB          | ITU          | 20  | 20  | 0.0631 |
| CHB          | LWK          | 20  | 20  | 0.1631 |
| CHB          | PEL          | 20  | 20  | 0.0794 |
| CHB          | PUR          | 20  | 20  | 0.1101 |
| CHB          | STU          | 20  | 20  | 0.0865 |
| CHB          | YRI          | 20  | 20  | 0.1631 |
| GBR          | ITU          | 20  | 20  | 0.0232 |
| GBR          | LWK          | 20  | 20  | 0.1198 |
| GBR          | PEL          | 20  | 20  | 0.0874 |
| GBR          | PUR          | 20  | 20  | 0.0267 |
| GBR          | STU          | 20  | 20  | 0.0350 |
| GBR          | YRI          | 20  | 20  | 0.1344 |
| ITU          | LWK          | 20  | 20  | 0.1057 |
| ITU          | PEL          | 20  | 20  | 0.0400 |
| ITU          | PUR          | 20  | 20  | 0.0269 |
| ITU          | STU          | 20  | 20  | 0.0052 |
| ITU          | YRI          | 20  | 20  | 0.1189 |
| LWK          | PEL          | 20  | 20  | 0.1737 |
| LWK          | PUR          | 20  | 20  | 0.1176 |
| LWK          | STU          | 20  | 20  | 0.1112 |
| LWK          | YRI          | 20  | 20  | 0.0052 |
| PEL          | PUR          | 20  | 20  | 0.0621 |
| PEL          | STU          | 20  | 20  | 0.0609 |
| PEL          | YRI          | 20  | 20  | 0.1799 |
| PUR          | STU          | 20  | 20  | 0.0339 |
| PUR          | YRI          | 20  | 20  | 0.1226 |
| STU          | YRI          | 20  | 20  | 0.1187 |



## 7 Genome scan FST outliers (top 20 windows by FST; see 39\_genome\_scan\_fst.tsv for the complete list)

| chromosome | window_start | window_end | n_snps | global_fst |
|------------|--------------|------------|--------|------------|
| 22         | 20800428     | 20850427   | 104    | 0.1718     |
| 22         | 20750428     | 20800427   | 213    | 0.1319     |
| 22         | 20850428     | 20900427   | 113    | 0.1118     |
| 22         | 20250428     | 20300427   | 102    | 0.1097     |
| 22         | 20900428     | 20950427   | 138    | 0.1076     |
| 22         | 20150428     | 20200427   | 196    | 0.1007     |
| 22         | 20200428     | 20250427   | 111    | 0.0952     |
| 22         | 20000428     | 20050427   | 160    | 0.0875     |
| 22         | 20700428     | 20750427   | 135    | 0.0845     |
| 22         | 20400428     | 20450427   | 24     | 0.0838     |
| 22         | 20100428     | 20150427   | 115    | 0.0694     |
| 22         | 20950428     | 21000427   | 193    | 0.0693     |
| 22         | 20350428     | 20400427   | 64     | 0.0666     |
| 22         | 20450428     | 20500427   | 48     | 0.0639     |
| 22         | 20050428     | 20100427   | 131    | 0.0512     |
| 22         | 20600428     | 20650427   | 38     | 0.0510     |
| 22         | 20500428     | 20550427   | 9      | 0.0403     |
| 22         | 20300428     | 20350427   | 44     | 0.0351     |
| 22         | 20650428     | 20700427   | 31     | 0.0194     |

## 8 Close relatives (up to 20 shown; see 36\_close\_relatives.tsv for the complete list)

| sample_1 | sample_2 | population_1 | population_2 | IBS0   | kinship | relationship_degree |
|----------|----------|--------------|--------------|--------|---------|---------------------|
| HG03873  | HG03998  | ITU          | STU          | 0.0000 | 0.4525  | duplicate/MZ twin   |
| NA19331  | NA19334  | LWK          | LWK          | 0.0028 | 0.4459  | duplicate/MZ twin   |
| HG01938  | HG02278  | PEL          | PEL          | 0.0056 | 0.2781  | 1st-degree          |
| HG00742  | HG01054  | PUR          | PUR          | 0.0140 | 0.2468  | 1st-degree          |
| NA19172  | NA19247  | YRI          | YRI          | 0.0168 | 0.2448  | 1st-degree          |
| NA19360  | NA18516  | LWK          | YRI          | 0.0112 | 0.2313  | 1st-degree          |
| HG01067  | HG01182  | PUR          | PUR          | 0.0140 | 0.2225  | 1st-degree          |
| HG01067  | HG01086  | PUR          | PUR          | 0.0196 | 0.2125  | 1st-degree          |
| HG00099  | HG00141  | GBR          | GBR          | 0.0112 | 0.2110  | 1st-degree          |
| HG01067  | HG01305  | PUR          | PUR          | 0.0168 | 0.2100  | 1st-degree          |
| HG00122  | HG01308  | GBR          | PUR          | 0.0140 | 0.1971  | 1st-degree          |
| NA18516  | NA19222  | YRI          | YRI          | 0.0140 | 0.1959  | 1st-degree          |
| HG01396  | HG03740  | PUR          | STU          | 0.0084 | 0.1944  | 1st-degree          |
| NA18516  | NA19210  | YRI          | YRI          | 0.0196 | 0.1910  | 1st-degree          |
| HG00253  | HG01067  | GBR          | PUR          | 0.0168 | 0.1900  | 1st-degree          |
| HG01938  | HG02345  | PEL          | PEL          | 0.0084 | 0.1875  | 1st-degree          |
| NA18516  | NA19247  | YRI          | YRI          | 0.0196 | 0.1856  | 1st-degree          |
| HG00250  | HG00253  | GBR          | GBR          | 0.0168 | 0.1852  | 1st-degree          |
| HG00141  | HG00240  | GBR          | GBR          | 0.0168 | 0.1825  | 1st-degree          |
| HG00253  | HG01396  | GBR          | PUR          | 0.0168 | 0.1818  | 1st-degree          |

## 9 Runs of homozygosity (highest FROH samples, up to 20 shown; see 38\_ROH\_sample\_summary.tsv for the complete list)

| sample  | population | n_runs | total_length_bp | mean_length_bp | longest_run_bp | froh   |
|---------|------------|--------|-----------------|----------------|----------------|--------|
| HG02146 | PEL        | 4      | 788356          | 197089.00      | 574060         | 0.7884 |
| HG01942 | PEL        | 4      | 595694          | 148923.50      | 235561         | 0.5957 |
| HG01933 | PEL        | 4      | 537370          | 134342.50      | 311897         | 0.5374 |
| HG01578 | PEL        | 6      | 528877          | 88146.17       | 194872         | 0.5289 |
| HG03989 | STU        | 2      | 469894          | 234947.00      | 265232         | 0.4699 |
| HG00151 | GBR        | 4      | 438830          | 109707.50      | 207650         | 0.4388 |
| HG01950 | PEL        | 4      | 438096          | 109524.00      | 217282         | 0.4381 |
| NA18611 | CHB        | 3      | 437054          | 145684.67      | 249970         | 0.4371 |
| HG00637 | PUR        | 4      | 407361          | 101840.25      | 155344         | 0.4074 |
| HG00123 | GBR        | 3      | 406869          | 135623.00      | 221114         | 0.4069 |
| NA18612 | CHB        | 3      | 400265          | 133421.67      | 226775         | 0.4003 |
| NA18616 | CHB        | 5      | 391340          | 78268.00       | 268046         | 0.3914 |
| HG01054 | PUR        | 3      | 379034          | 126344.67      | 228087         | 0.3790 |
| HG01086 | PUR        | 3      | 364023          | 121341.00      | 162866         | 0.3640 |
| HG03836 | STU        | 2      | 359185          | 179592.50      | 284841         | 0.3592 |
| HG03990 | STU        | 3      | 349620          | 116540.00      | 175138         | 0.3496 |
| NA18647 | CHB        | 3      | 339679          | 113226.33      | 182839         | 0.3397 |
| HG01917 | PEL        | 3      | 337023          | 112341.00      | 261589         | 0.3370 |
| HG01927 | PEL        | 2      | 330751          | 165375.50      | 191744         | 0.3308 |
| NA18564 | CHB        | 3      | 323282          | 107760.67      | 181407         | 0.3233 |

## 10 Discriminant analysis of principal components

| K  | n_pca | n_da | BIC | assignment_accuracy | mean_success | calinski_harabasz | davies_bouldin | replicate_max_rmse |
|----|-------|------|-----|---------------------|--------------|-------------------|----------------|--------------------|
| 2  | 40    | 1    | NA  | 0.2500              | 0.9958       | 15.6223           | 2.7833         | 0.0971             |
| 3  | 10    | 2    | NA  | 0.2938              | 0.9966       | 13.3767           | 3.0860         | 0.0000             |
| 4  | 10    | 3    | NA  | 0.3562              | 0.9648       | 11.6552           | 2.8675         | 0.3841             |
| 5  | 30    | 4    | NA  | 0.4062              | 0.9361       | 10.2898           | 2.6773         | 0.3512             |
| 6  | 10    | 5    | NA  | 0.3938              | 0.9306       | 9.5564            | 2.5154         | 0.2944             |
| 7  | 10    | 6    | NA  | 0.4812              | 0.9407       | 9.0366            | 2.4878         | 0.0727             |
| 8  | 10    | 7    | NA  | 0.4625              | 0.9058       | 8.3942            | 2.4244         | 0.1614             |
| 9  | 10    | 8    | NA  | 0.4688              | 0.9239       | 7.8689            | 2.6212         | 0.1830             |
| 10 | 10    | 9    | NA  | 0.3812              | 0.8975       | 7.3769            | 2.6614         | 0.2373             |

## 11 DAPC SNP loadings (top 20 by contribution across all K; see 22f\_DAPC\_loadings\_K.tsv for the complete per-K top-N lists)

| K | axis | snp_id | chromosome | position | contribution |
|---|------|--------|------------|----------|--------------|
| 3 | LD2  | 10525  | 22         | 20376408 | 0.0202       |
| 3 | LD2  | 10527  | 22         | 20376436 | 0.0199       |
| 3 | LD2  | 10765  | 22         | 20396584 | 0.0184       |
| 4 | LD2  | 10525  | 22         | 20376408 | 0.0184       |
| 3 | LD2  | 10675  | 22         | 20388698 | 0.0184       |
| 3 | LD2  | 10757  | 22         | 20394994 | 0.0182       |
| 3 | LD2  | 10807  | 22         | 20401090 | 0.0182       |
| 4 | LD2  | 10527  | 22         | 20376436 | 0.0181       |
| 4 | LD2  | 10807  | 22         | 20401090 | 0.0181       |
| 3 | LD2  | 10806  | 22         | 20401066 | 0.0180       |
| 3 | LD2  | 10758  | 22         | 20395128 | 0.0180       |
| 4 | LD2  | 10765  | 22         | 20396584 | 0.0179       |
| 4 | LD2  | 10758  | 22         | 20395128 | 0.0179       |
| 4 | LD2  | 10823  | 22         | 20402404 | 0.0179       |
| 3 | LD2  | 10937  | 22         | 20437004 | 0.0178       |
| 4 | LD2  | 10806  | 22         | 20401066 | 0.0177       |
| 3 | LD2  | 10823  | 22         | 20402404 | 0.0177       |
| 3 | LD2  | 10561  | 22         | 20378905 | 0.0175       |
| 4 | LD2  | 10757  | 22         | 20394994 | 0.0174       |
| 3 | LD2  | 10812  | 22         | 20401719 | 0.0174       |

## 12 Analysis of molecular variance

| component                                    | Sigma    | %        |
|----------------------------------------------|----------|----------|
| Variations Between population                | 27.1959  | 9.0406   |
| Variations Between samples Within population | 12.8319  | 4.2657   |
| Variations Within samples                    | 260.7906 | 86.6937  |
| Total variations                             | 300.8184 | 100.0000 |

| statistic              | Phi    |
|------------------------|--------|
| Phi-samples-total      | 0.1331 |
| Phi-samples-population | 0.0469 |
| Phi-population-total   | 0.0904 |

## 13 Mantel test and isolation by distance

| mantel_r | mantel_p | slope  | r_squared |
|----------|----------|--------|-----------|
| 0.3129   | 0.001    | 0.0028 | 0.1366    |

## 14 Admixture-model cross-validation

| K | cv_error |
|---|----------|
| 2 | 0.47688  |
| 3 | 0.46089  |
| 4 | 0.45539  |
| 5 | 0.45427  |
| 6 | 0.45225  |
| 7 | 0.45211  |
| 8 | 0.45249  |
| 9 | 0.45470  |

## 15 Chromosome-specific results

| chromosome | qc_snps | ld_snps | global_fst | pc1_percent |
|------------|---------|---------|------------|-------------|
| 22         | 1969    | 357     | 0.0915     | 50.9697     |

## 16 Figure gallery

The gallery contains one version of every analysis figure. The HTML report prefers browser-friendly SVG or PNG files. The PDF report prefers the compact vector PDF source for each figure. Original files remain available in the sibling `figures/` directory.

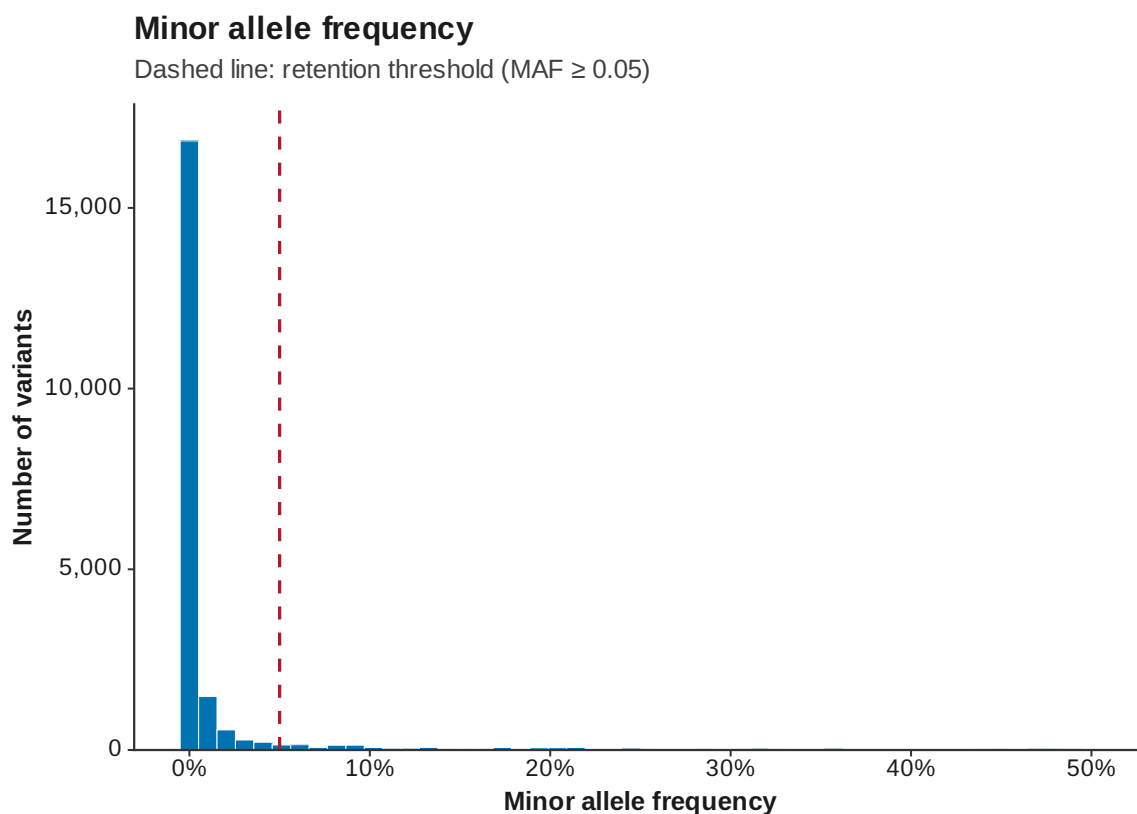


Figure 1: MAF

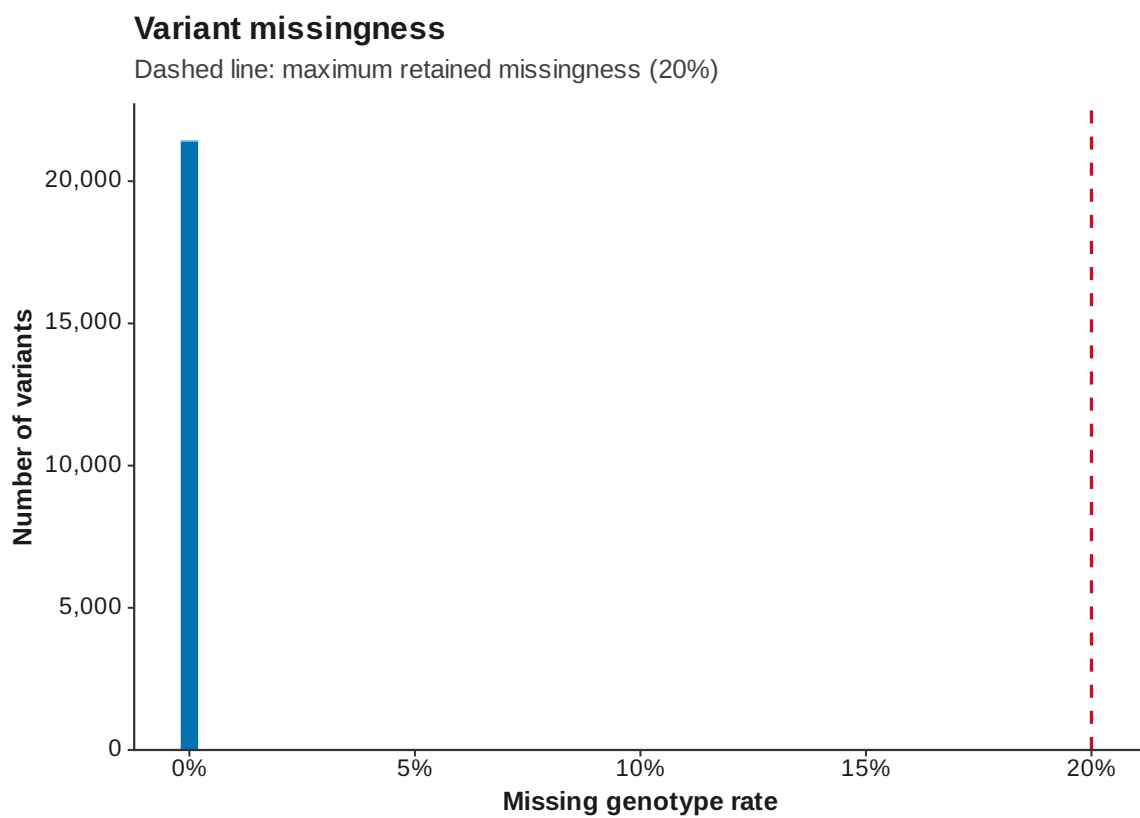


Figure 2: Variant missingness

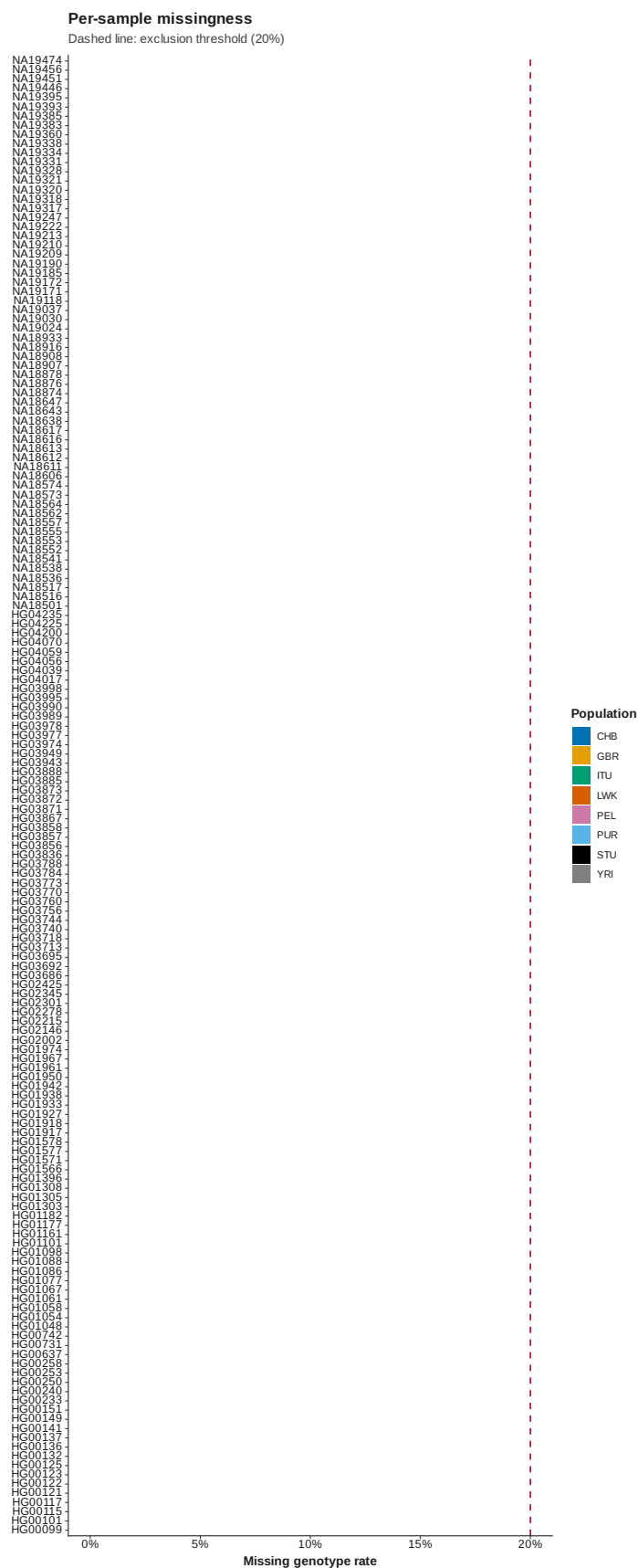


Figure 3: Sample missingness

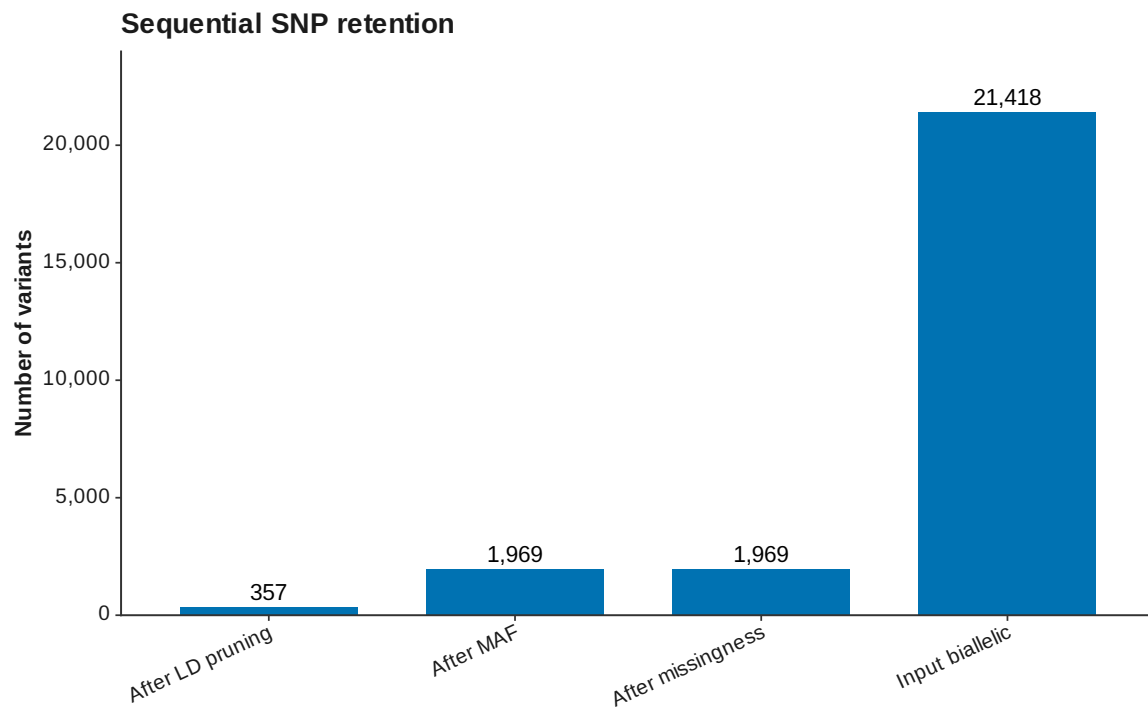


Figure 4: SNP retention

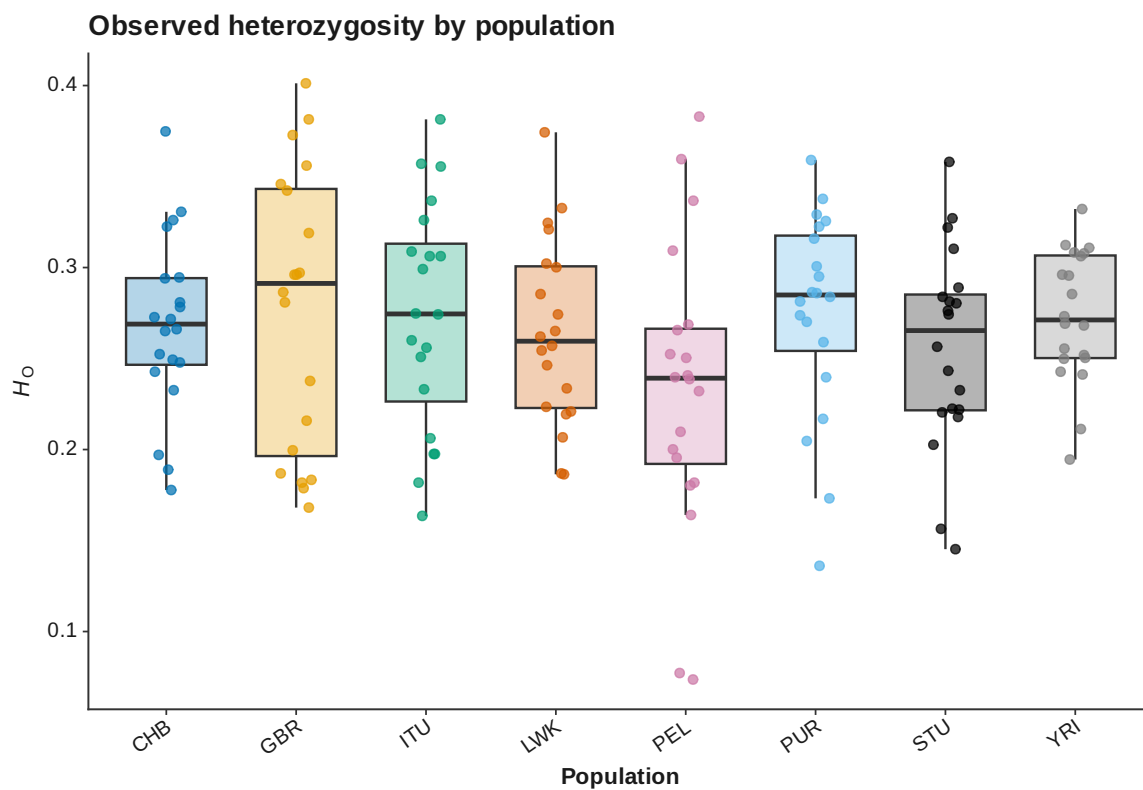


Figure 5: Sample heterozygosity



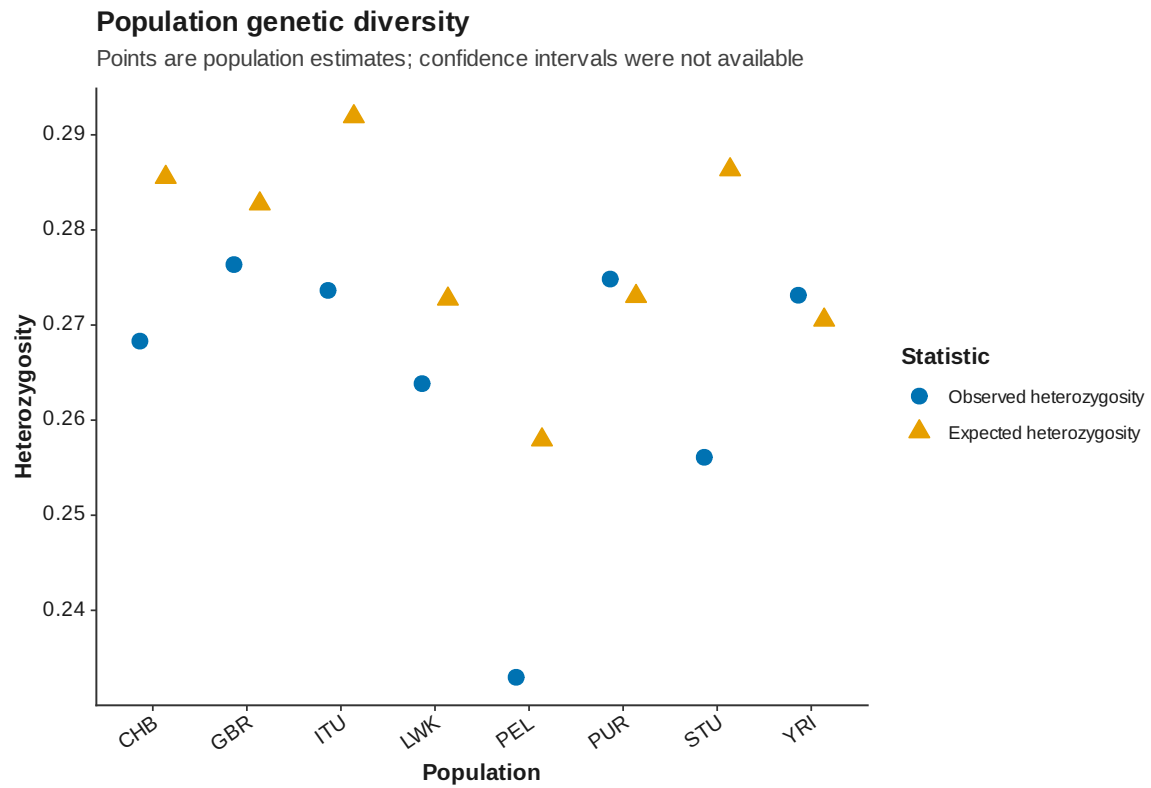


Figure 6: Population diversity

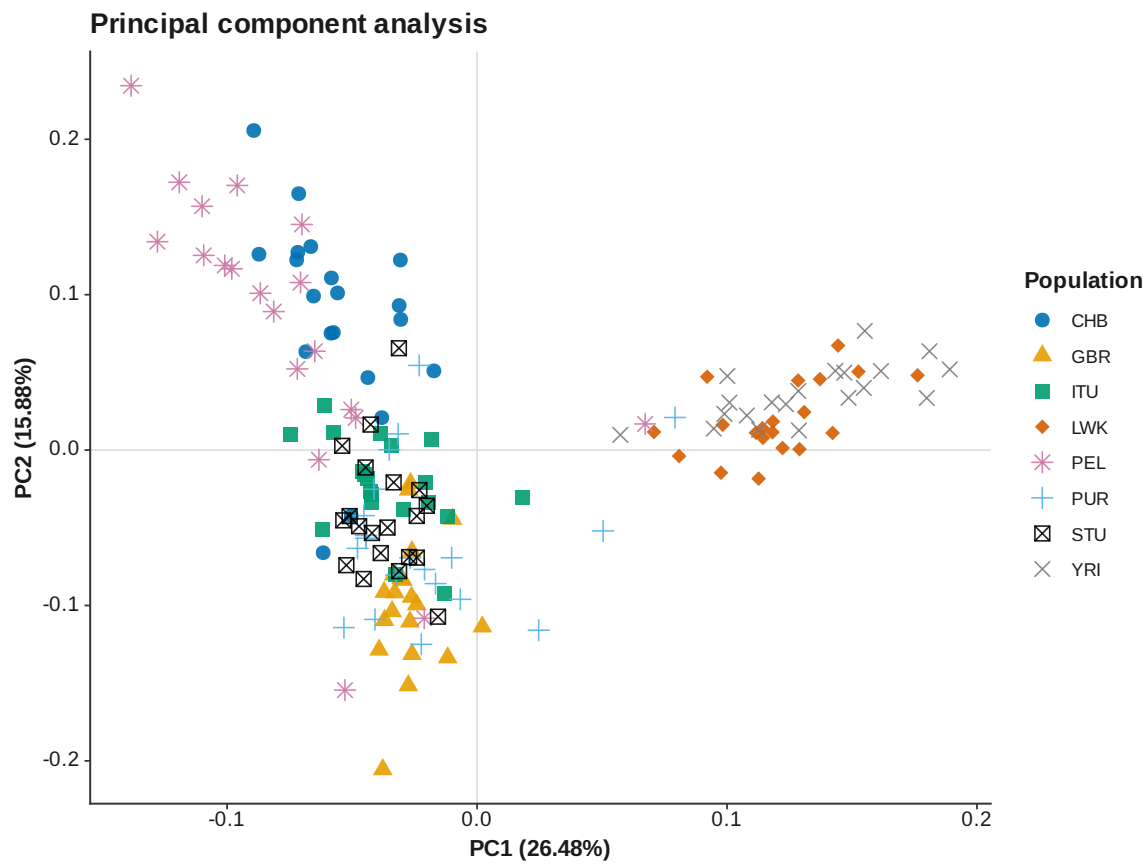


Figure 7: PCA PC 1 PC 2

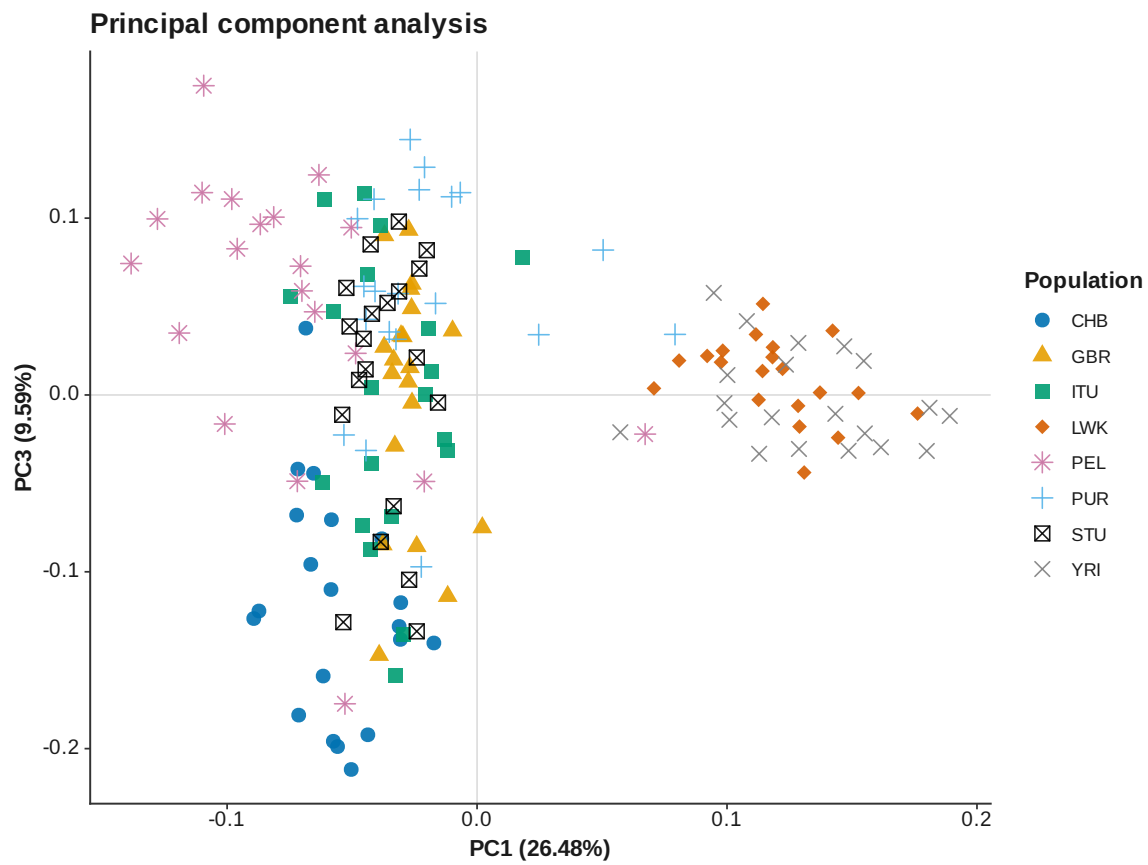


Figure 8: PCA PC 1 PC 3

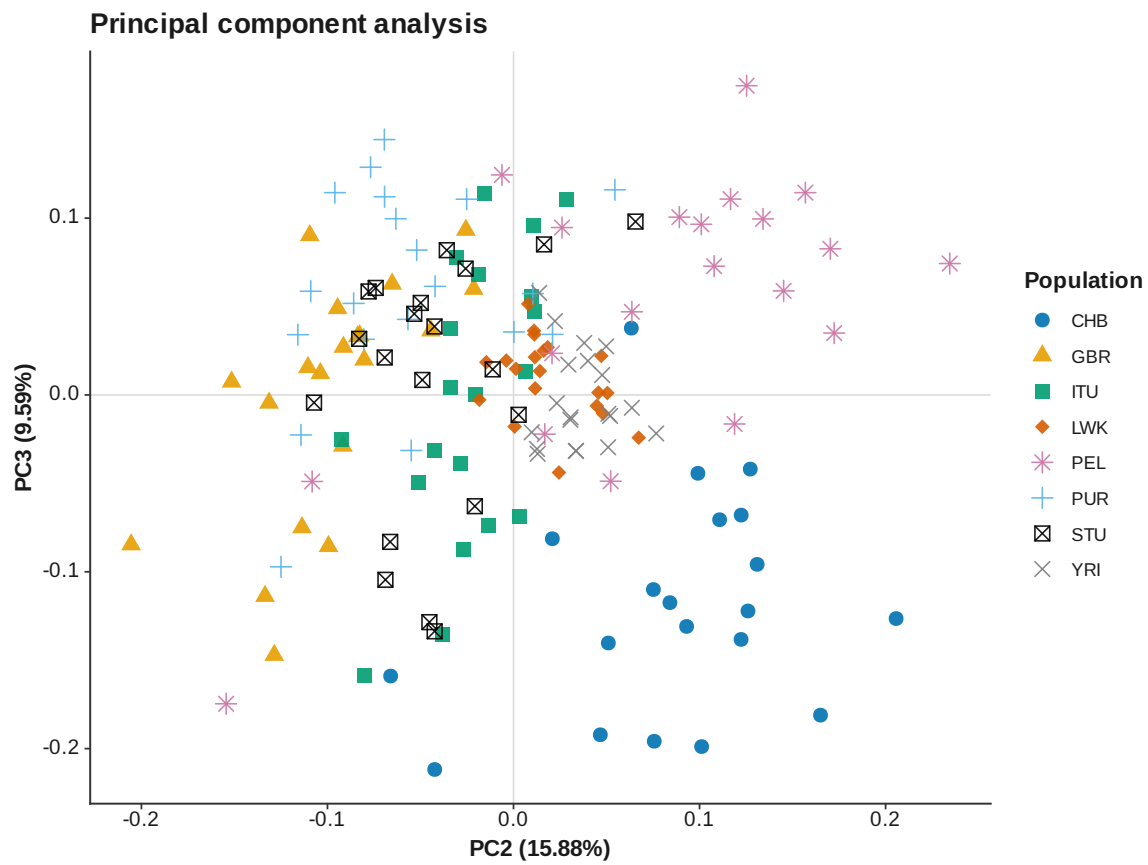


Figure 9: PCA PC 2 PC 3

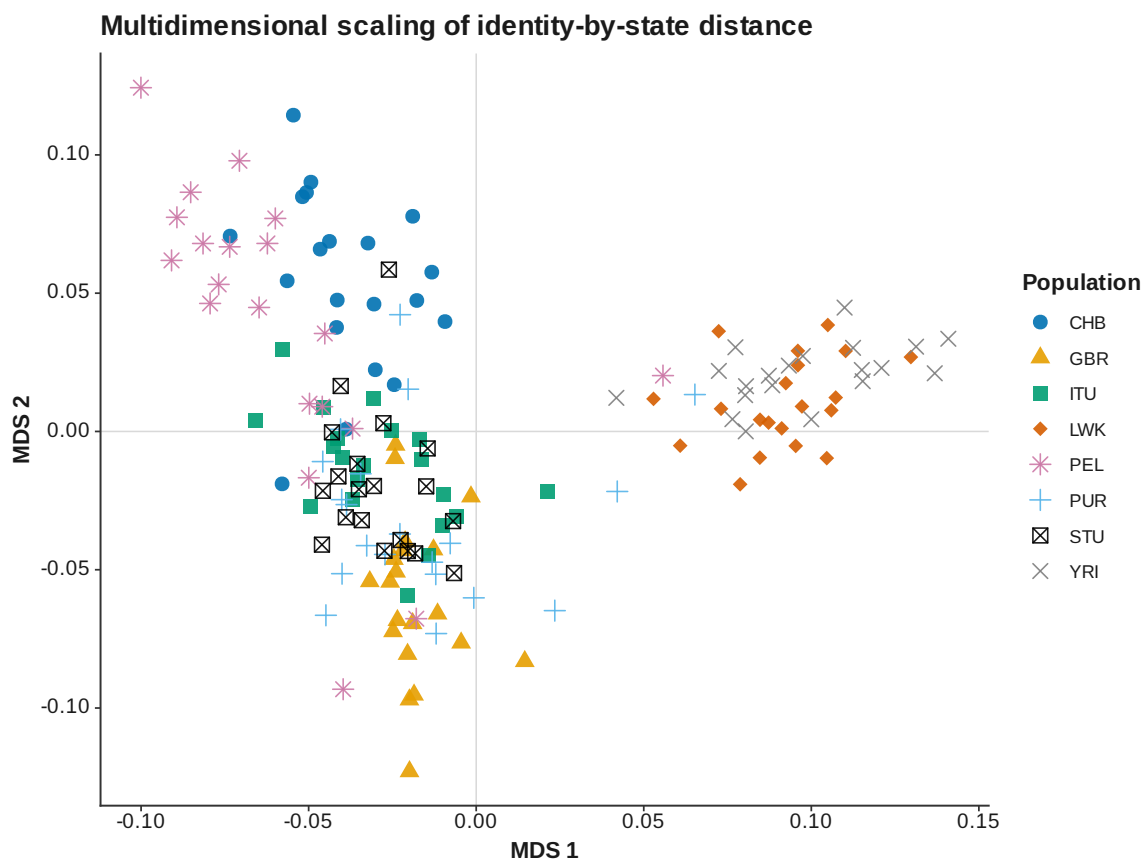


Figure 10: IBS MDS

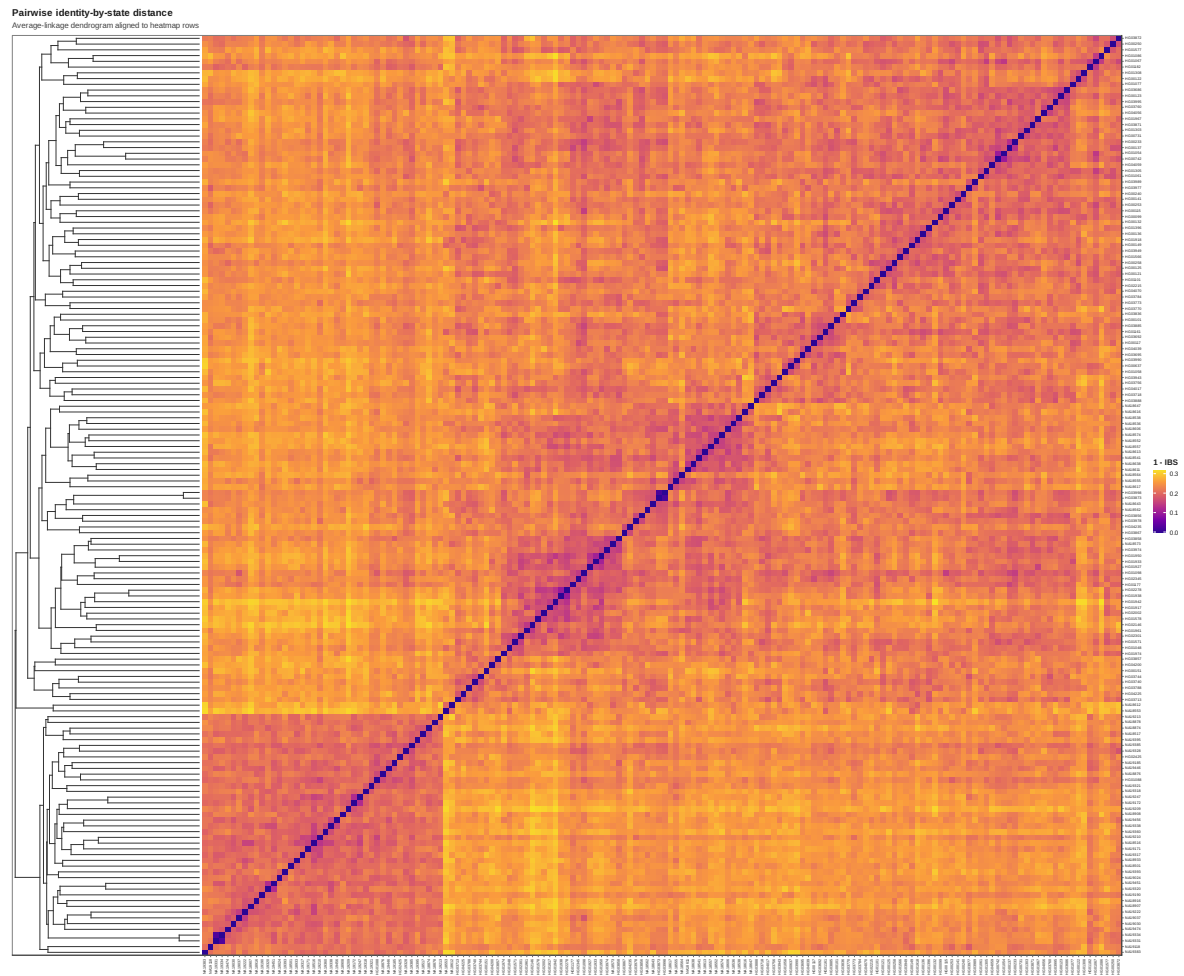


Figure 11: IBS heatmap

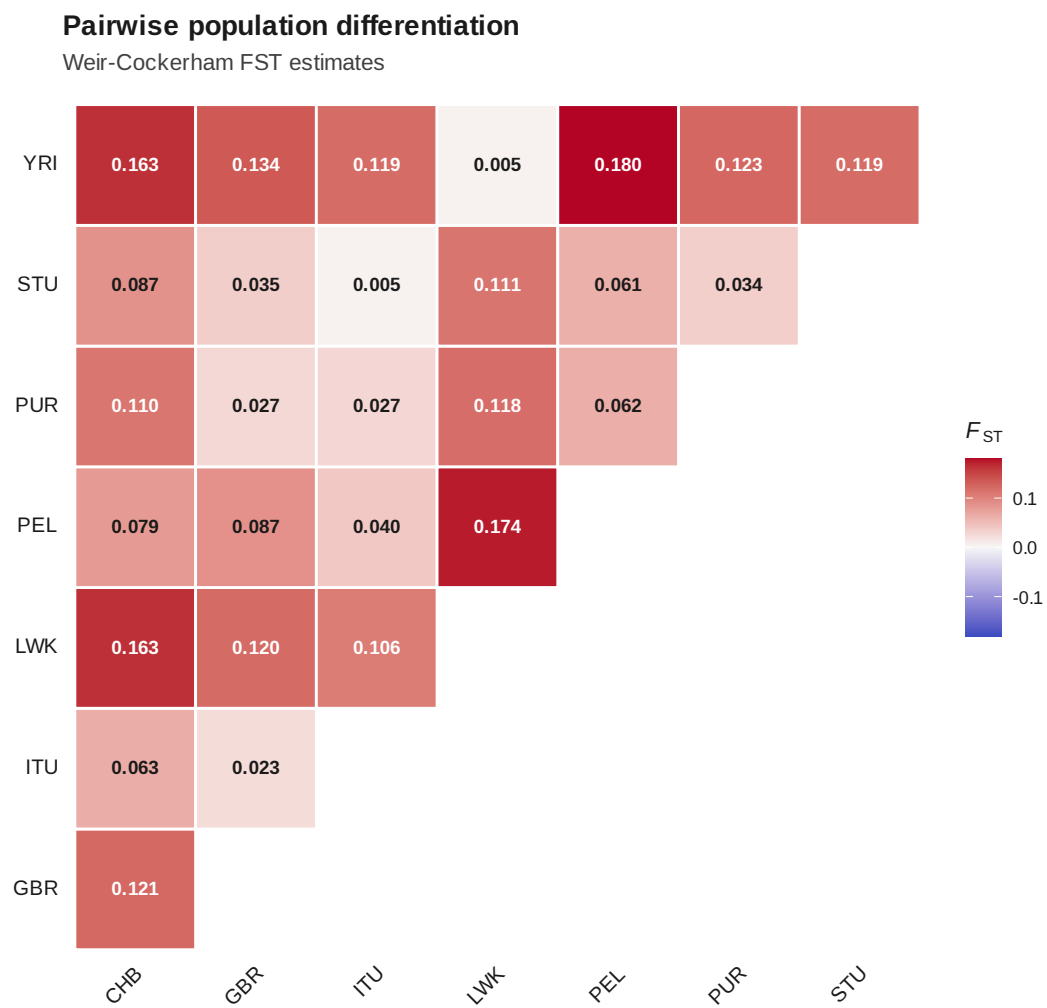


Figure 12: Pairwise  $F_{ST}$

### Discriminant analysis of principal components (K = 3)

Replicate membership RMSE = 4.679e-17 (stability threshold = 0.05).

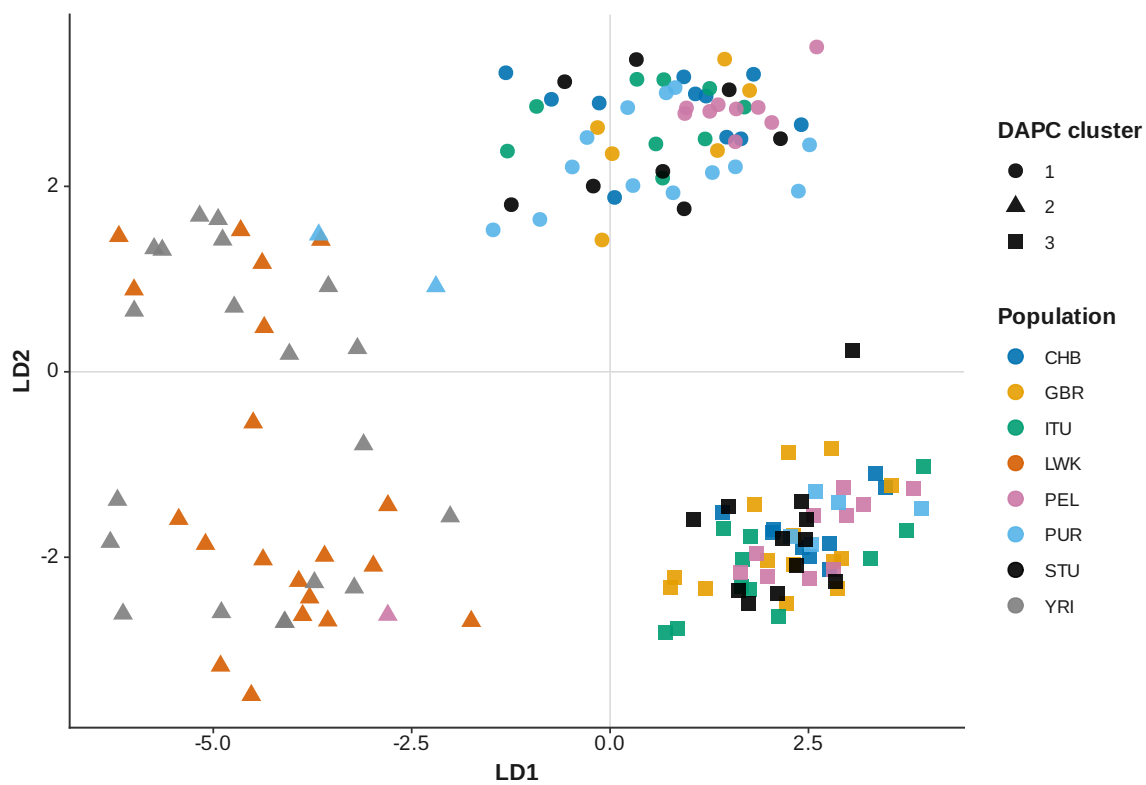


Figure 13: DAPC K 3



### Discriminant analysis of principal components (K = 4)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.3841 > 0.05).  
Avoid interpreting these assignments.**

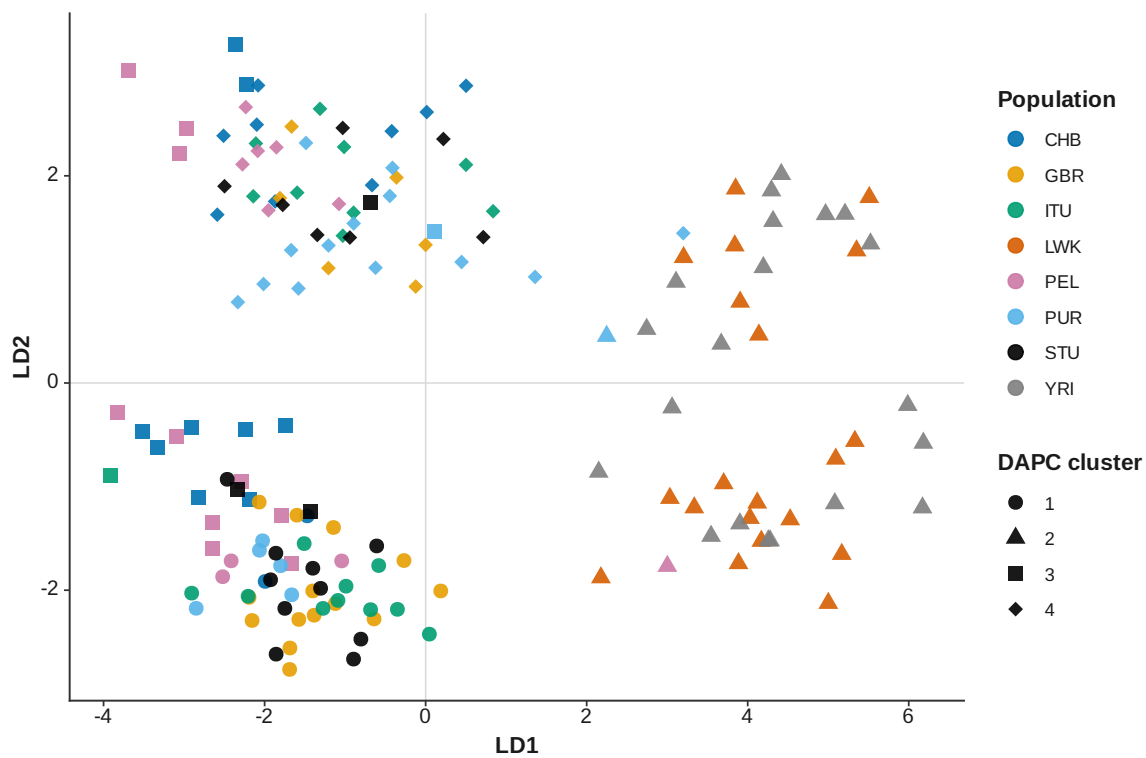


Figure 14: DAPC K 4

### Discriminant analysis of principal components (K = 5)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.3512 > 0.05).  
Avoid interpreting these assignments.**

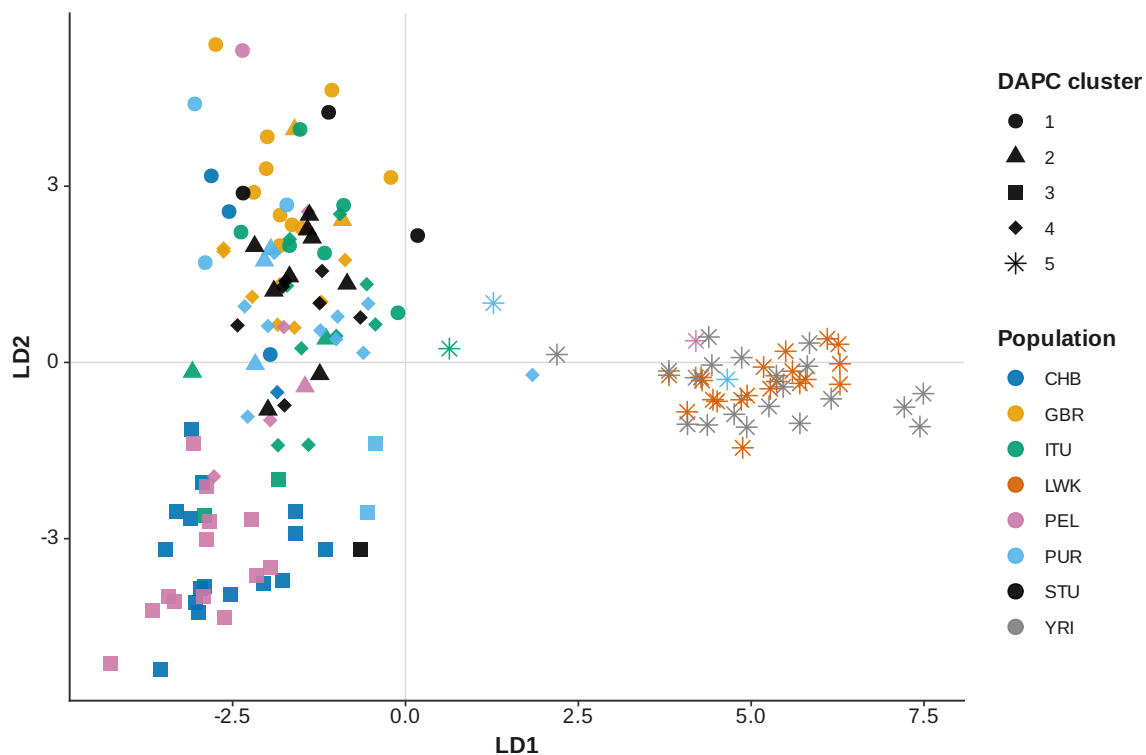


Figure 15: DAPC K 5

### Discriminant analysis of principal components (K = 6)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.2944 > 0.05).  
Avoid interpreting these assignments.**

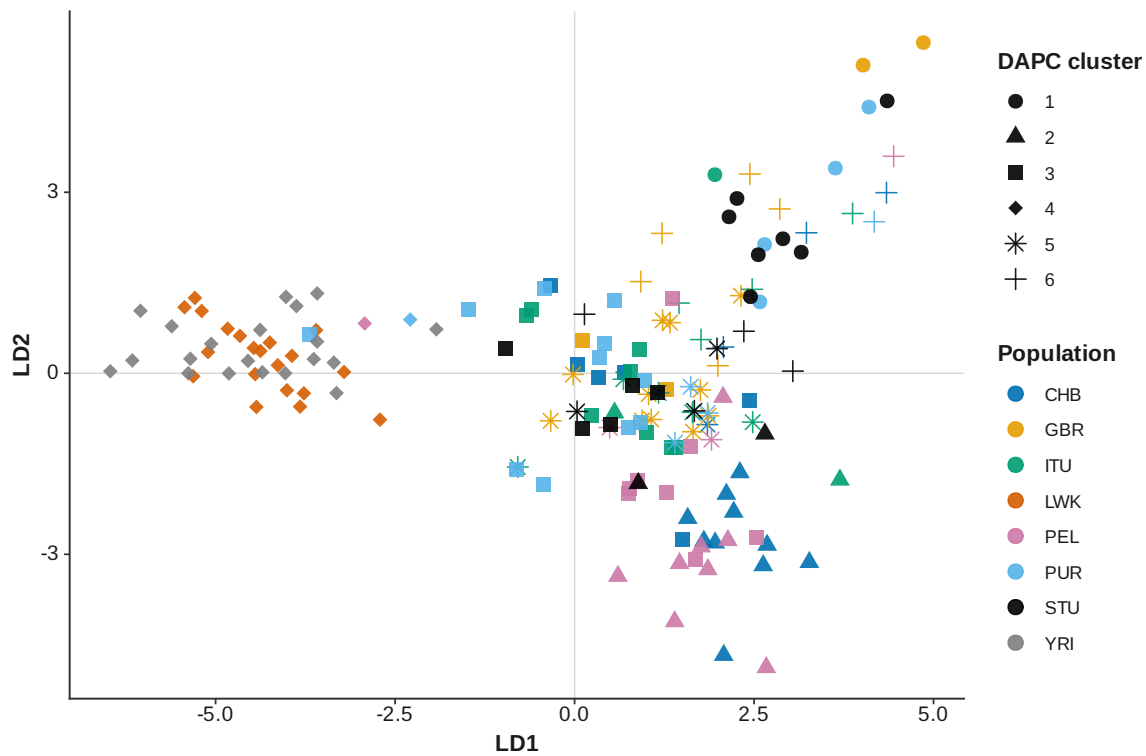


Figure 16: DAPC K 6

### Discriminant analysis of principal components (K = 7)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.07268 > 0.05).  
Avoid interpreting these assignments.**

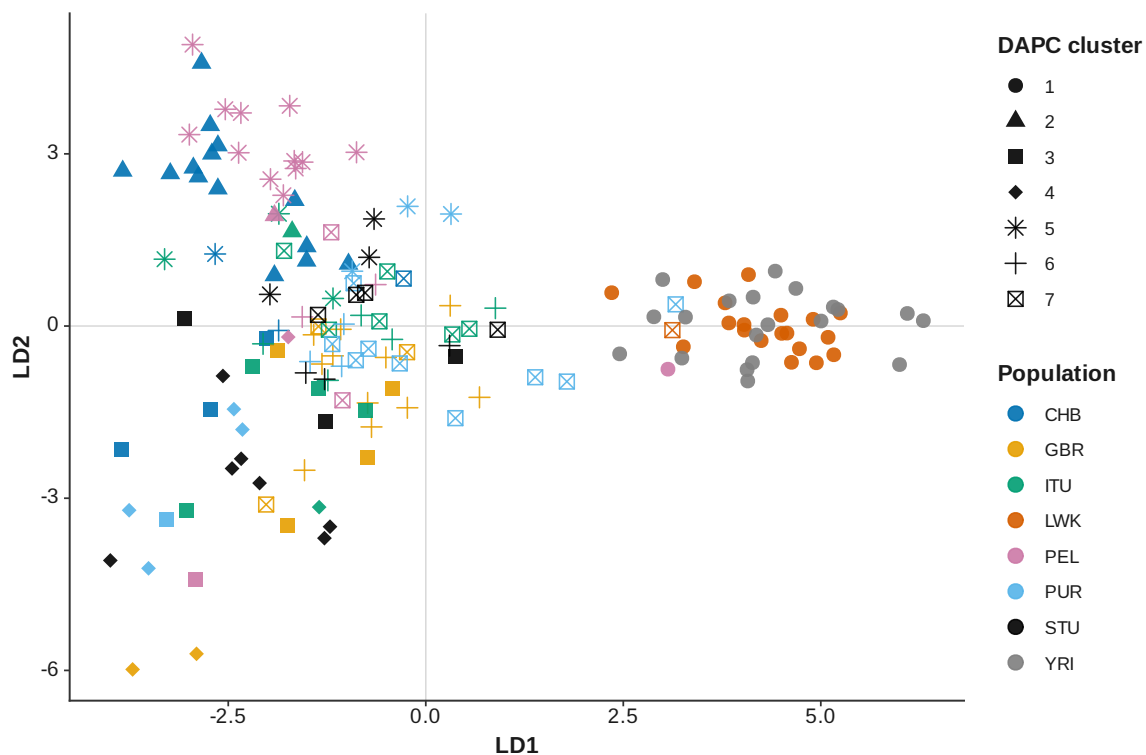


Figure 17: DAPC K 7

### Discriminant analysis of principal components (K = 8)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.1614 > 0.05).  
Avoid interpreting these assignments.**

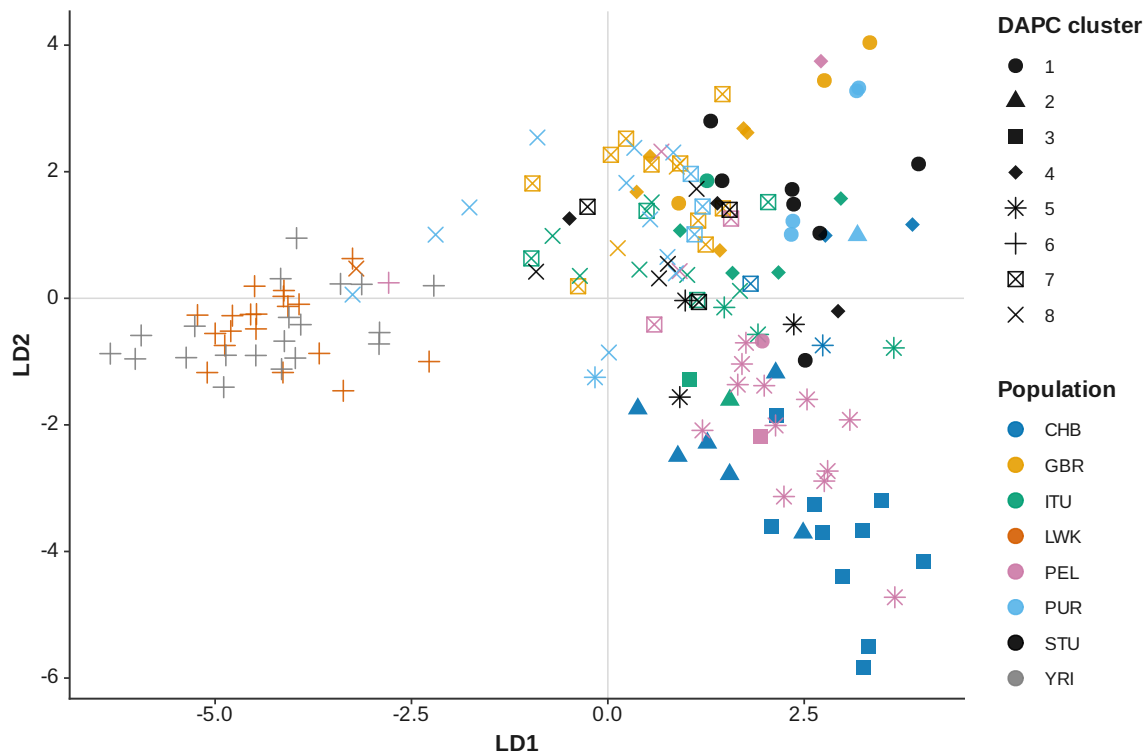


Figure 18: DAPC K 8

### Discriminant analysis of principal components (K = 9)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.183 > 0.05).  
Avoid interpreting these assignments.**

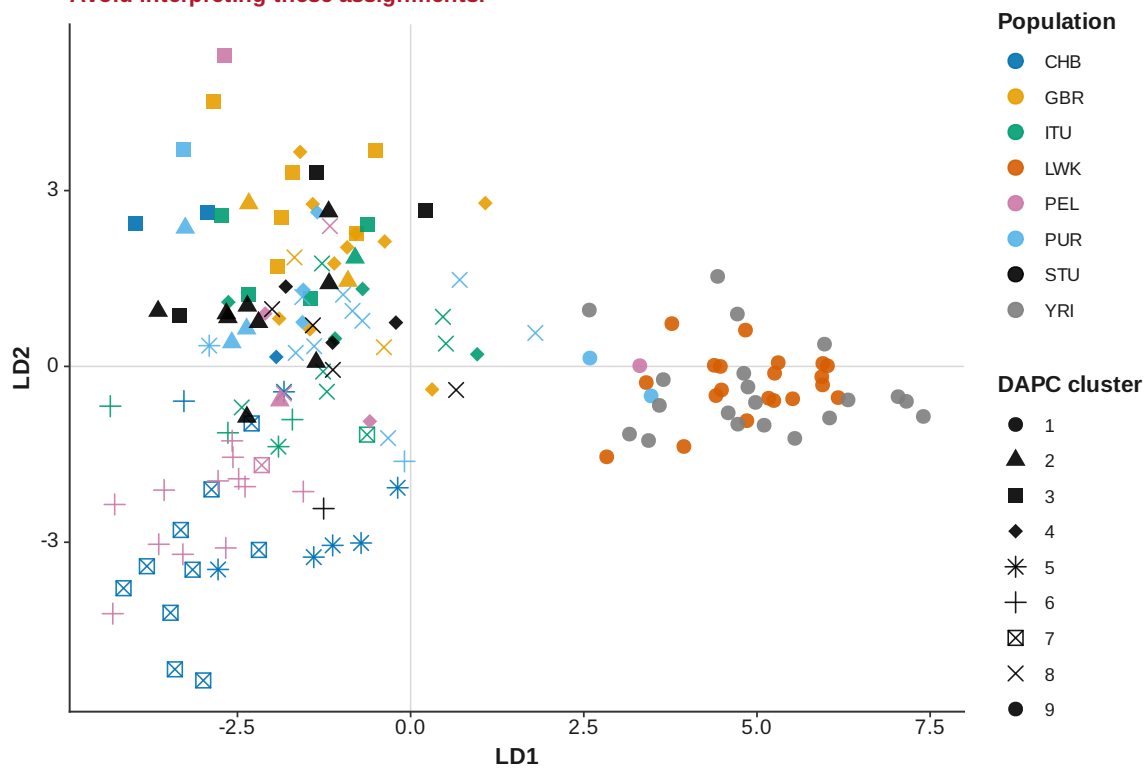


Figure 19: DAPC K 9

# Discriminant analysis of principal components (K = 10)

**WARNING: DAPC replicate membership is unstable (RMSE = 0.2373 > 0.05).  
Avoid interpreting these assignments.**

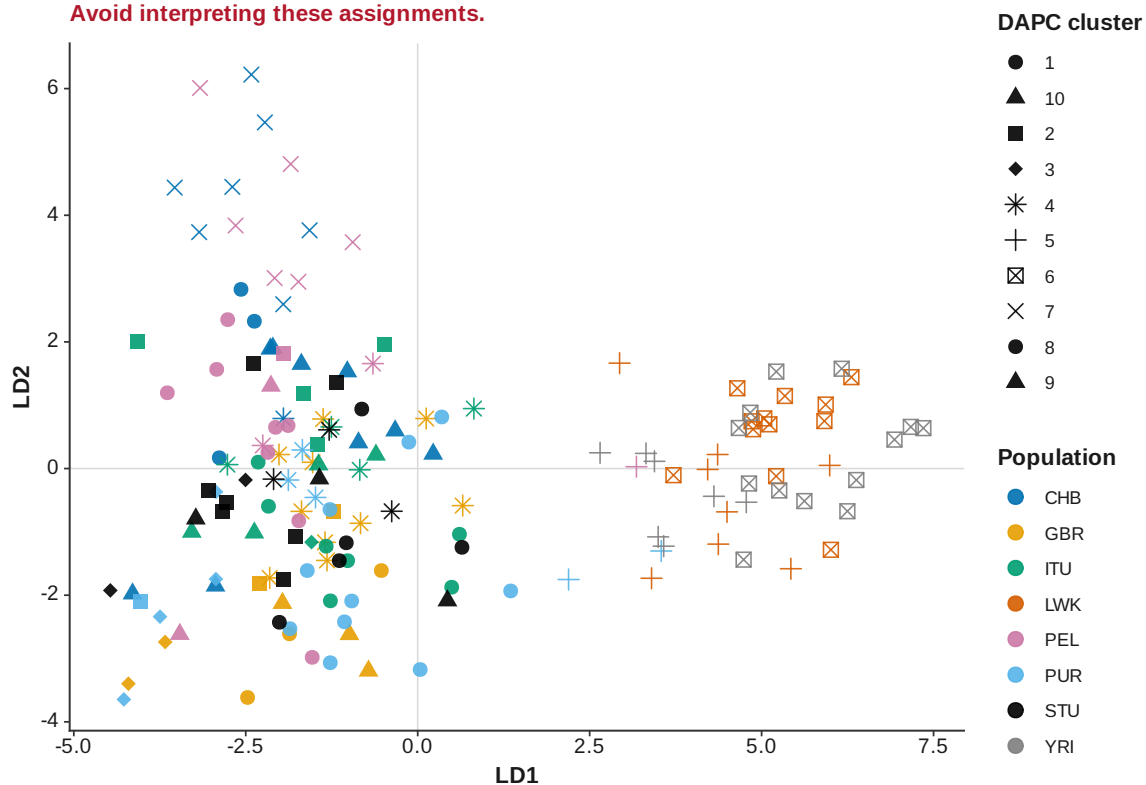


Figure 20: DAPC K 10

### Discriminant analysis of principal components cluster-number selection

Consensus number of clusters = 3; 5 of 12 method votes agree

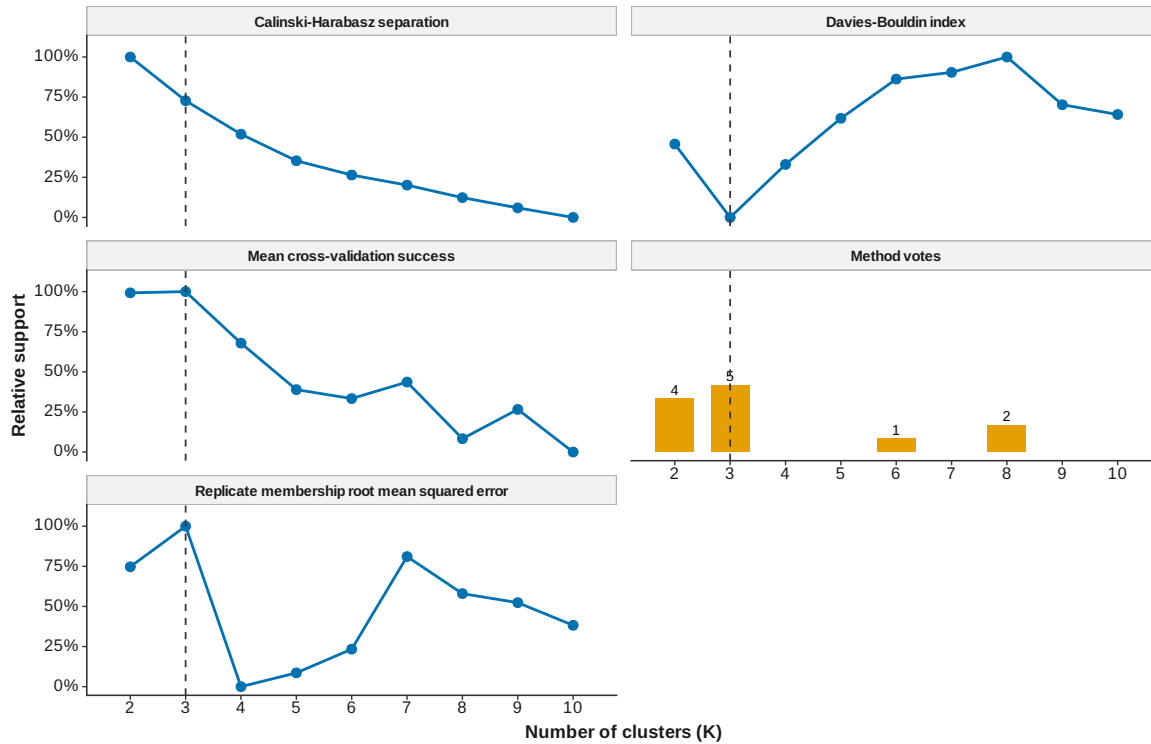


Figure 21: DAPC cluster number selection



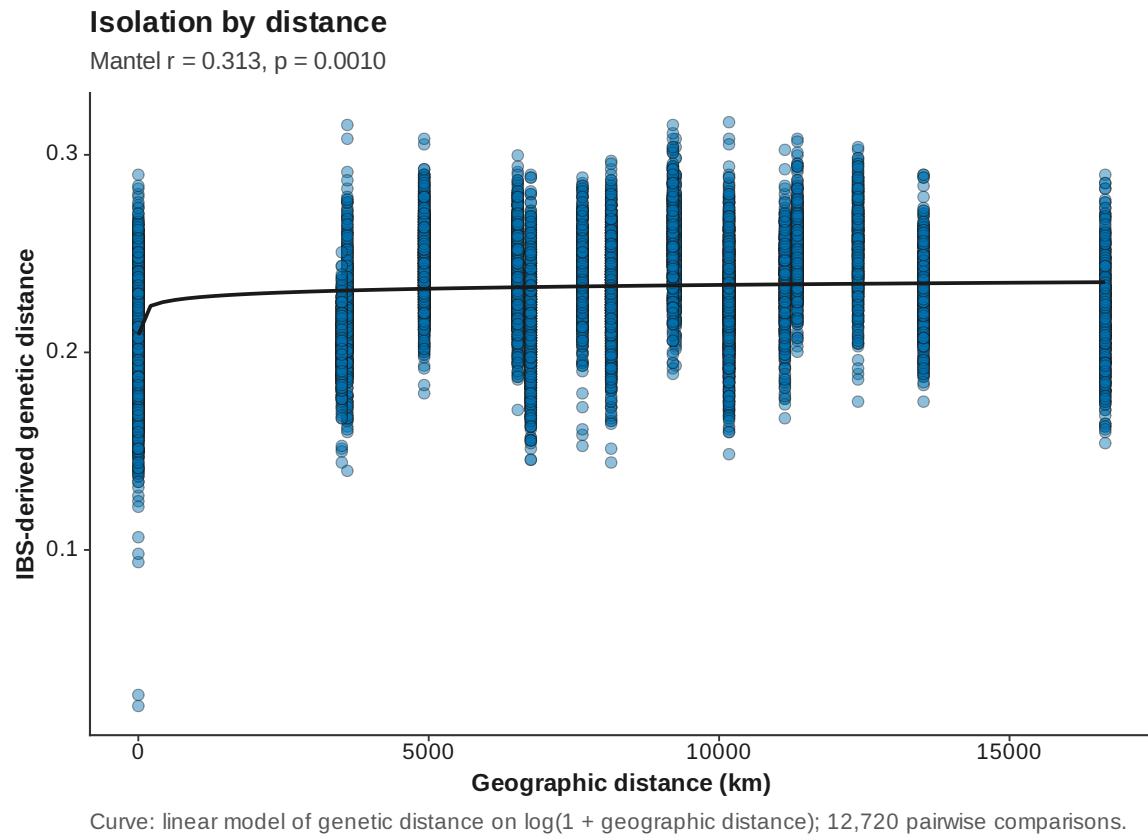


Figure 22: Isolation by distance

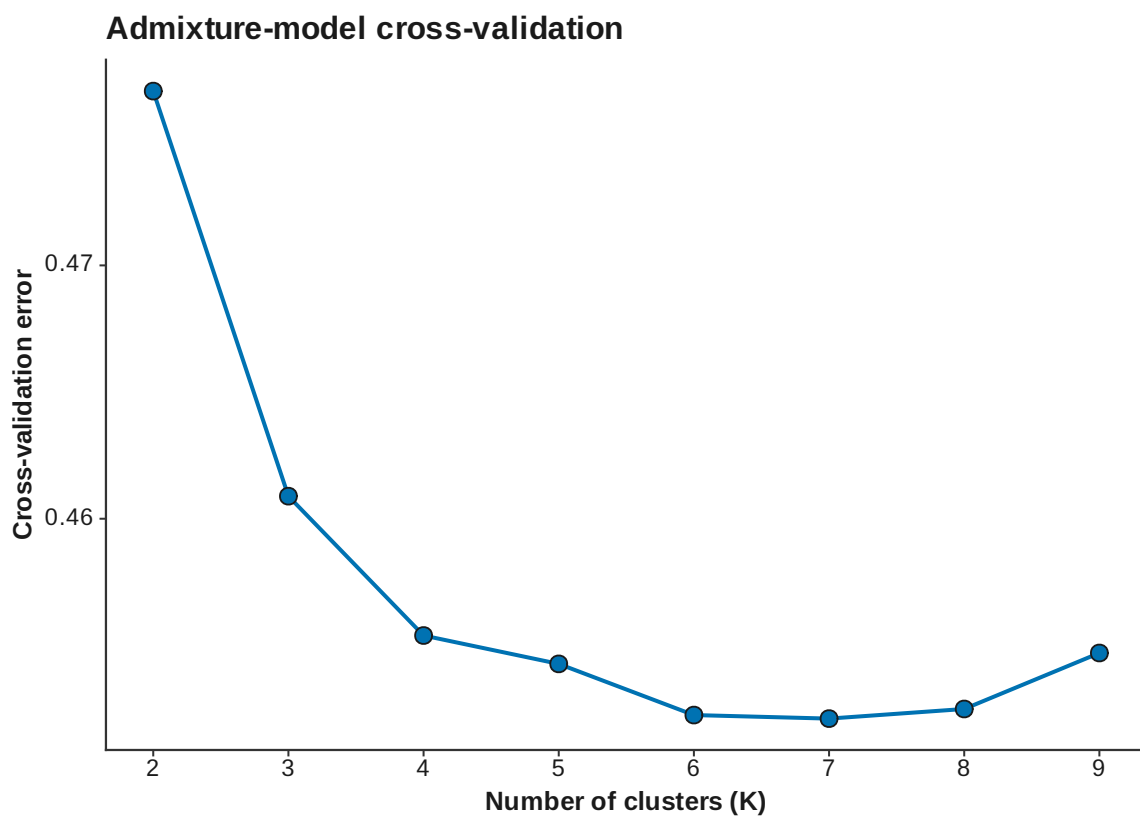


Figure 23: ADMIXTURE CV

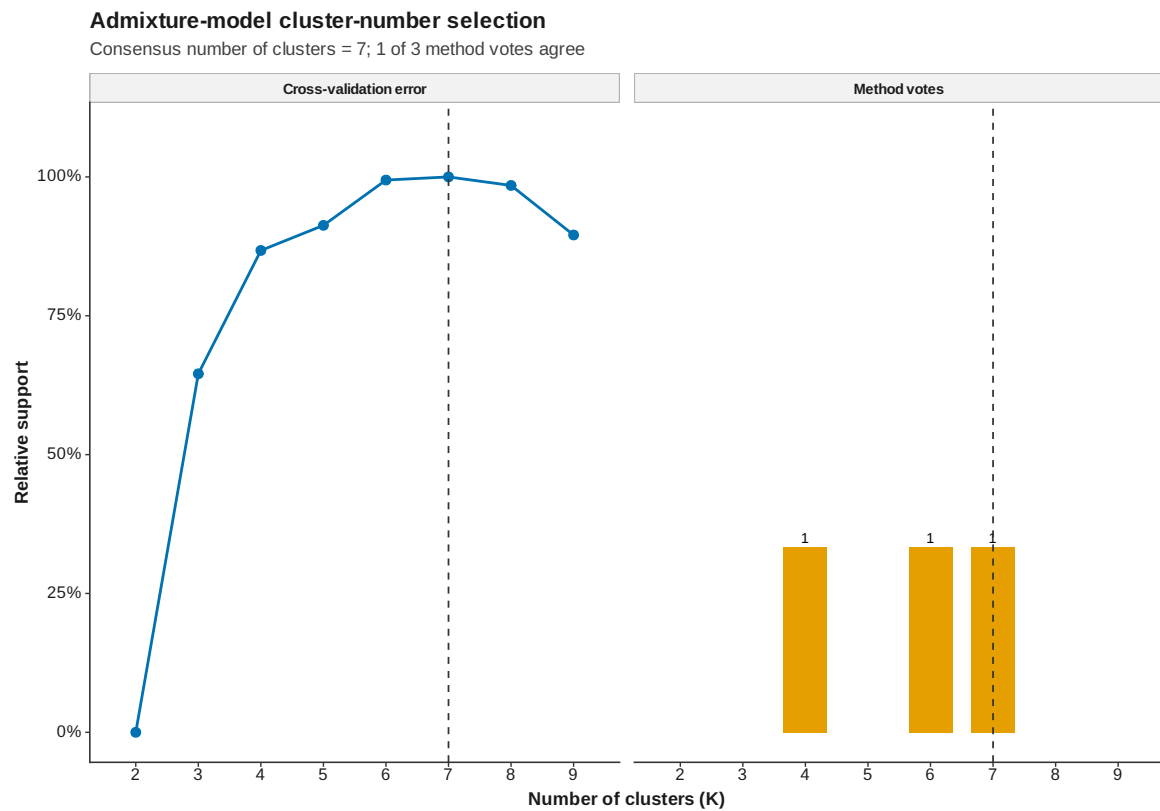


Figure 24: ADMIXTURE cluster number selection

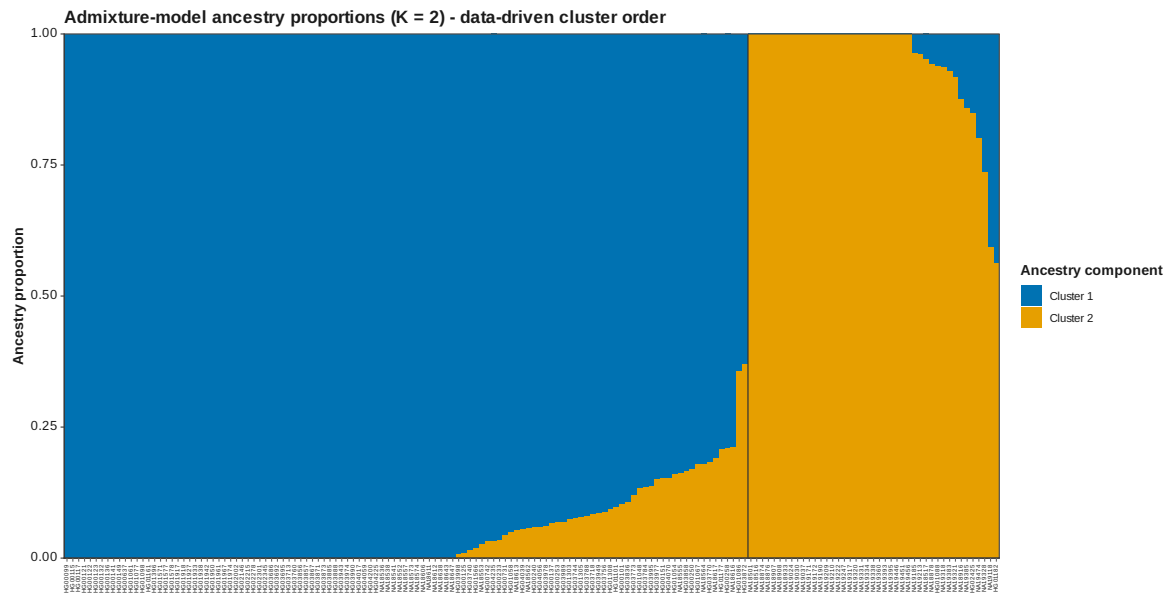


Figure 25: ADMIXTURE Q data driven K 2

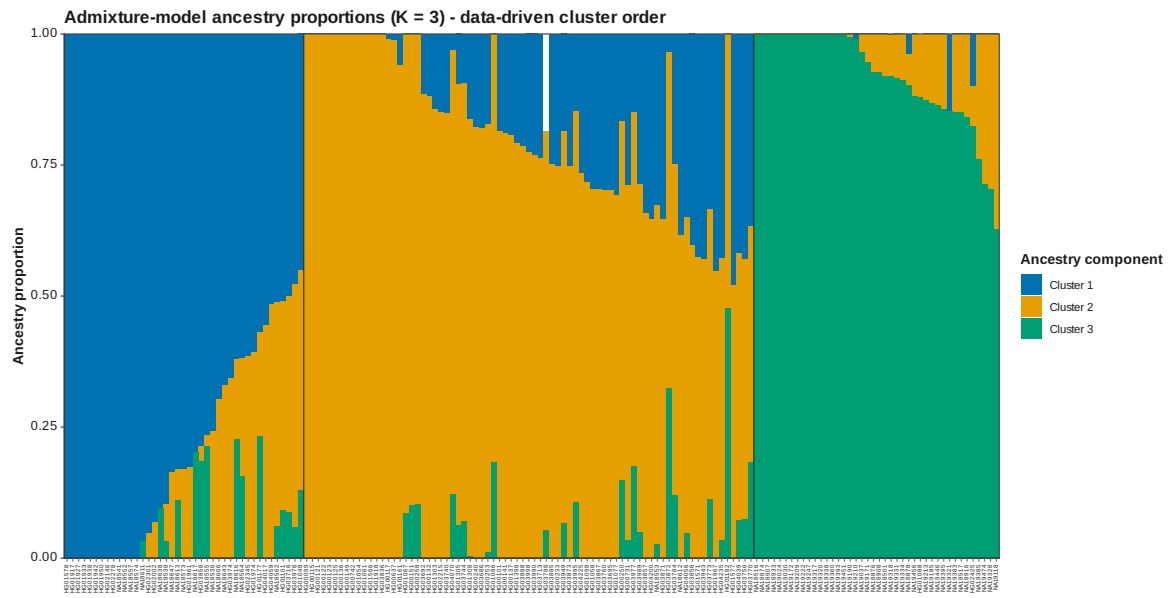


Figure 26: ADMIXTURE Q data driven K 3

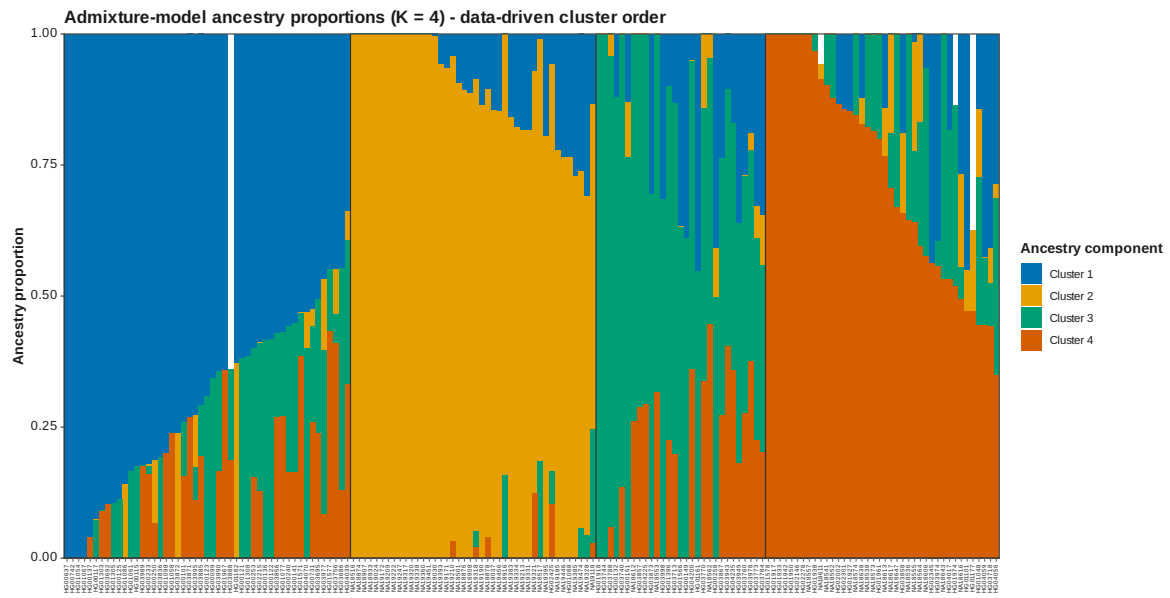


Figure 27: ADMIXTURE Q data driven K 4

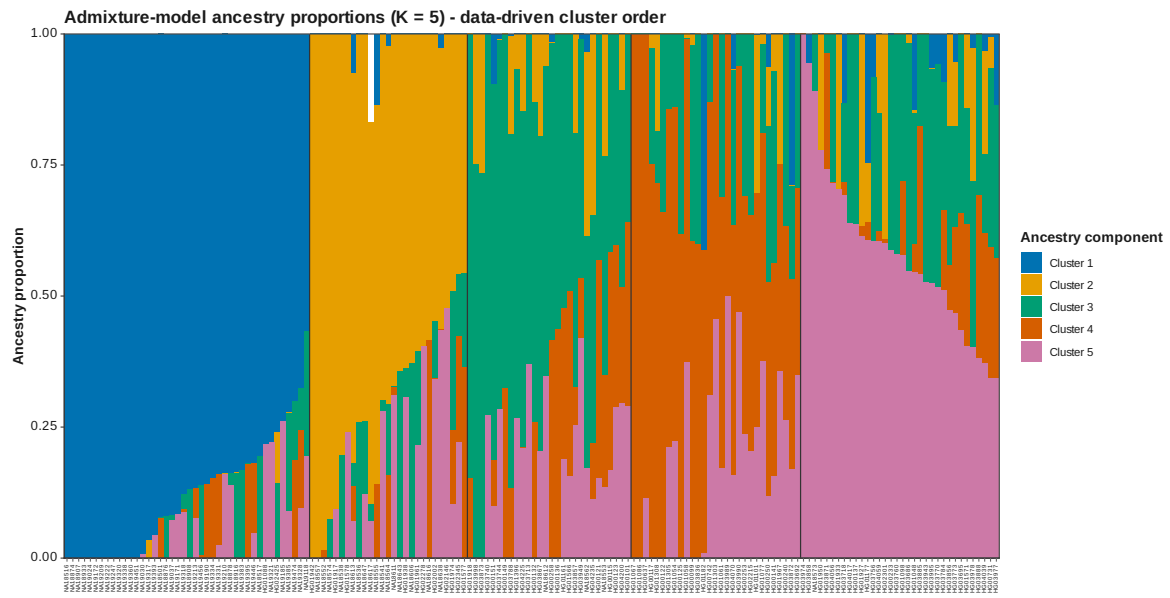


Figure 28: ADMIXTURE Q data driven K 5

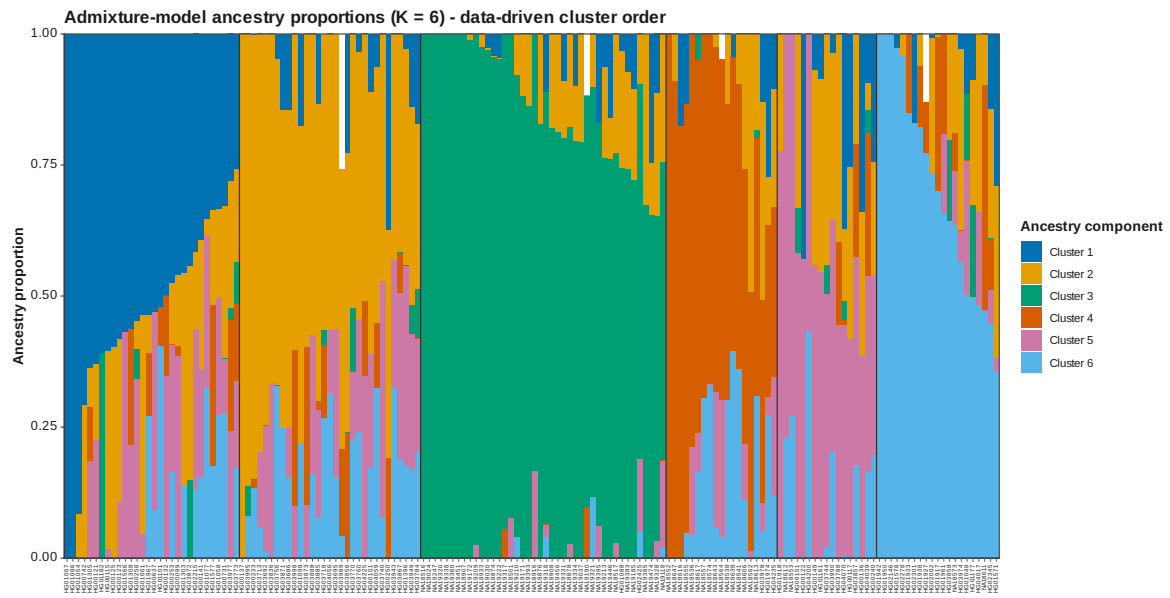


Figure 29: ADMIXTURE Q data driven K 6



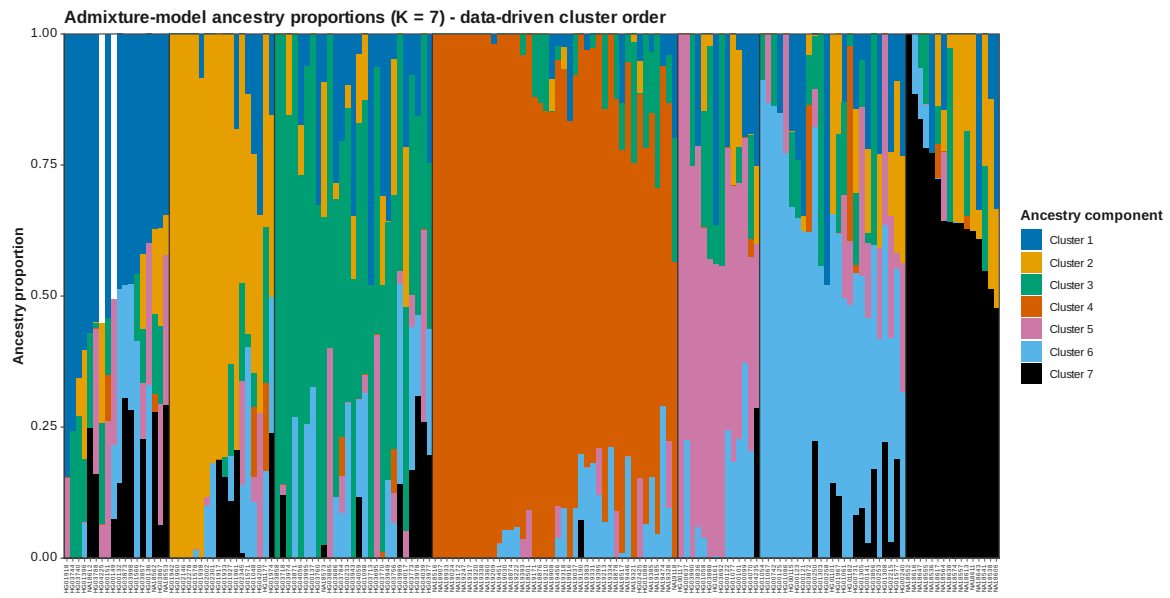


Figure 30: ADMIXTURE Q data driven K 7

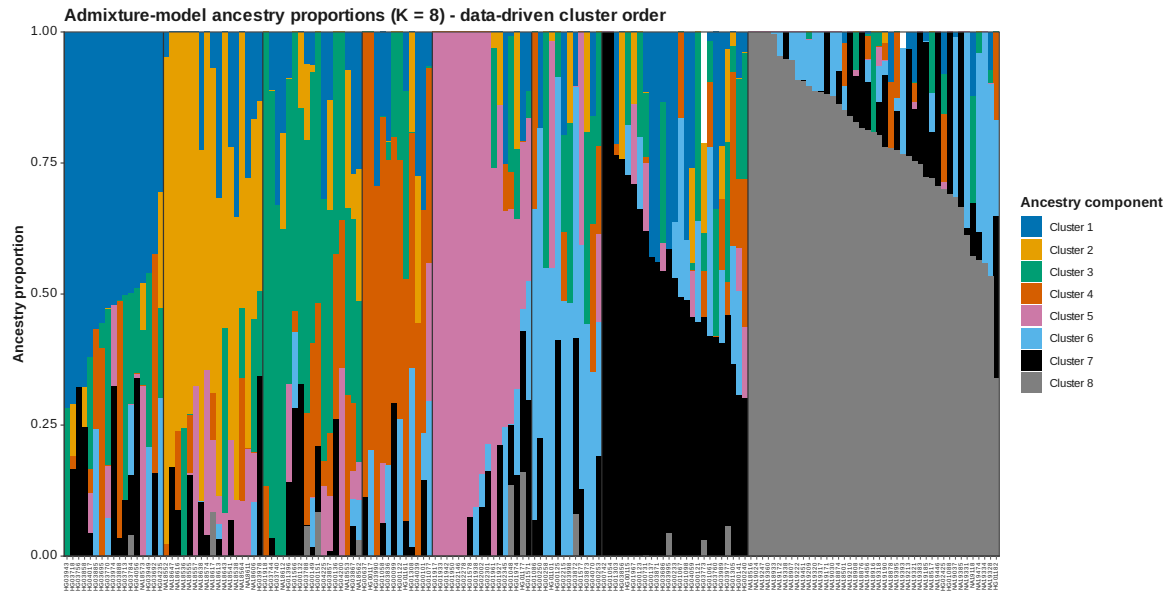


Figure 31: ADMIXTURE Q data driven K 8

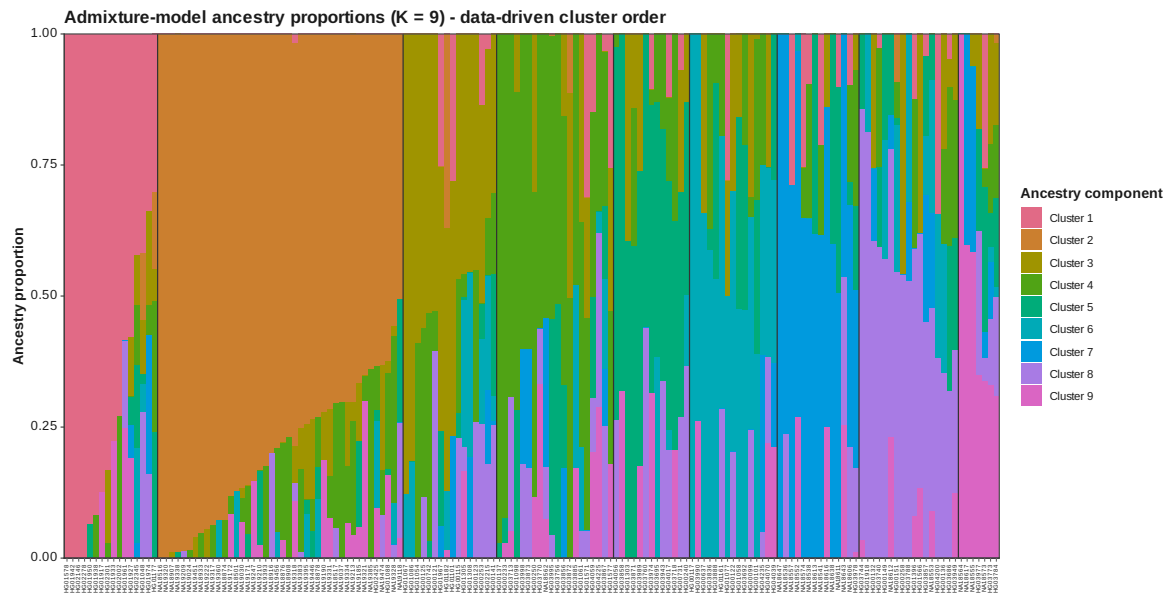


Figure 32: ADMIXTURE Q data driven K 9

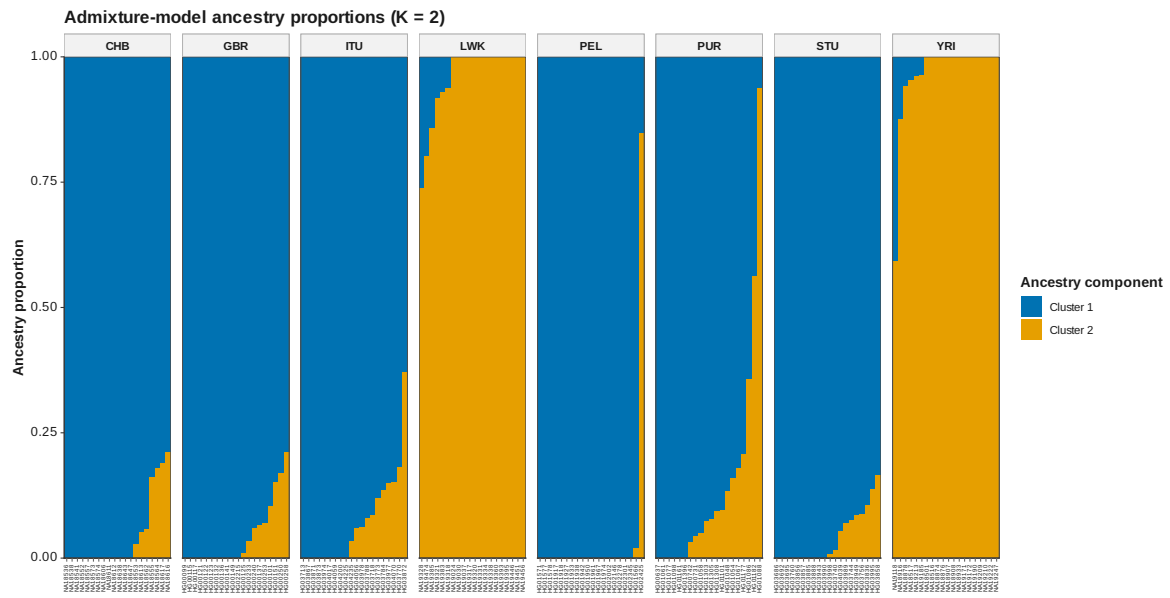


Figure 33: ADMIXTURE Q K 2

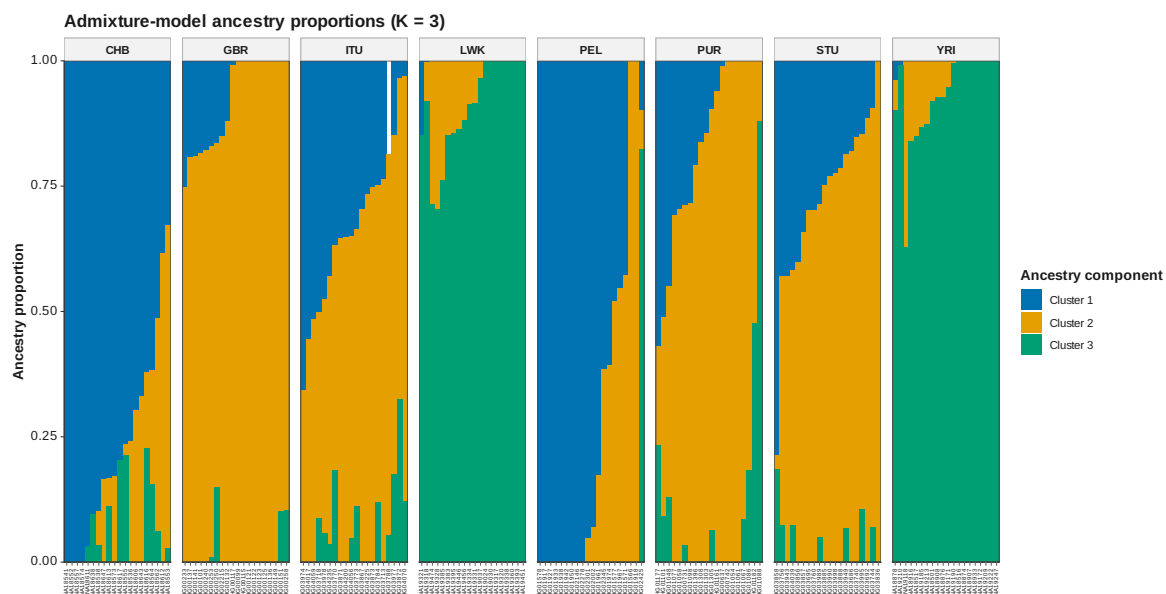


Figure 34: ADMIXTURE Q K 3

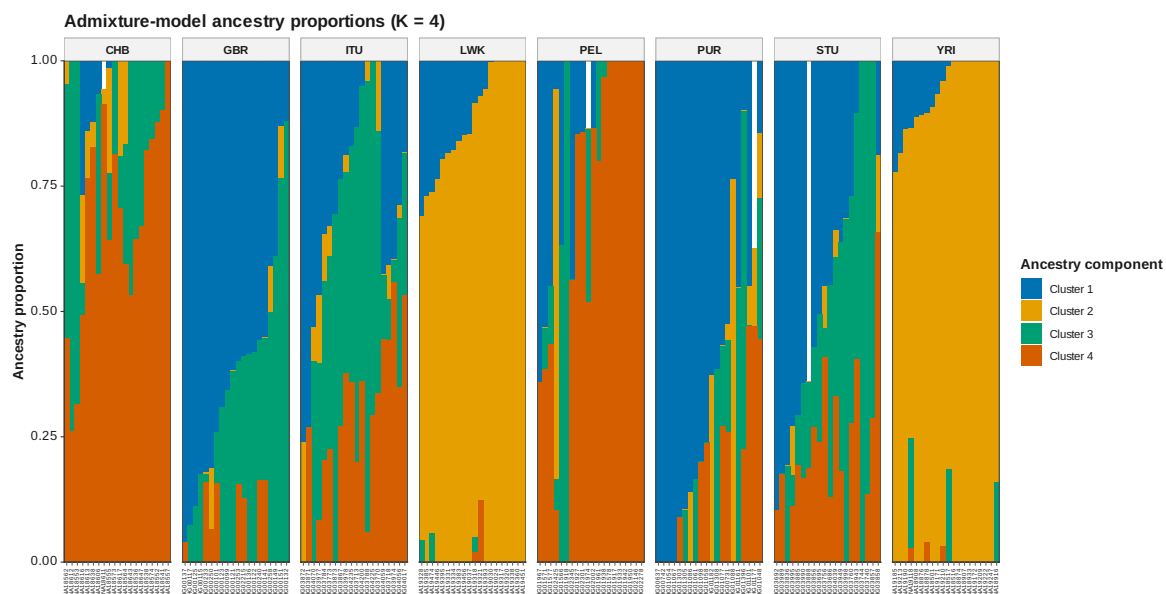


Figure 35: ADMIXTURE Q K 4

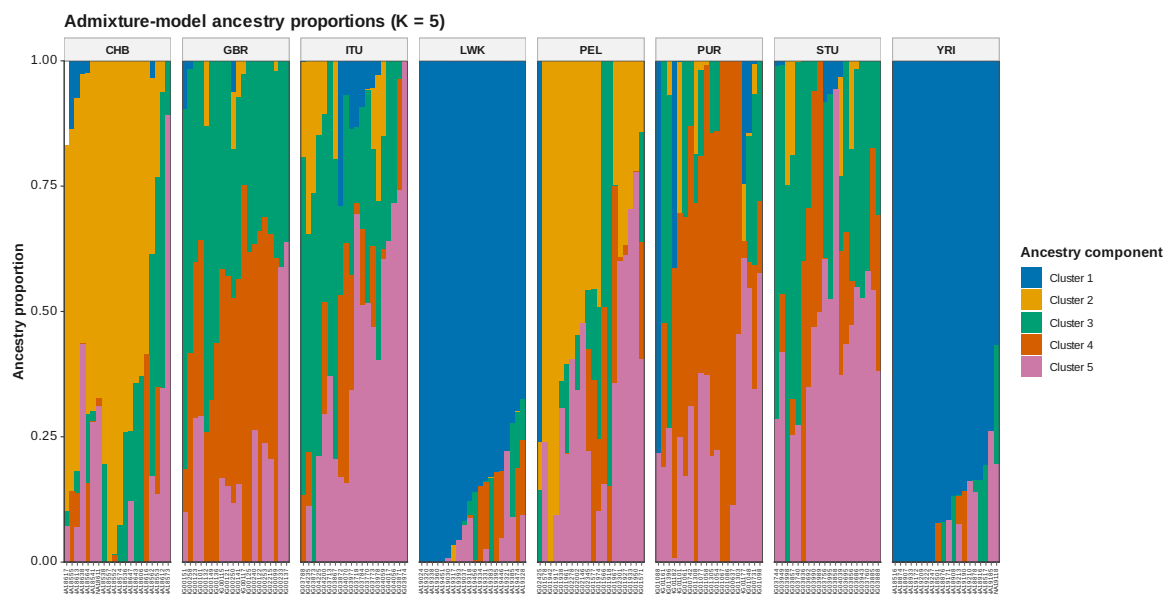


Figure 36: ADMIXTURE Q K 5

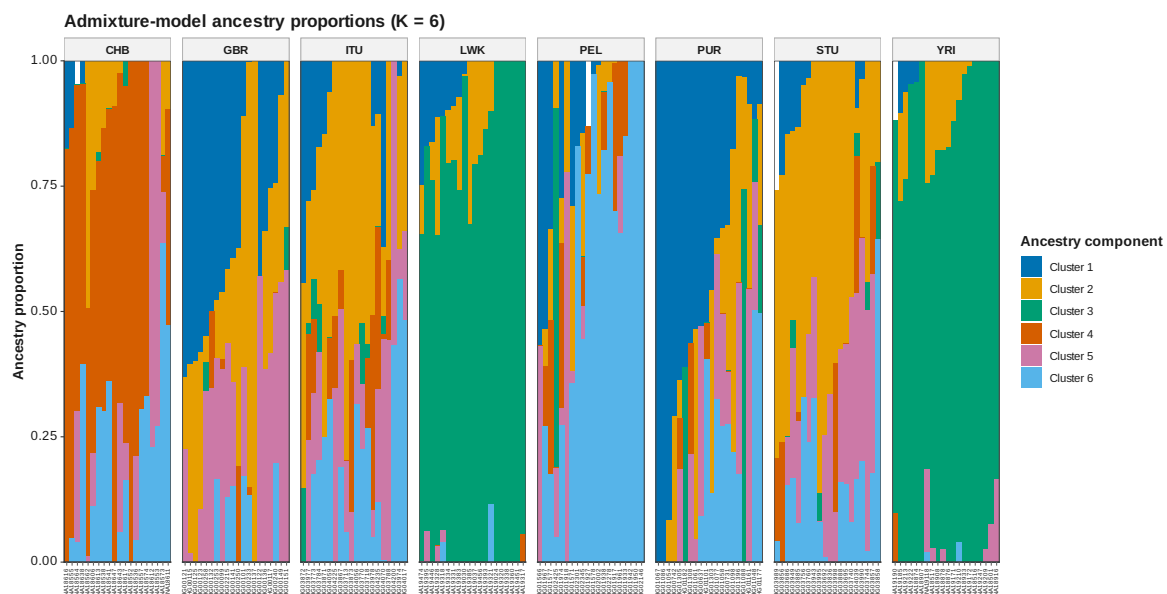


Figure 37: ADMIXTURE Q K 6



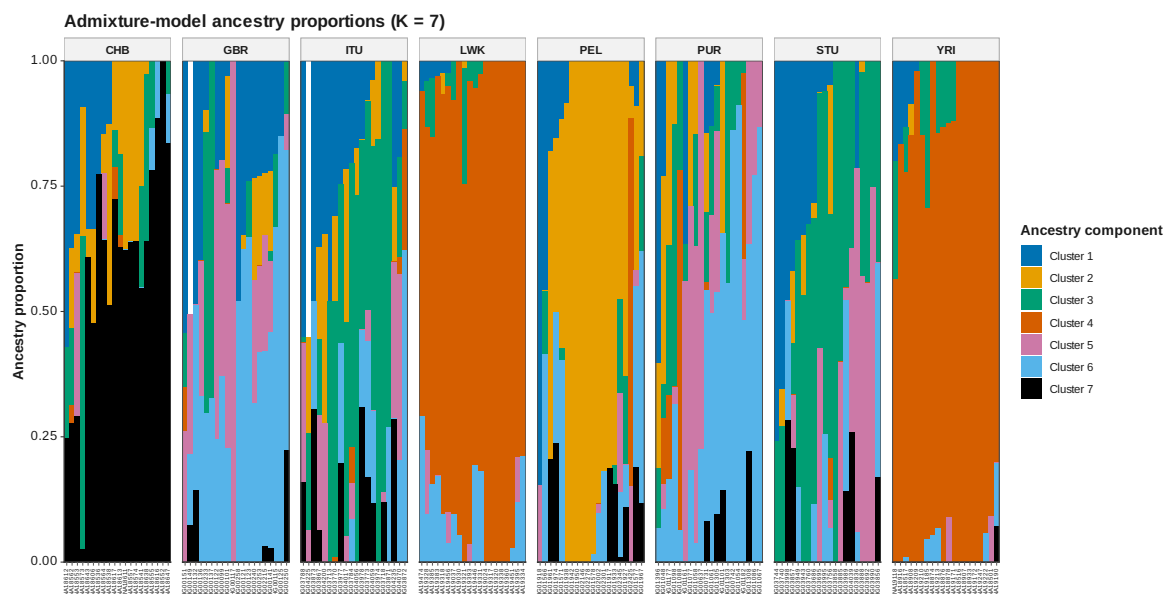


Figure 38: ADMIXTURE Q K 7

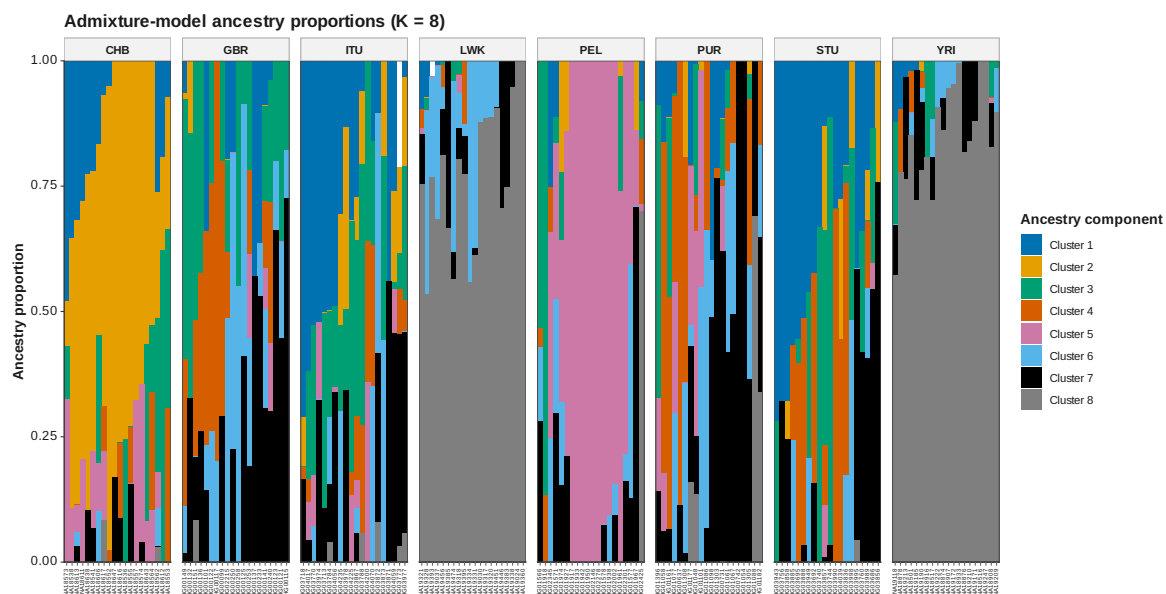


Figure 39: ADMIXTURE Q K 8

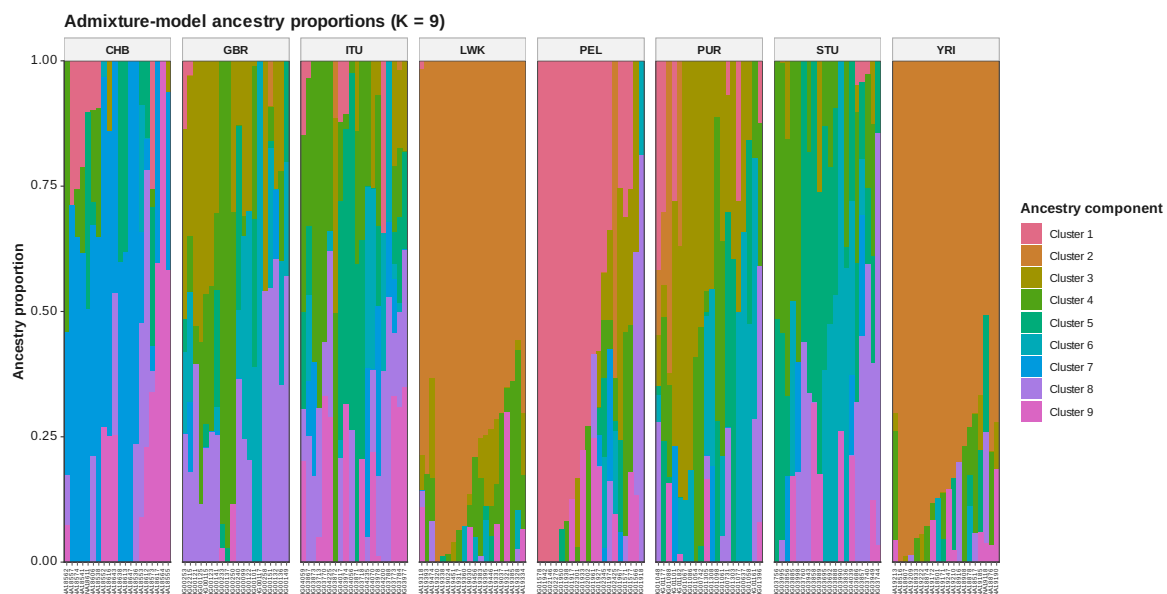


Figure 40: ADMIXTURE Q K 9

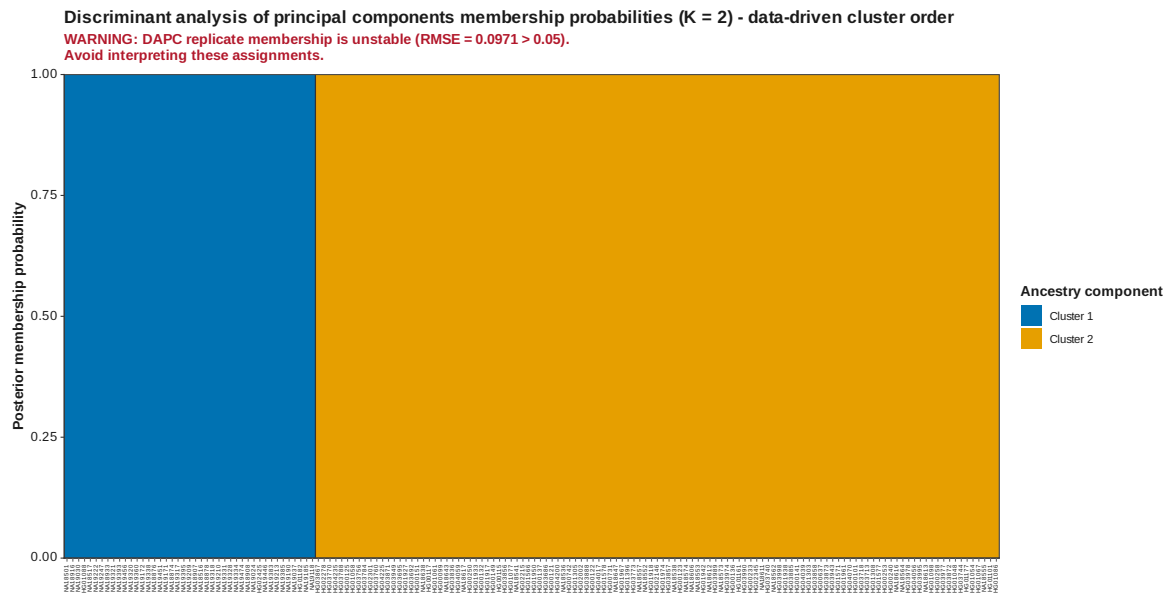


Figure 41: DAPC membership data driven K 2

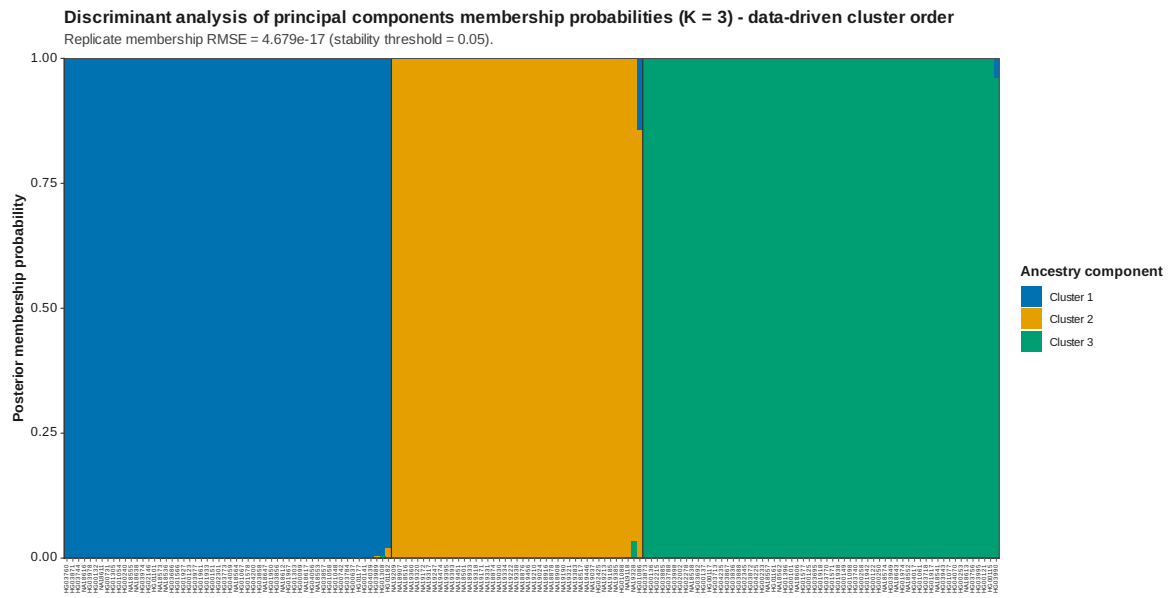


Figure 42: DAPC membership data driven K 3

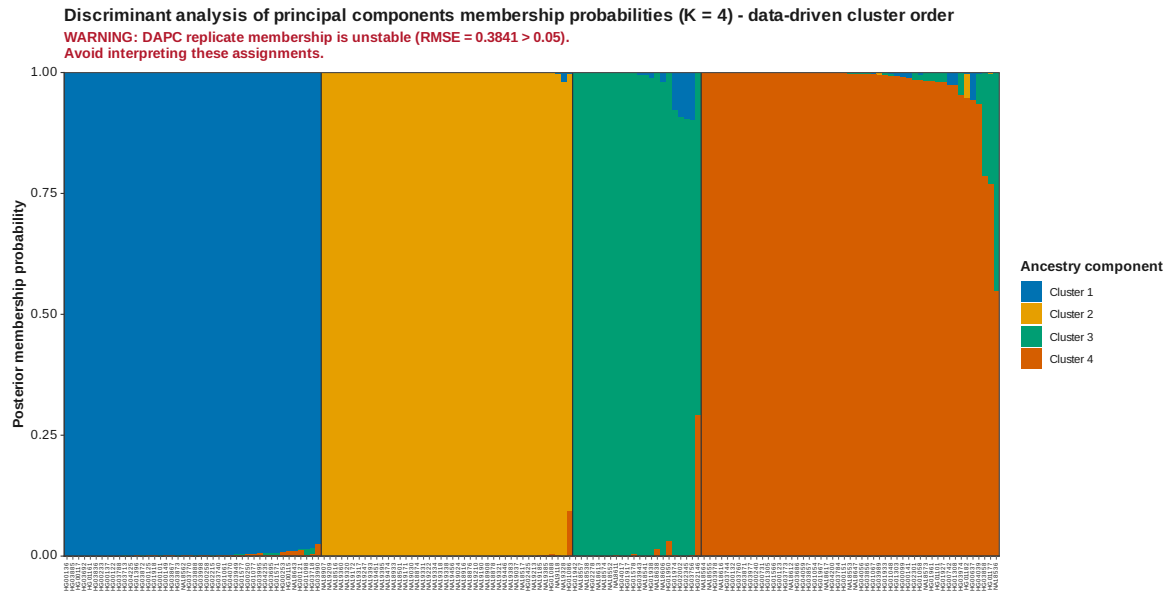


Figure 43: DAPC membership data driven K 4

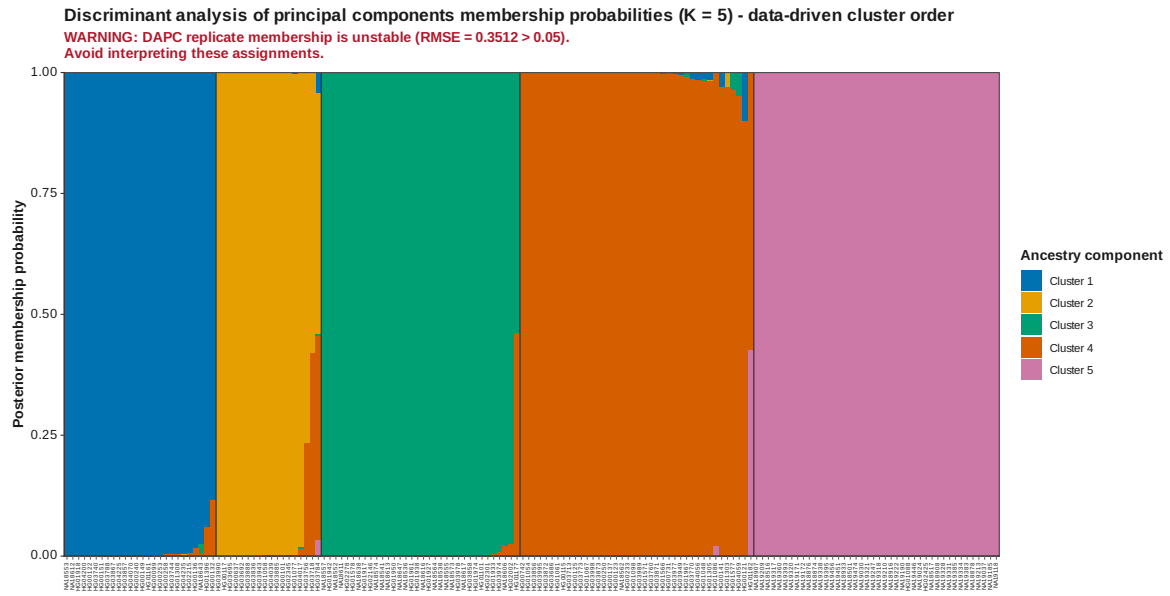


Figure 44: DAPC membership data driven K 5

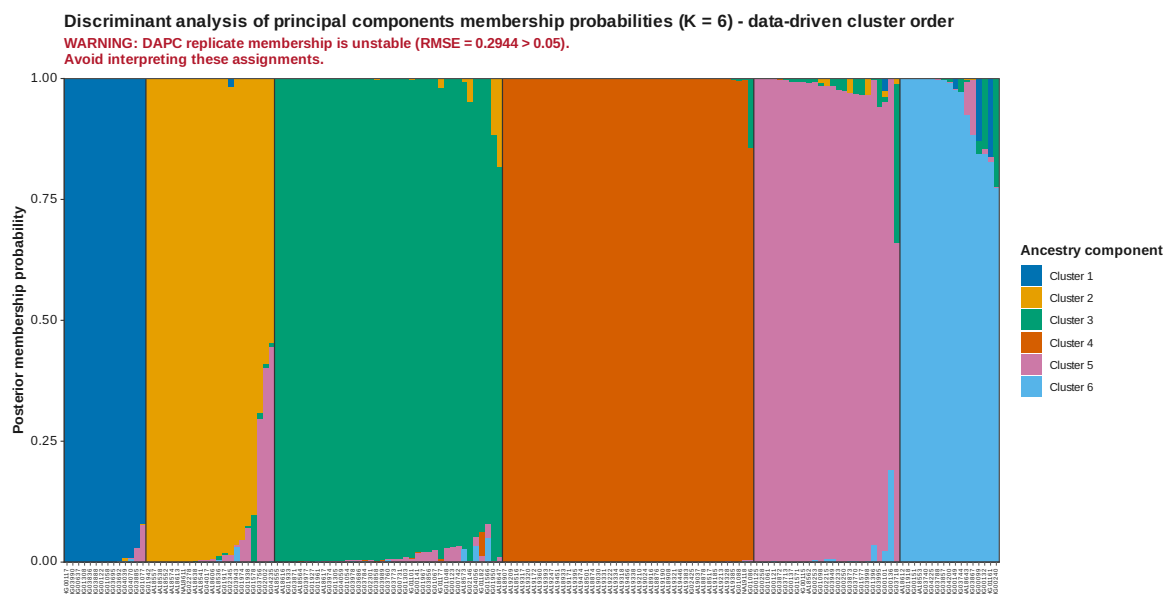


Figure 45: DAPC membership data driven K 6



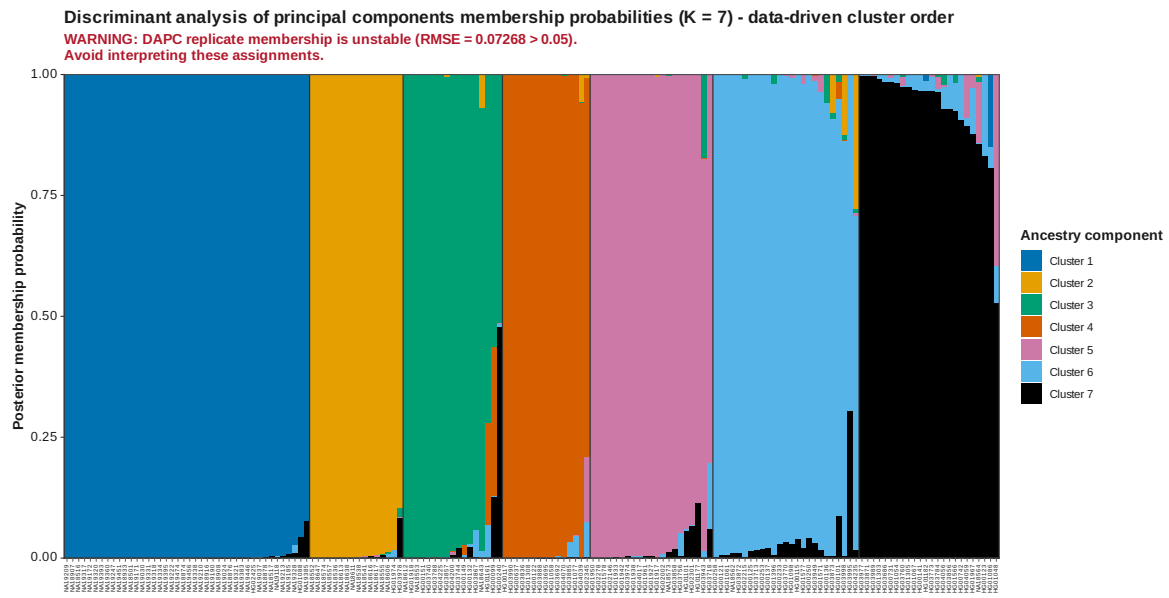


Figure 46: DAPC membership data driven K 7

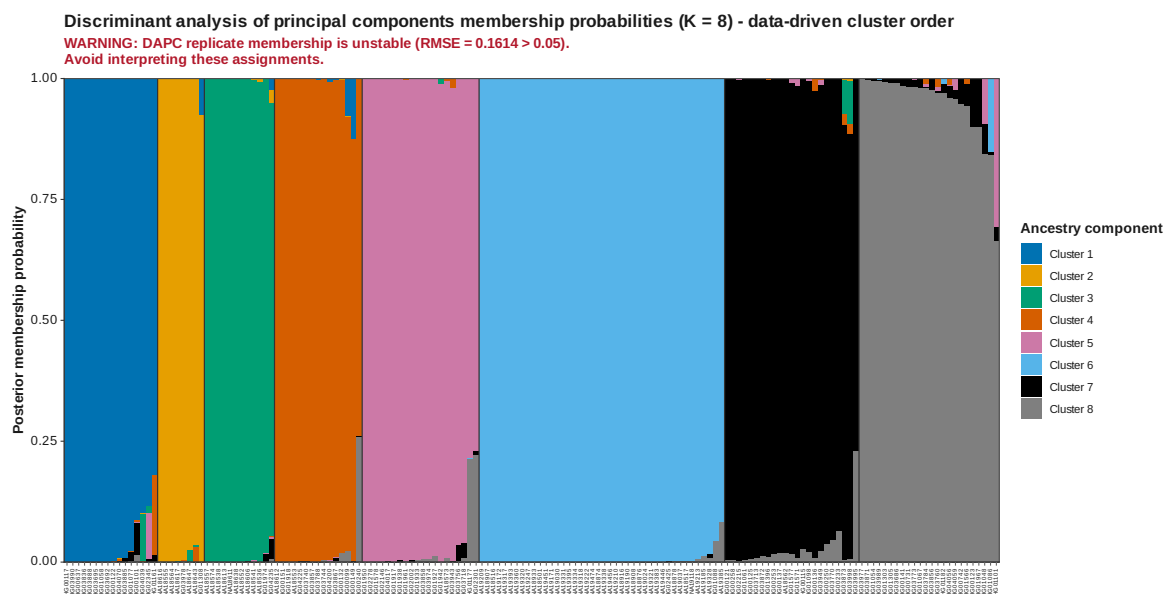


Figure 47: DAPC membership data driven K 8

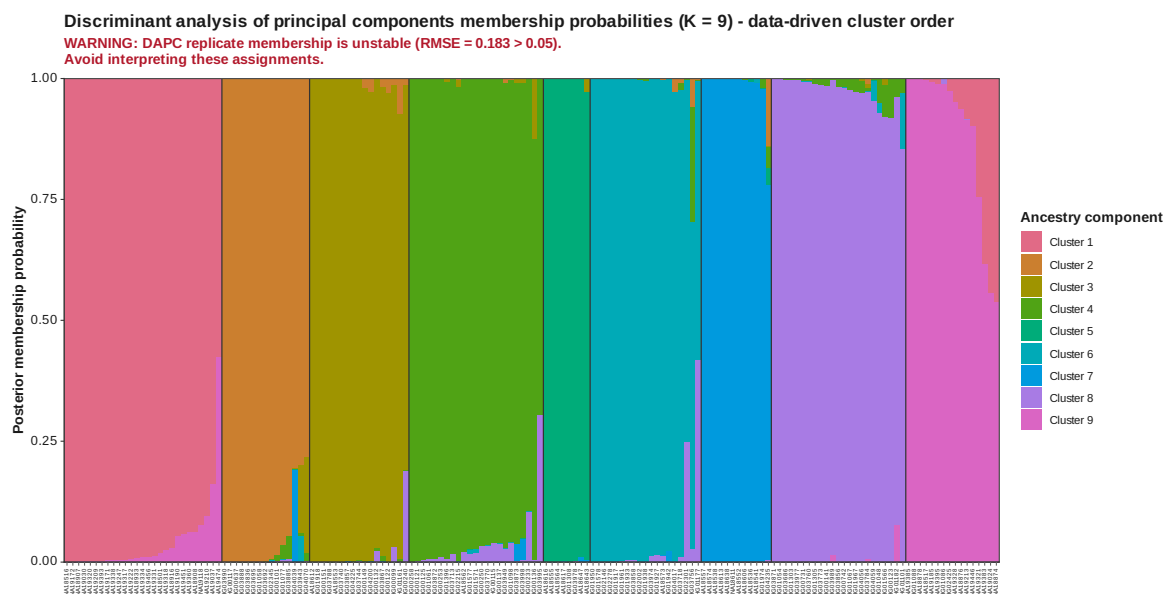


Figure 48: DAPC membership data driven K 9

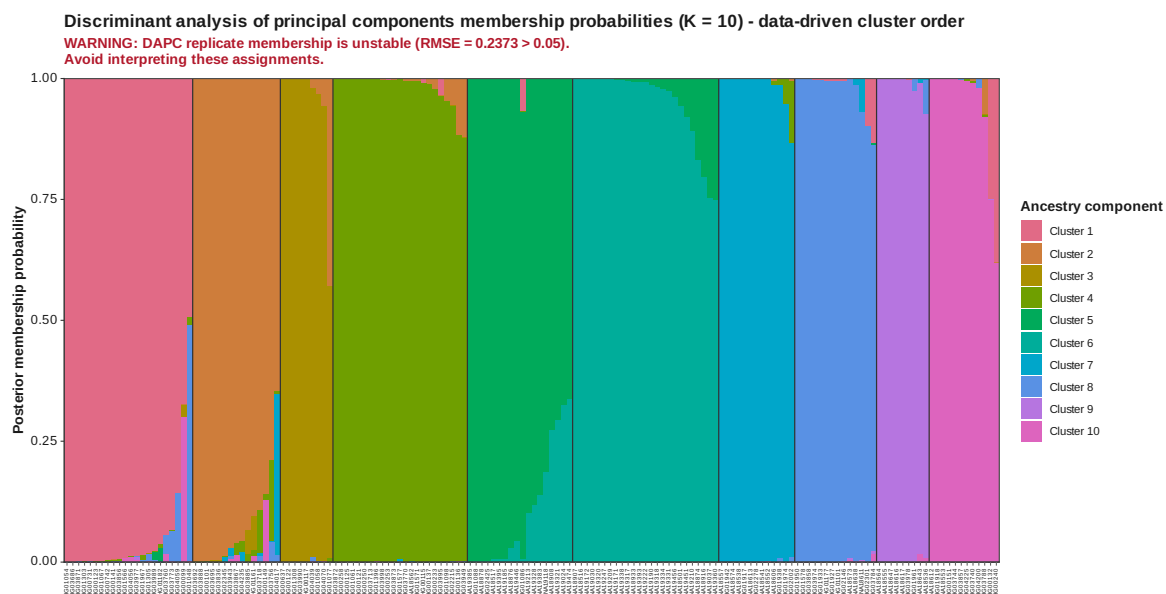


Figure 49: DAPC membership data driven K 10

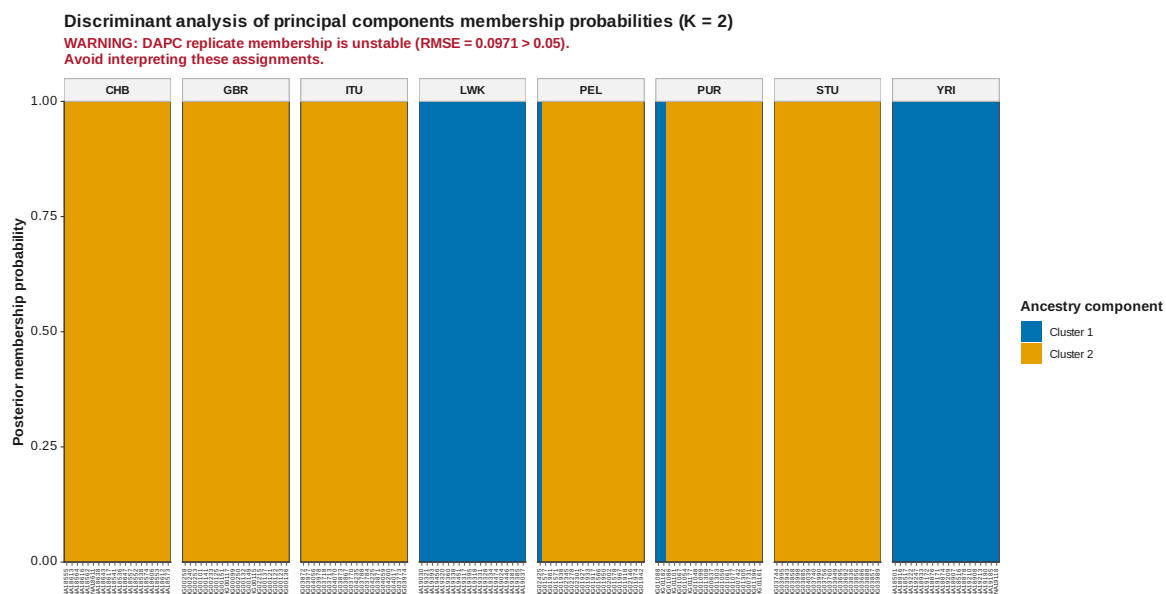


Figure 50: DAPC membership K 2

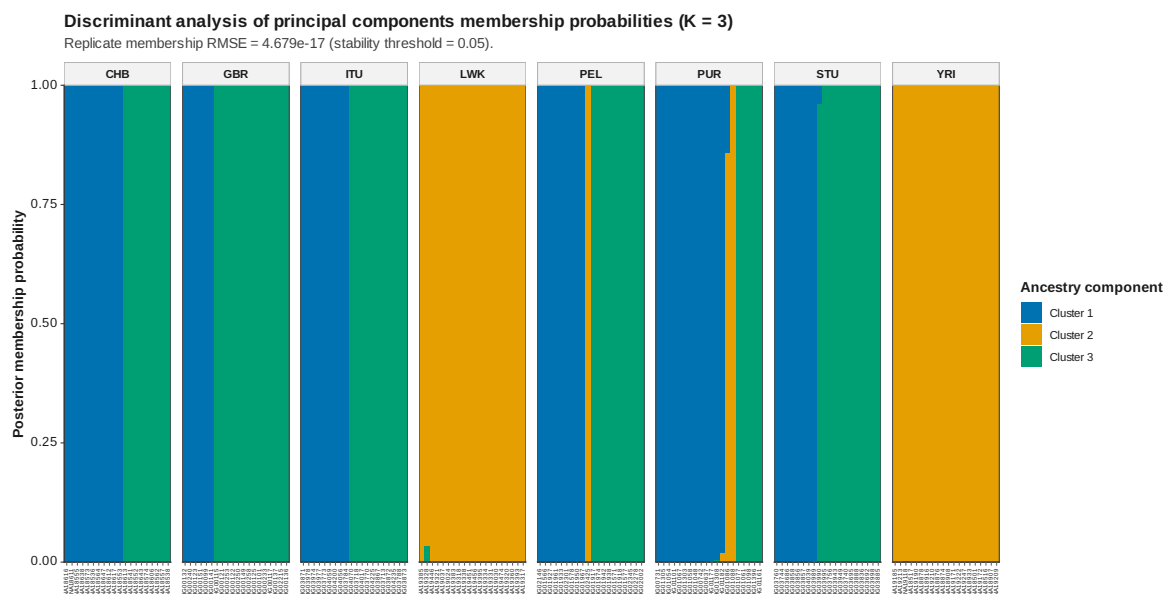


Figure 51: DAPC membership K 3

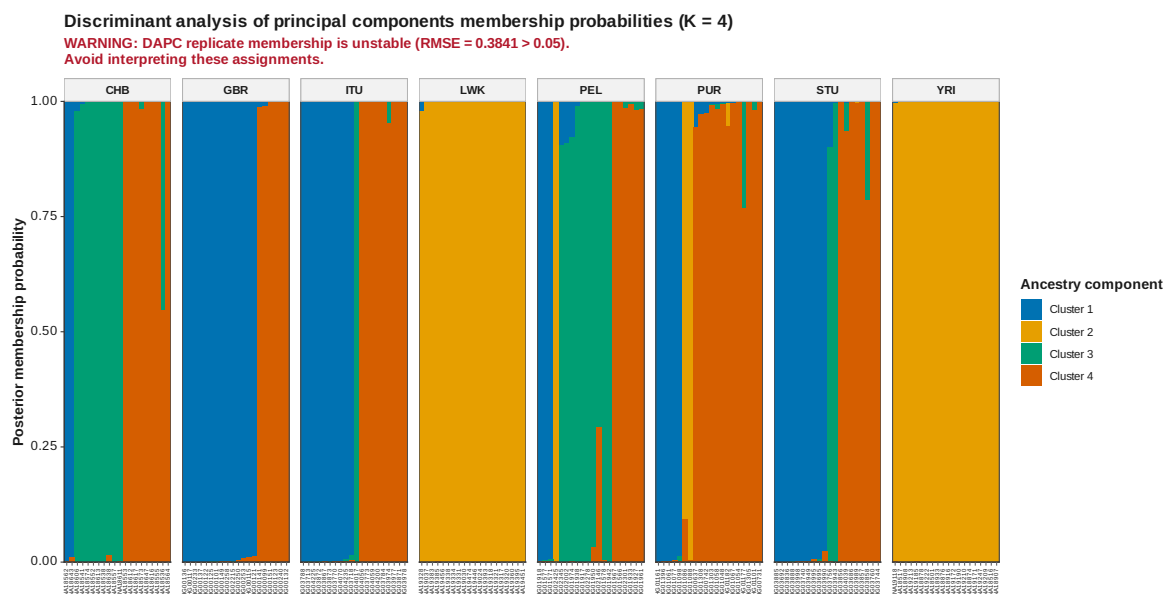


Figure 52: DAPC membership K 4

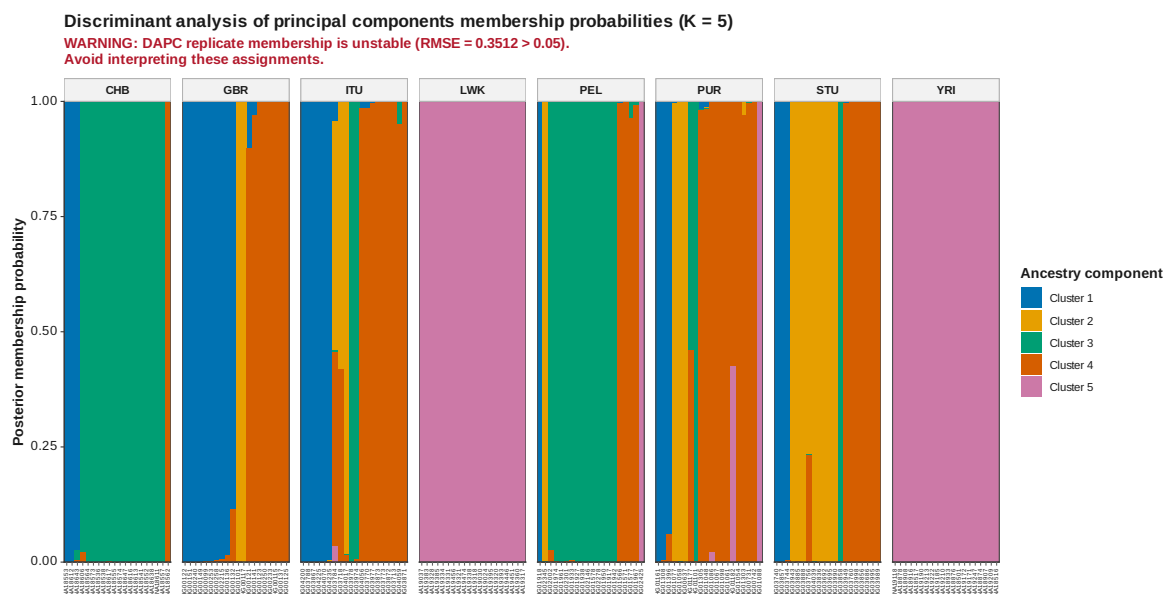


Figure 53: DAPC membership K 5



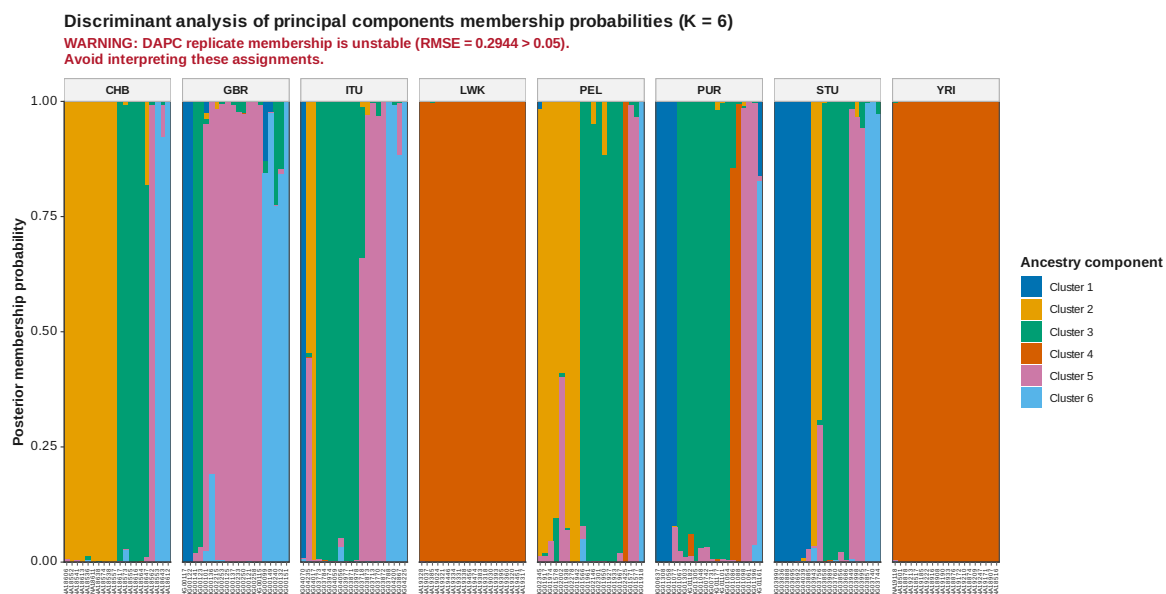


Figure 54: DAPC membership K 6

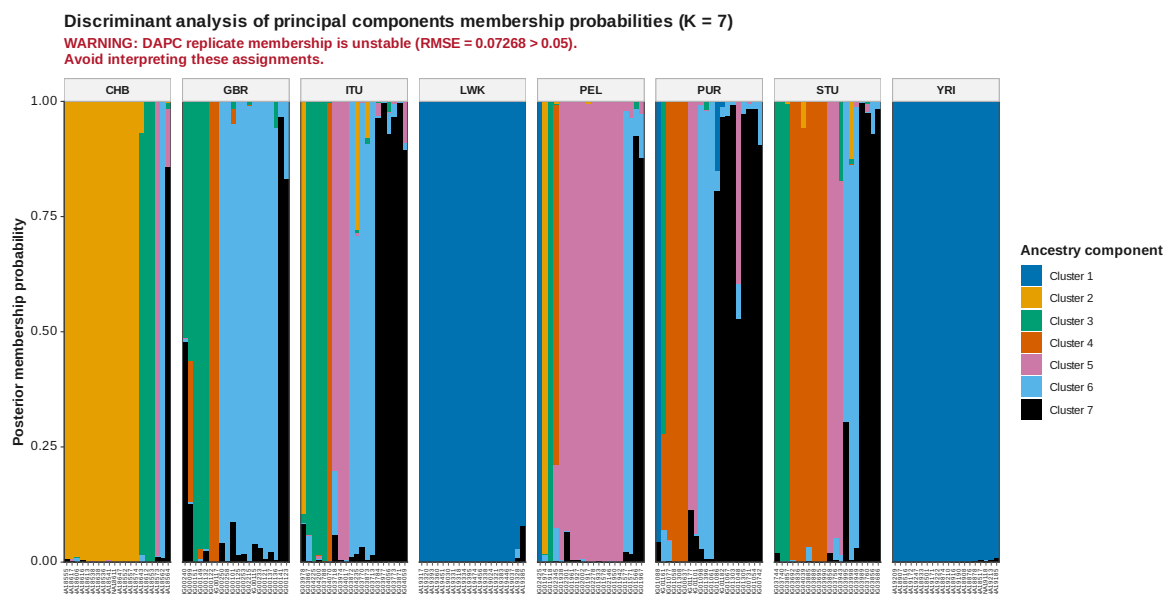


Figure 55: DAPC membership K 7

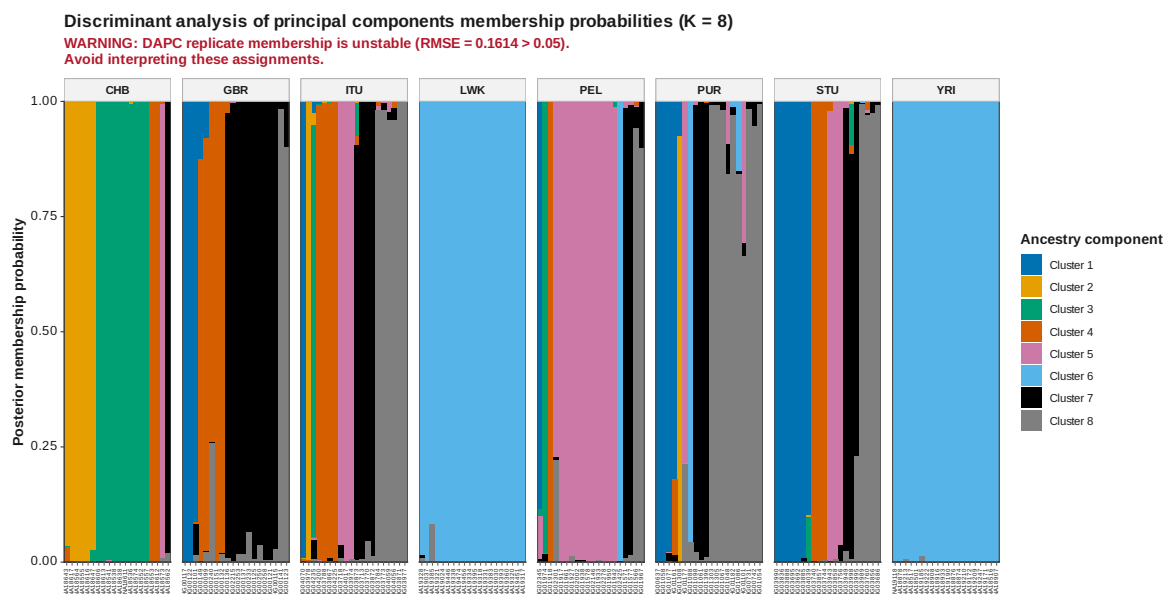


Figure 56: DAPC membership K 8

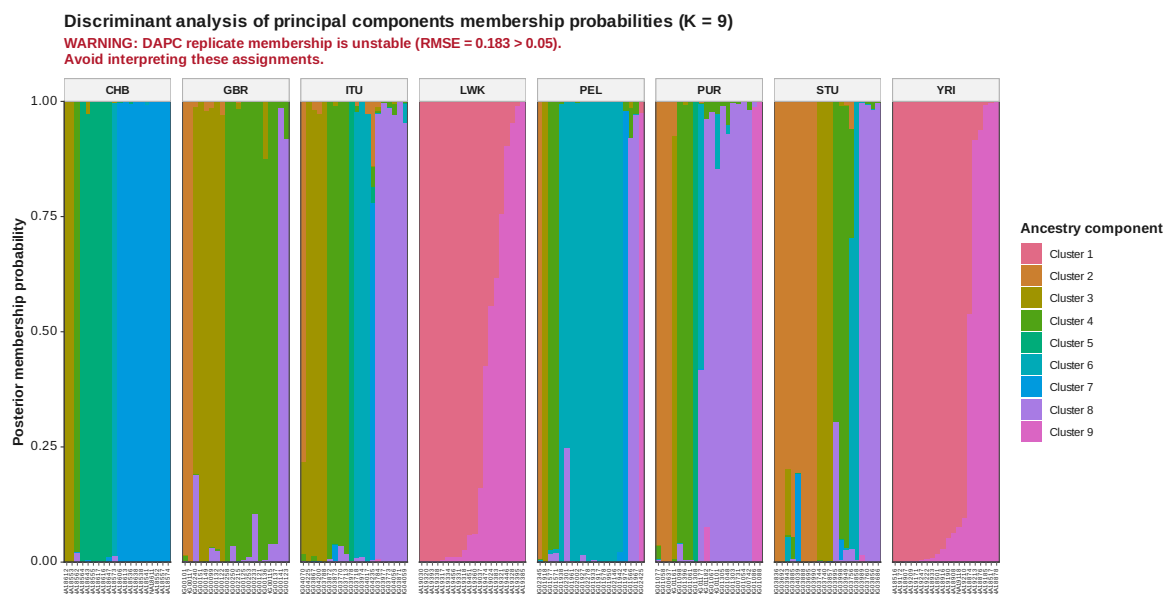


Figure 57: DAPC membership K 9

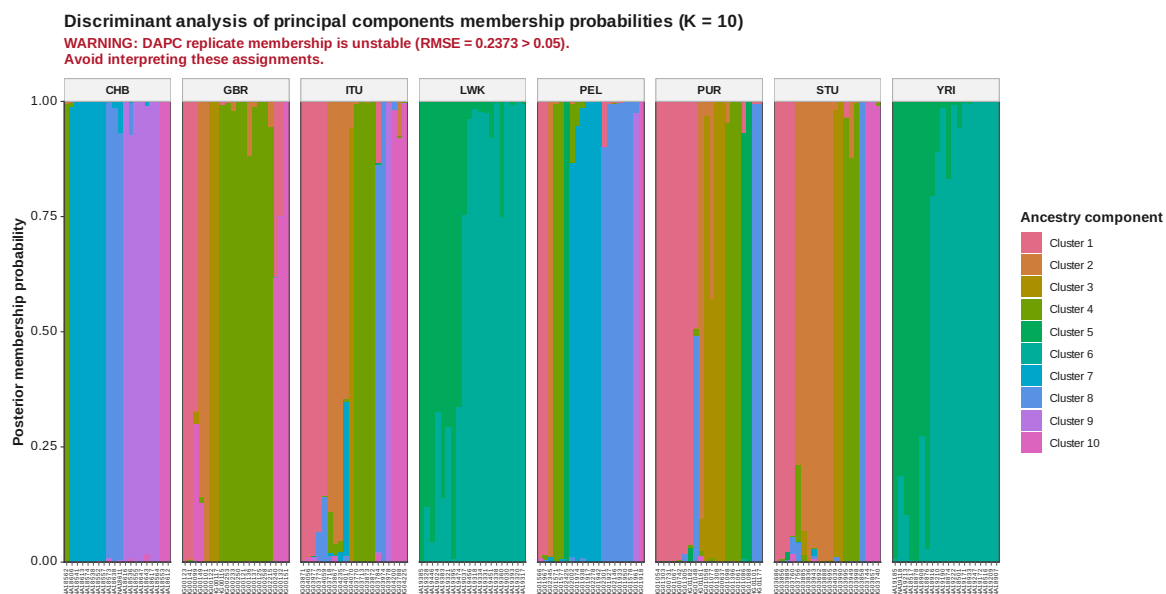


Figure 58: DAPC membership K 10

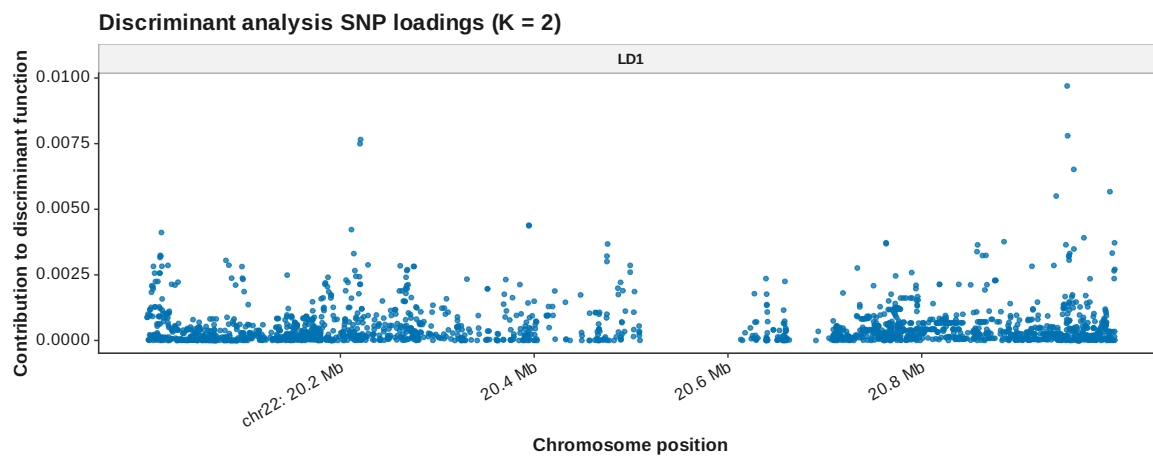


Figure 59: DAPC loadings manhattan K 2

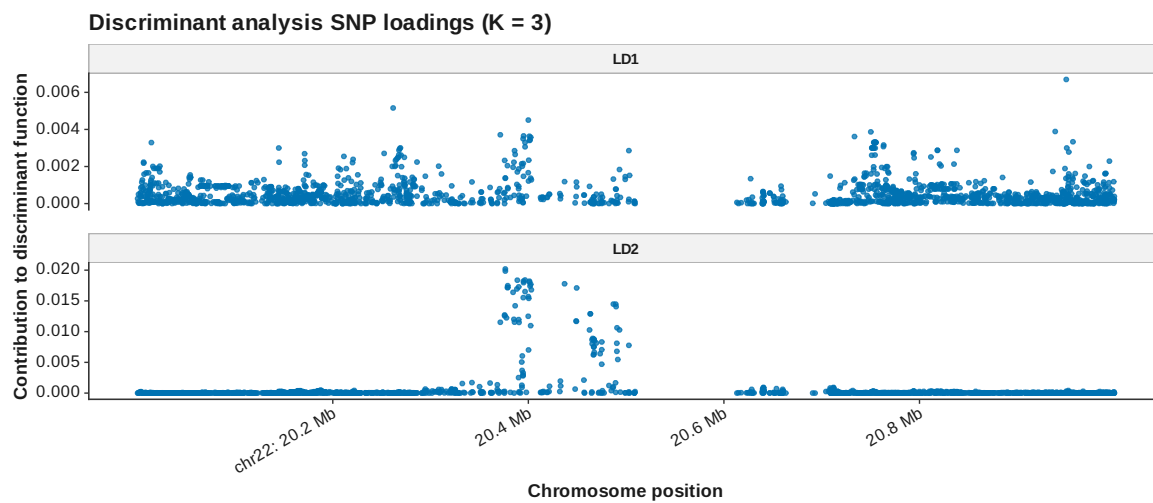


Figure 60: DAPC loadings manhattan K 3

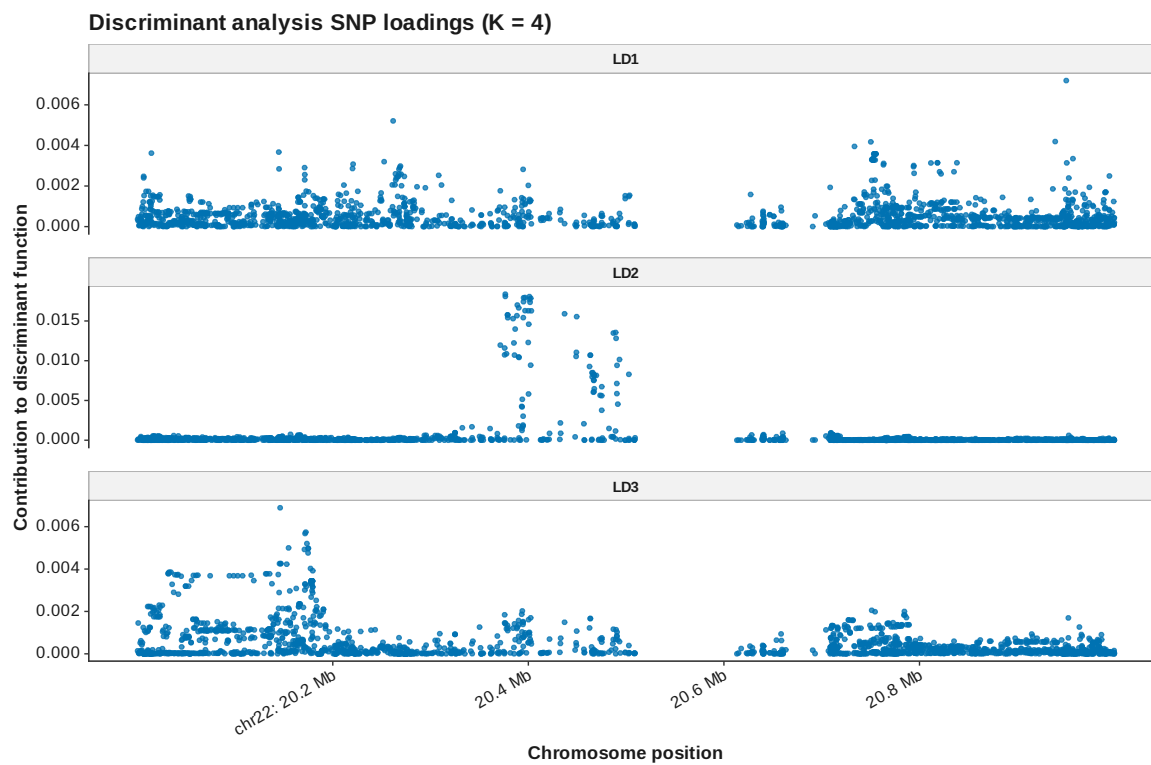


Figure 61: DAPC loadings manhattan K 4



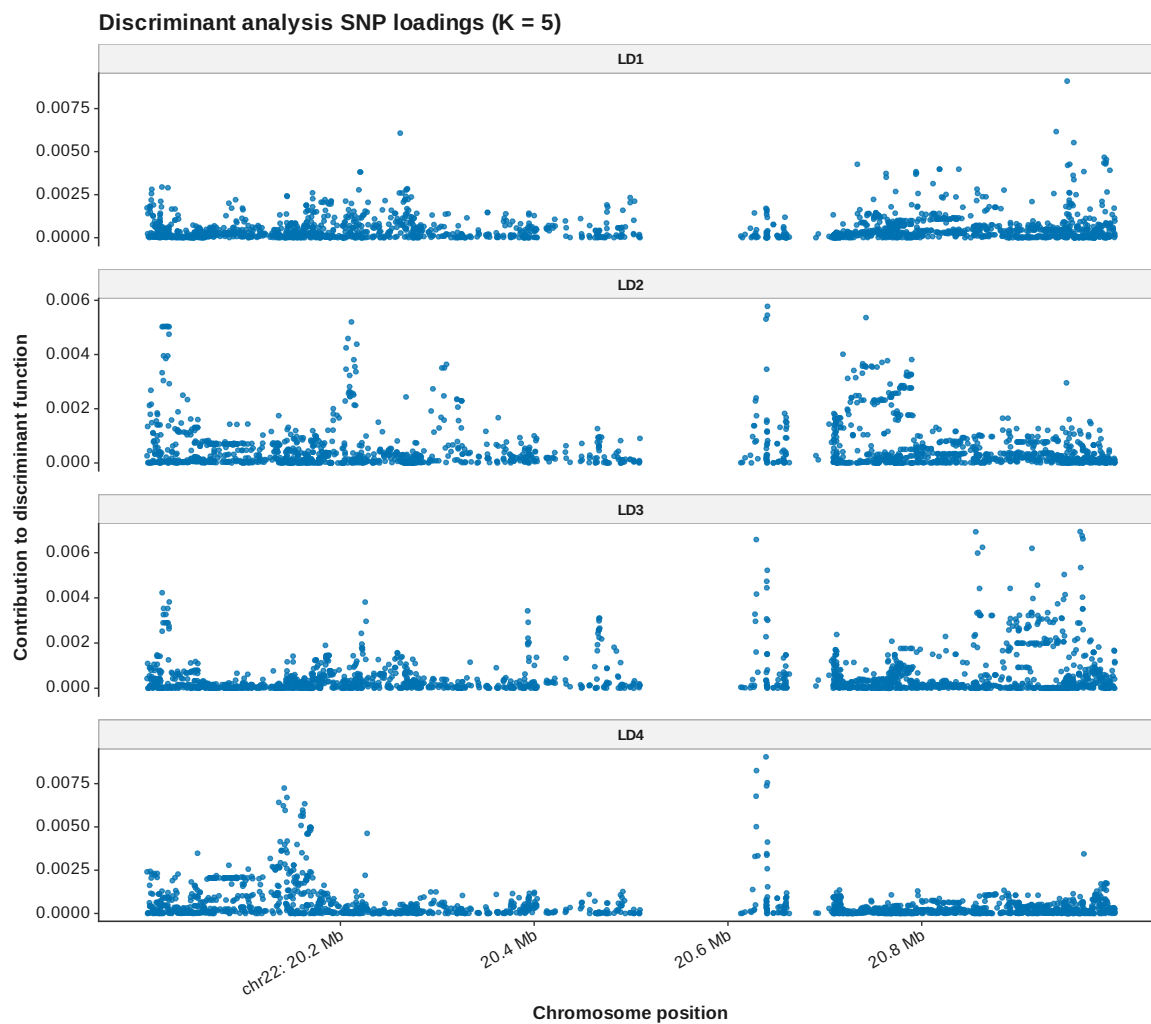


Figure 62: DAPC loadings manhattan K 5

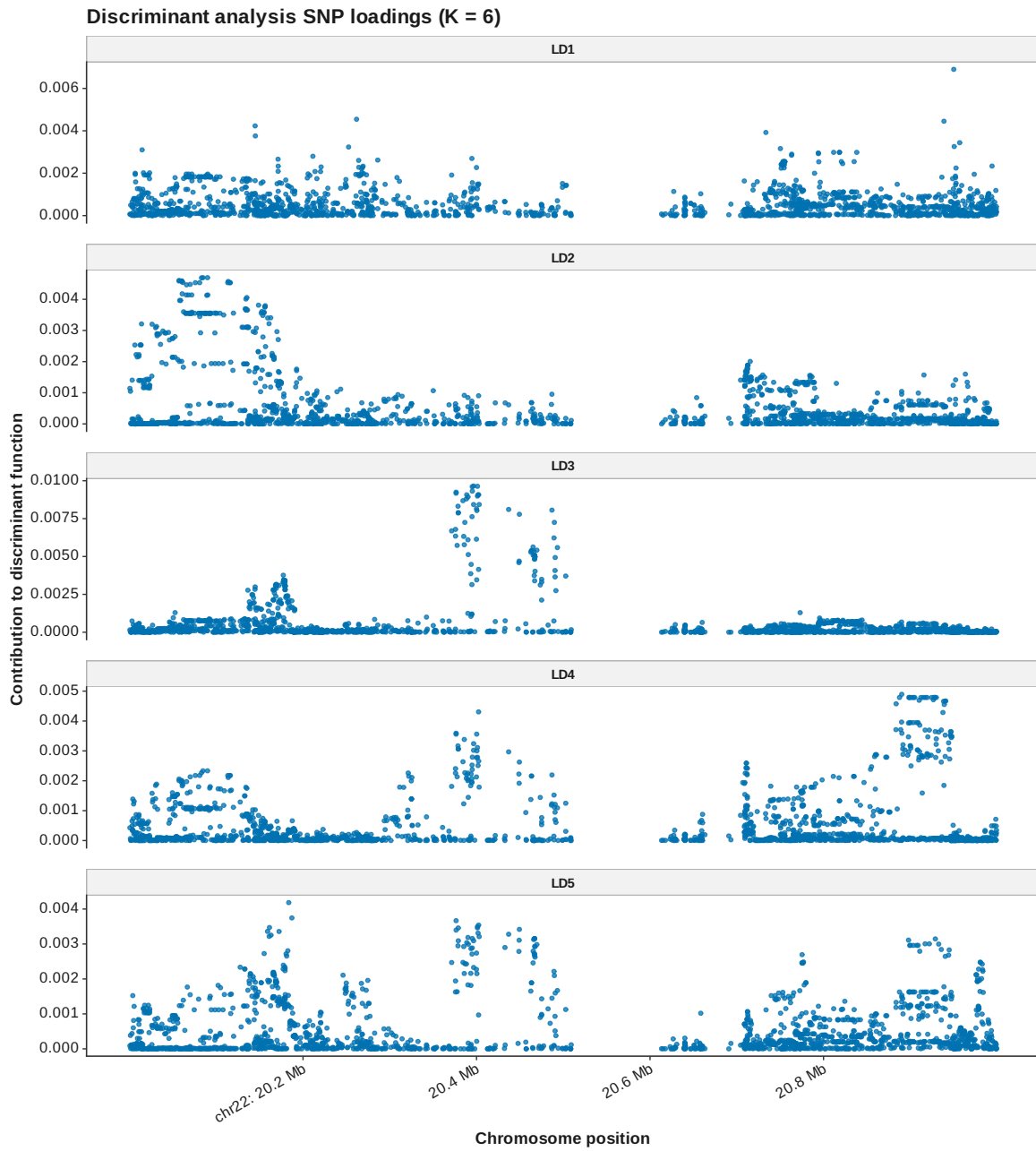


Figure 63: DAPC loadings manhattan K 6

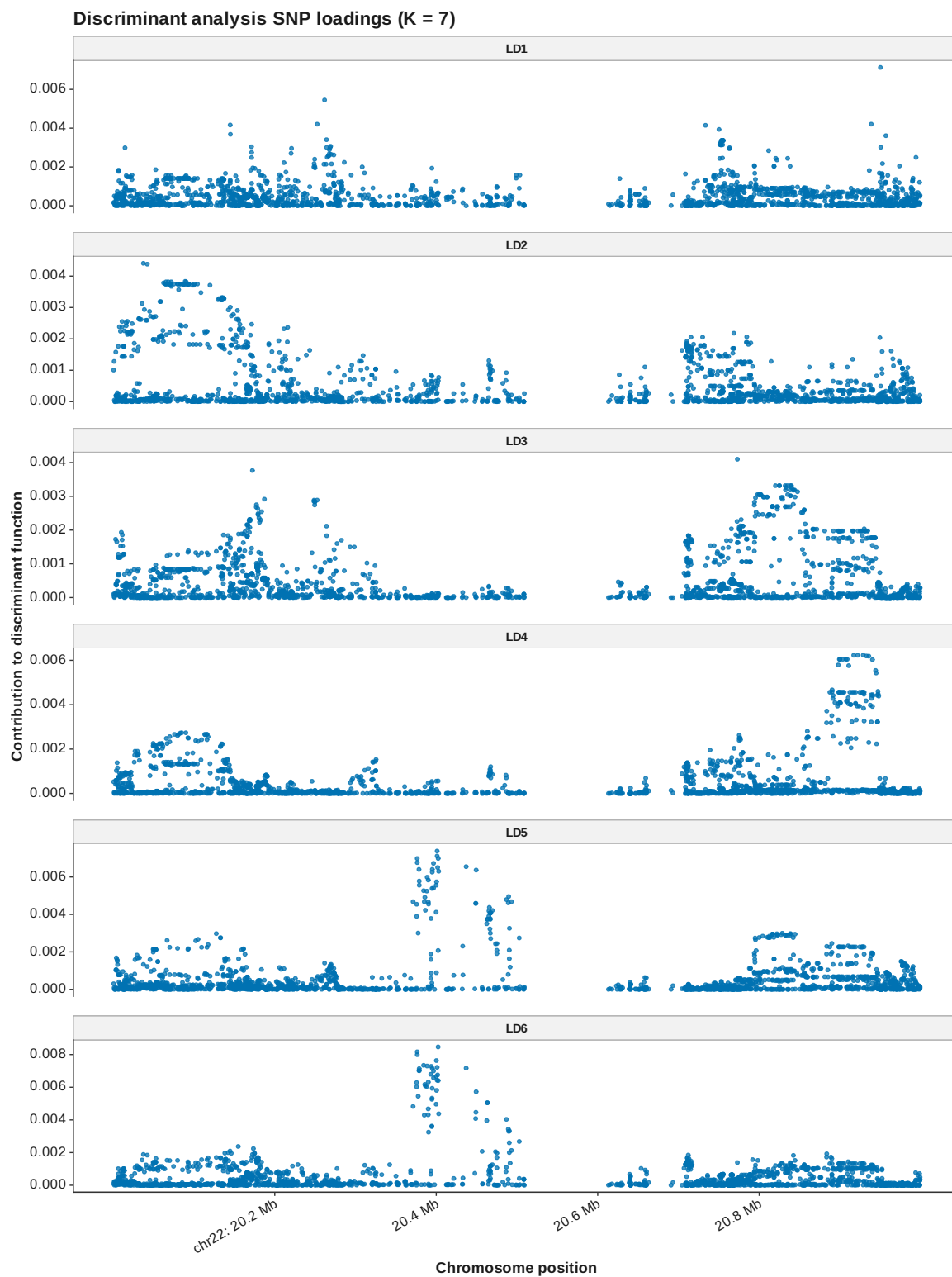


Figure 64: DAPC loadings manhattan K 7

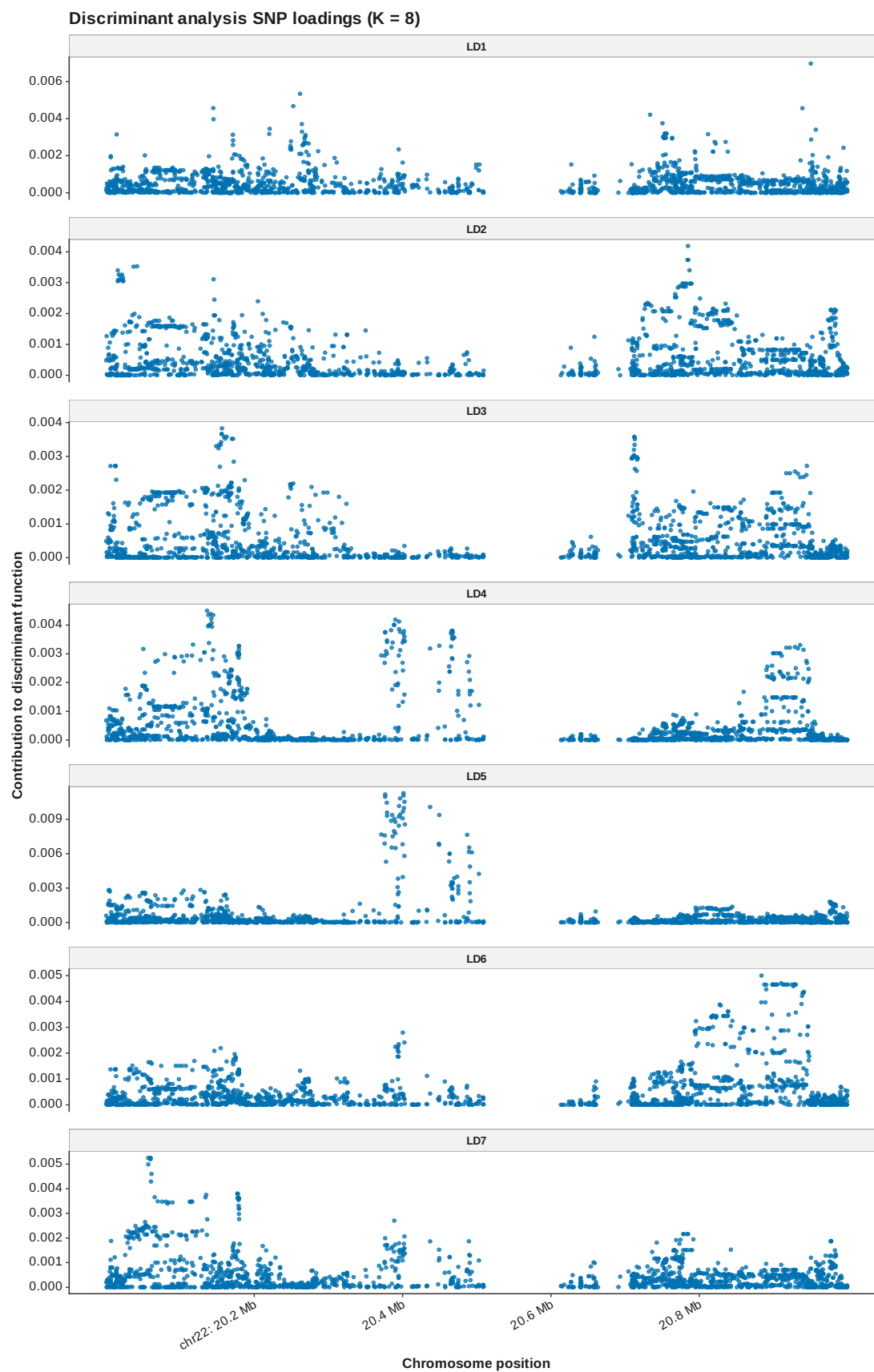


Figure 65: DAPC loadings manhattan K 8

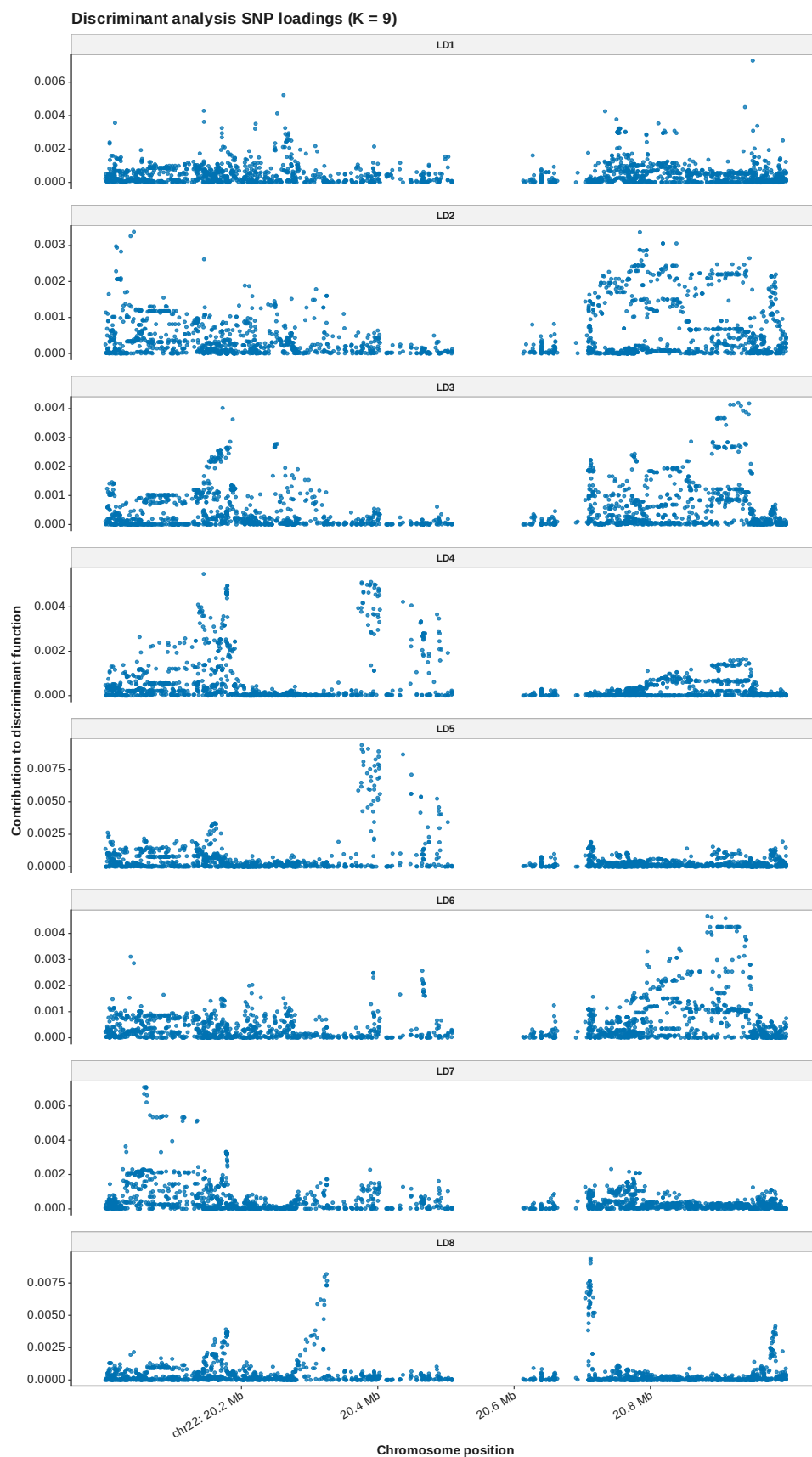


Figure 66: DAPC loadings manhattan K 9

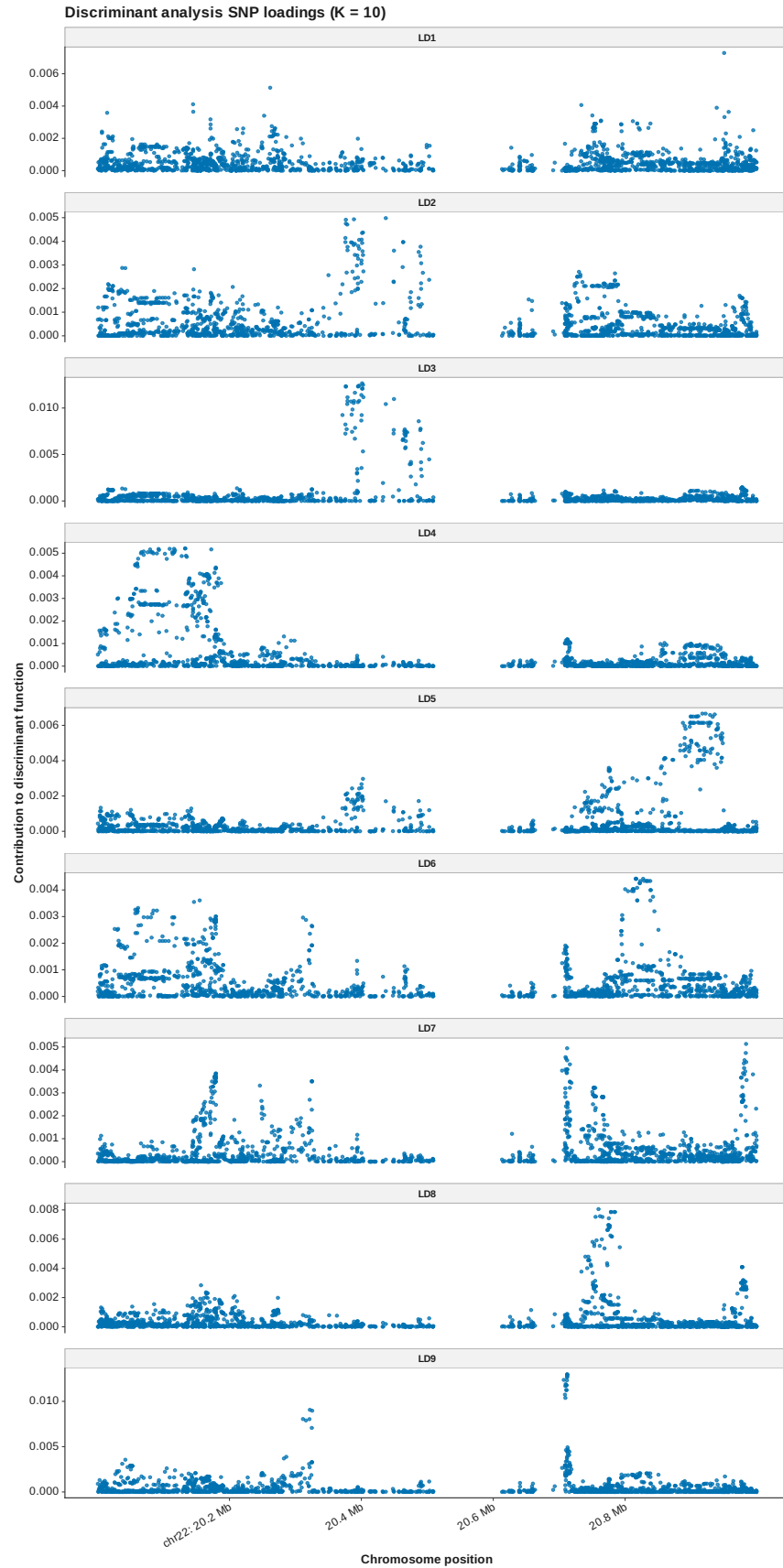


Figure 67: DAPC loadings manhattan K 10

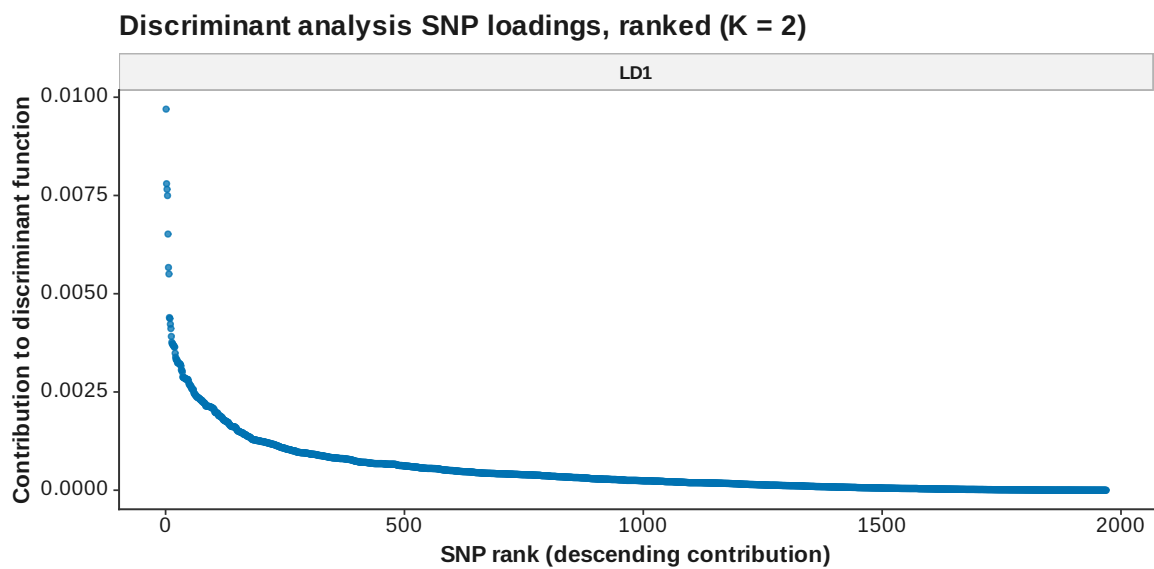


Figure 68: DAPC loadings ranked K 2

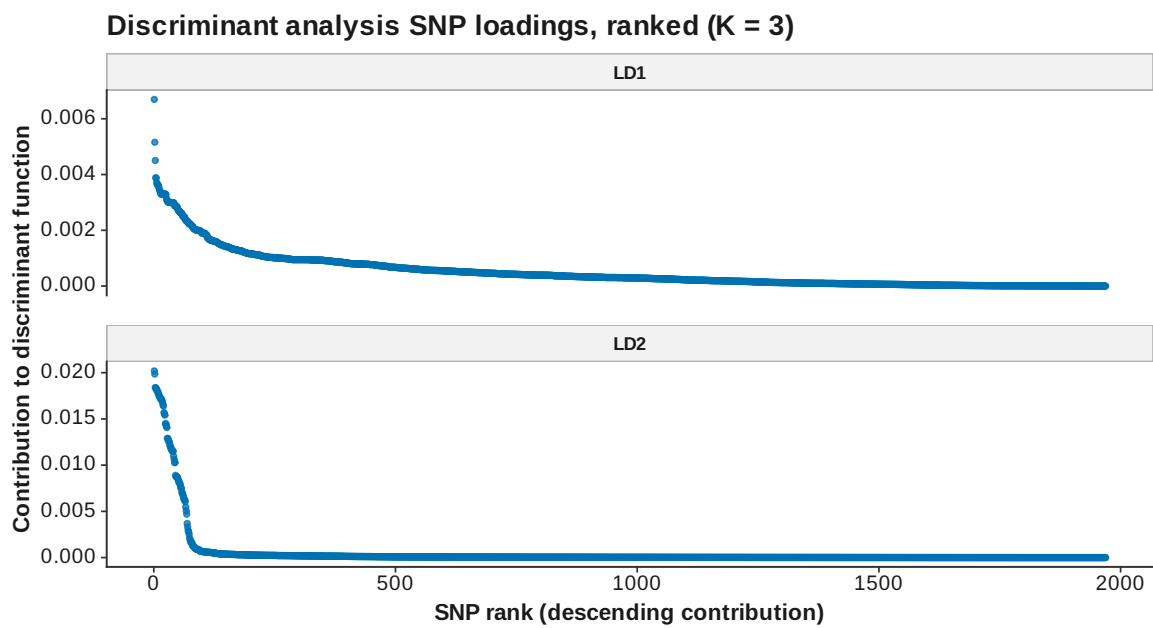


Figure 69: DAPC loadings ranked K 3



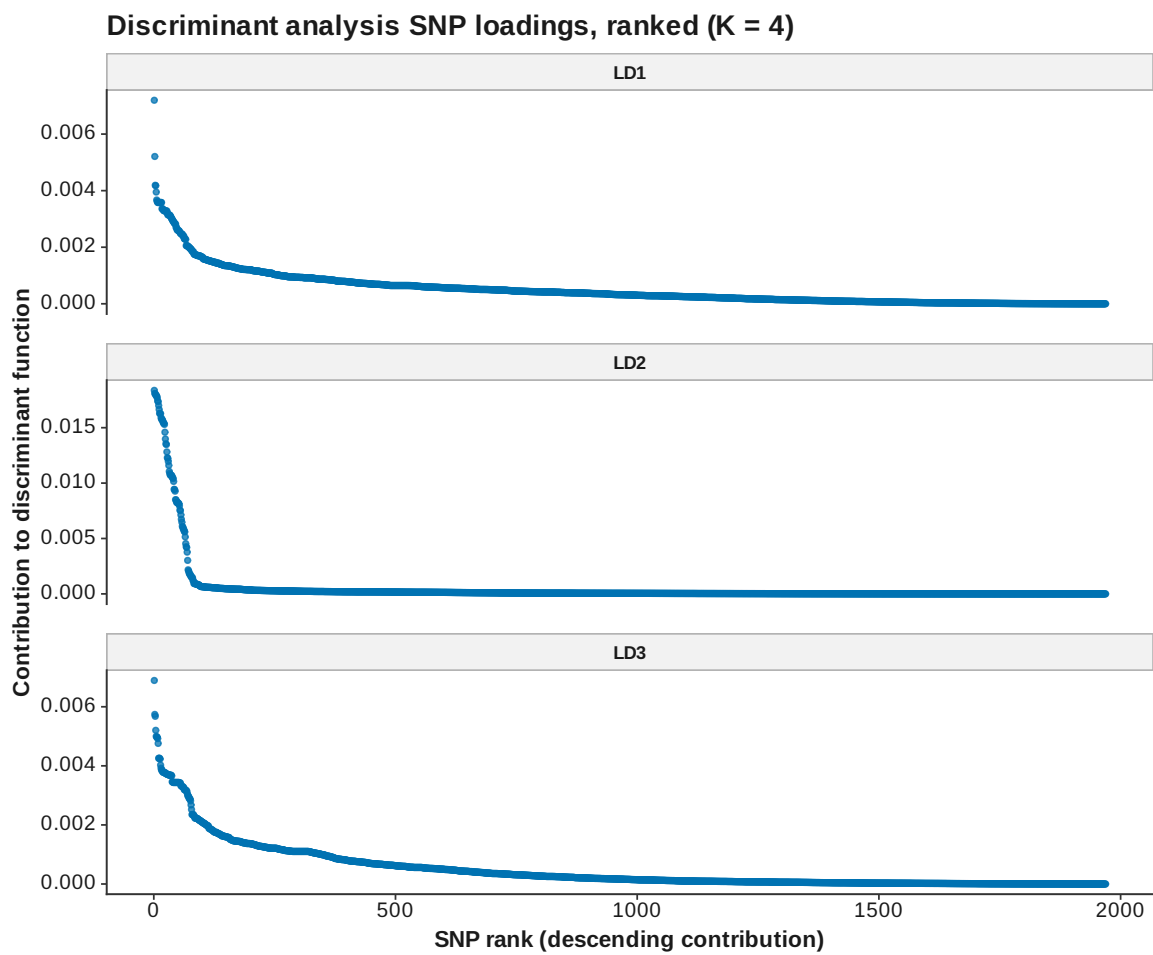


Figure 70: DAPC loadings ranked K 4

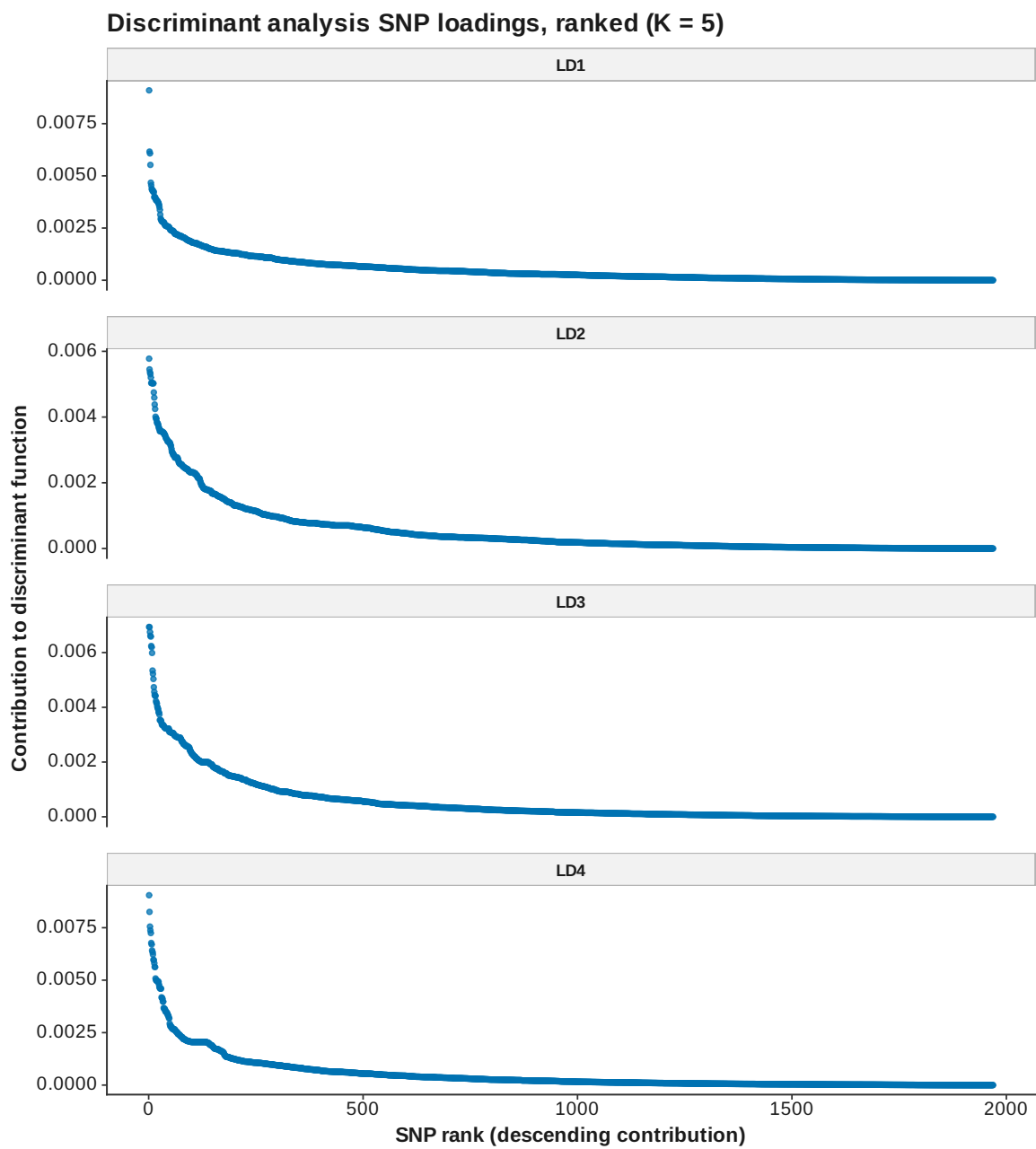


Figure 71: DAPC loadings ranked K 5

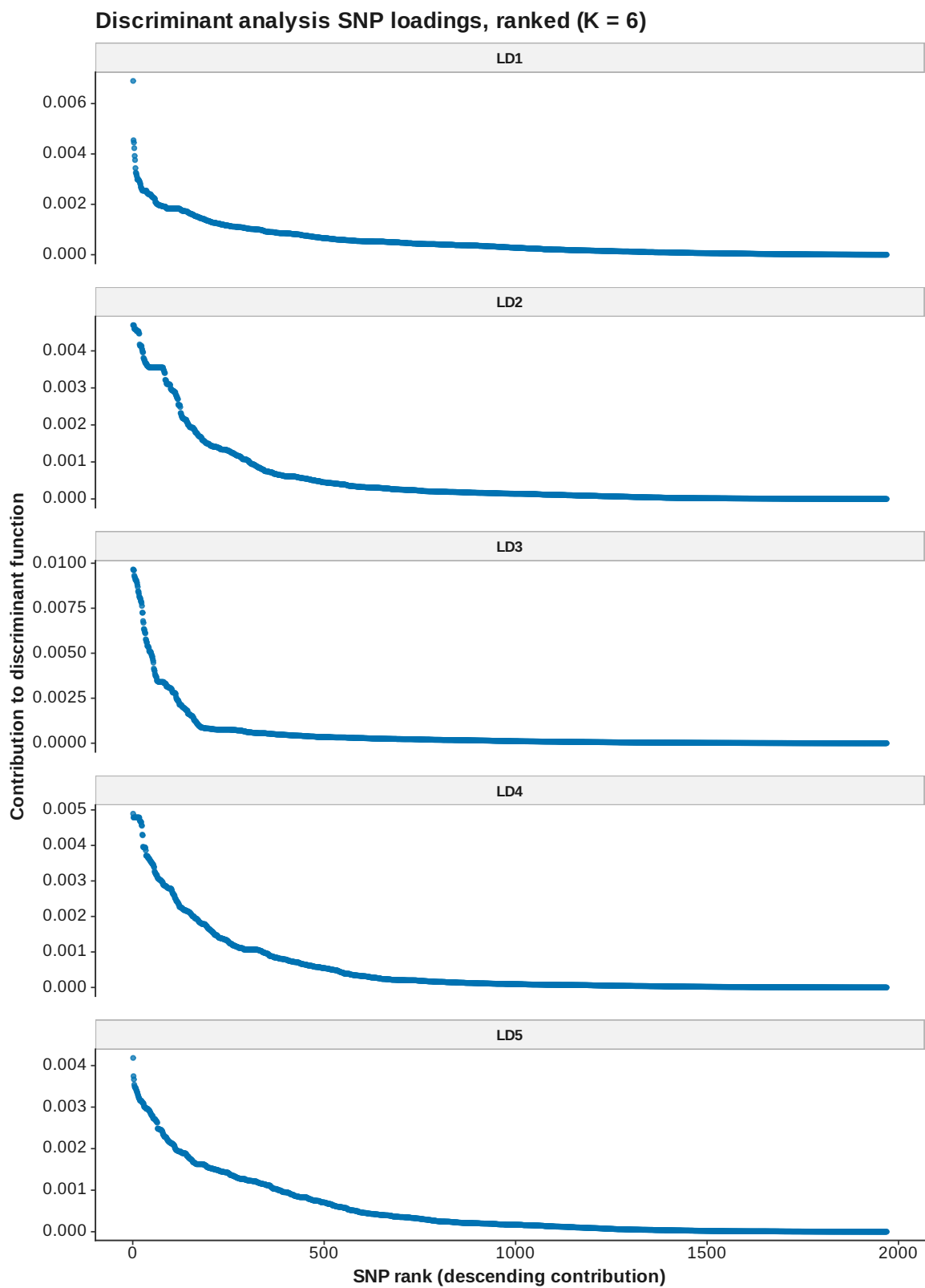


Figure 72: DAPC loadings ranked K 6

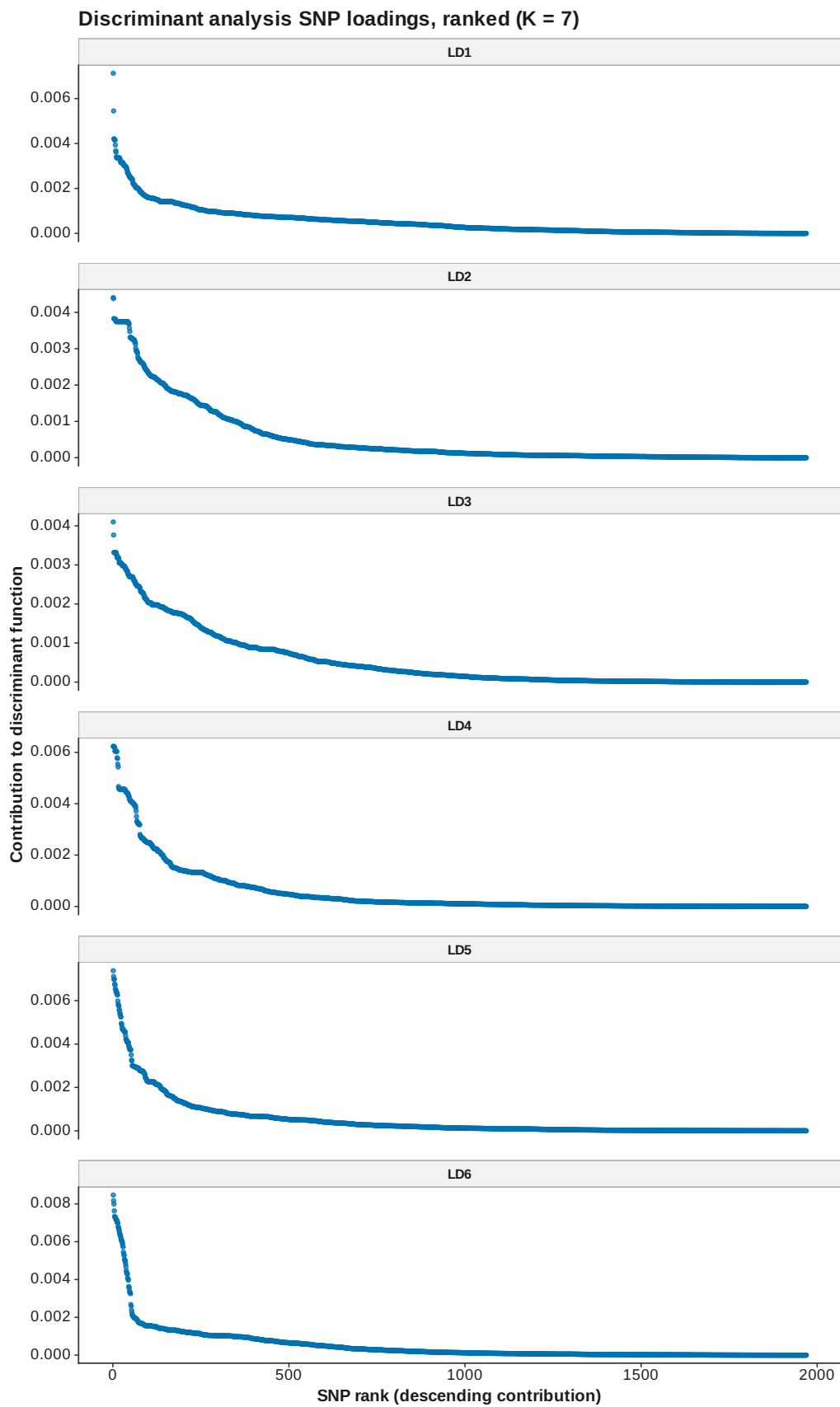


Figure 73: DAPC loadings ranked K 7

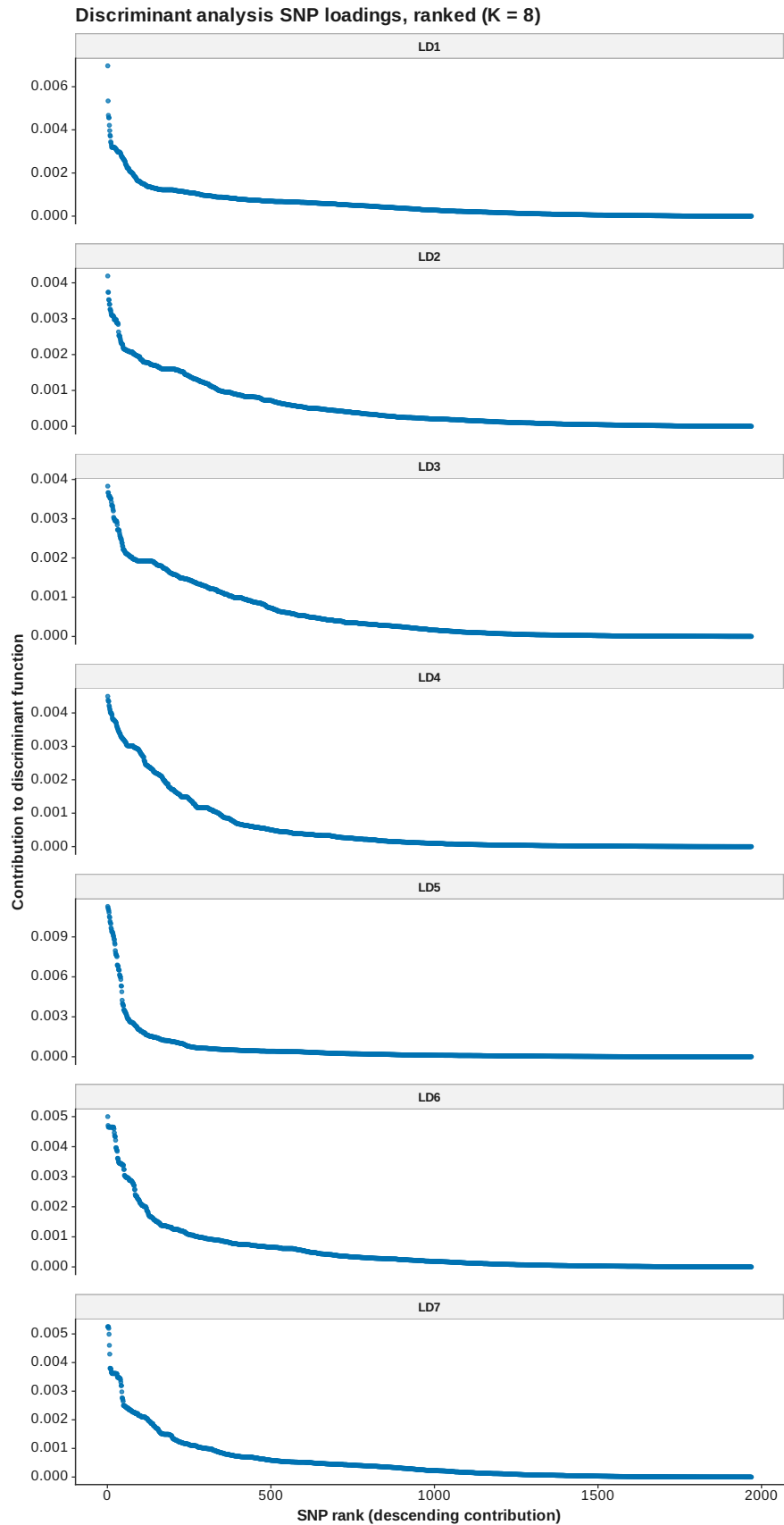


Figure 74: DAPC loadings ranked K 8

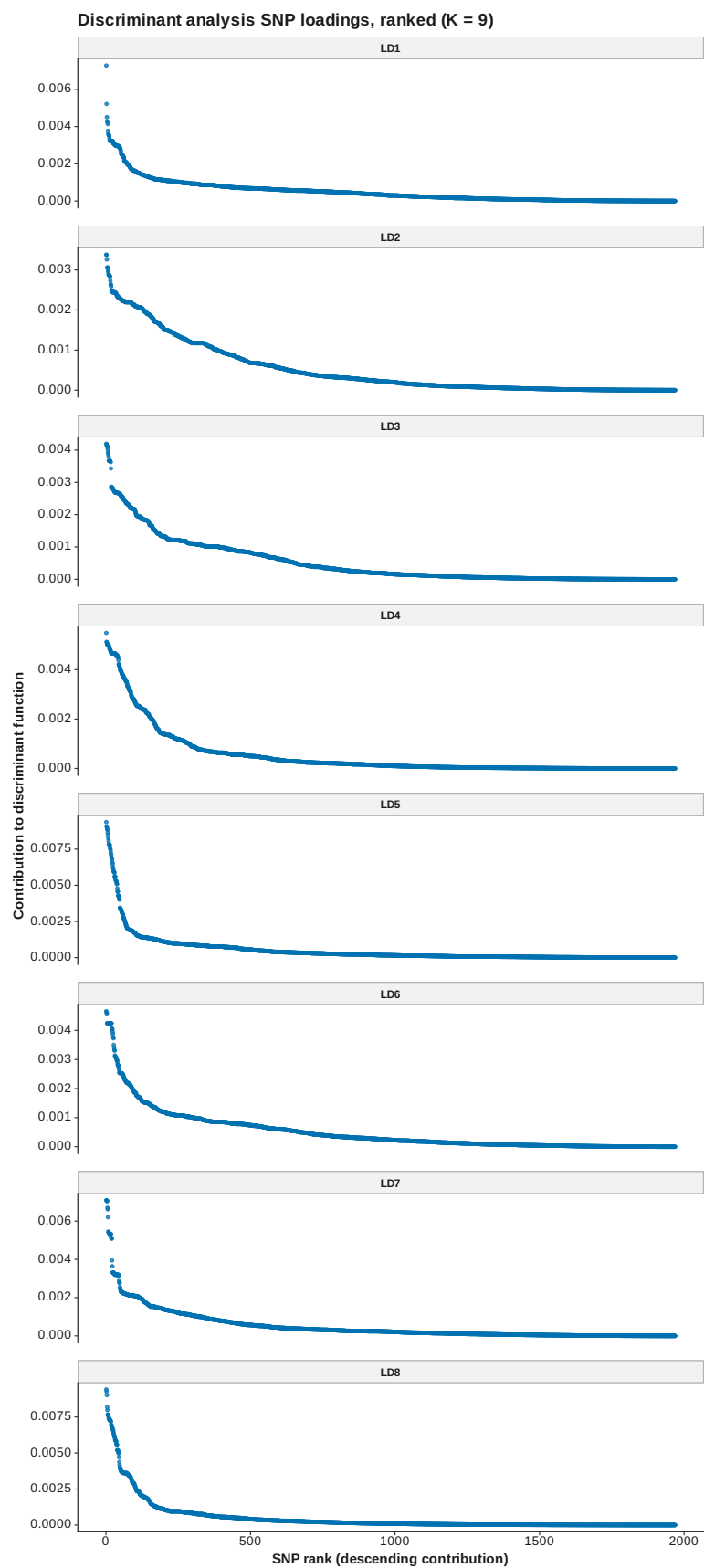


Figure 75: DAPC loadings ranked K 9

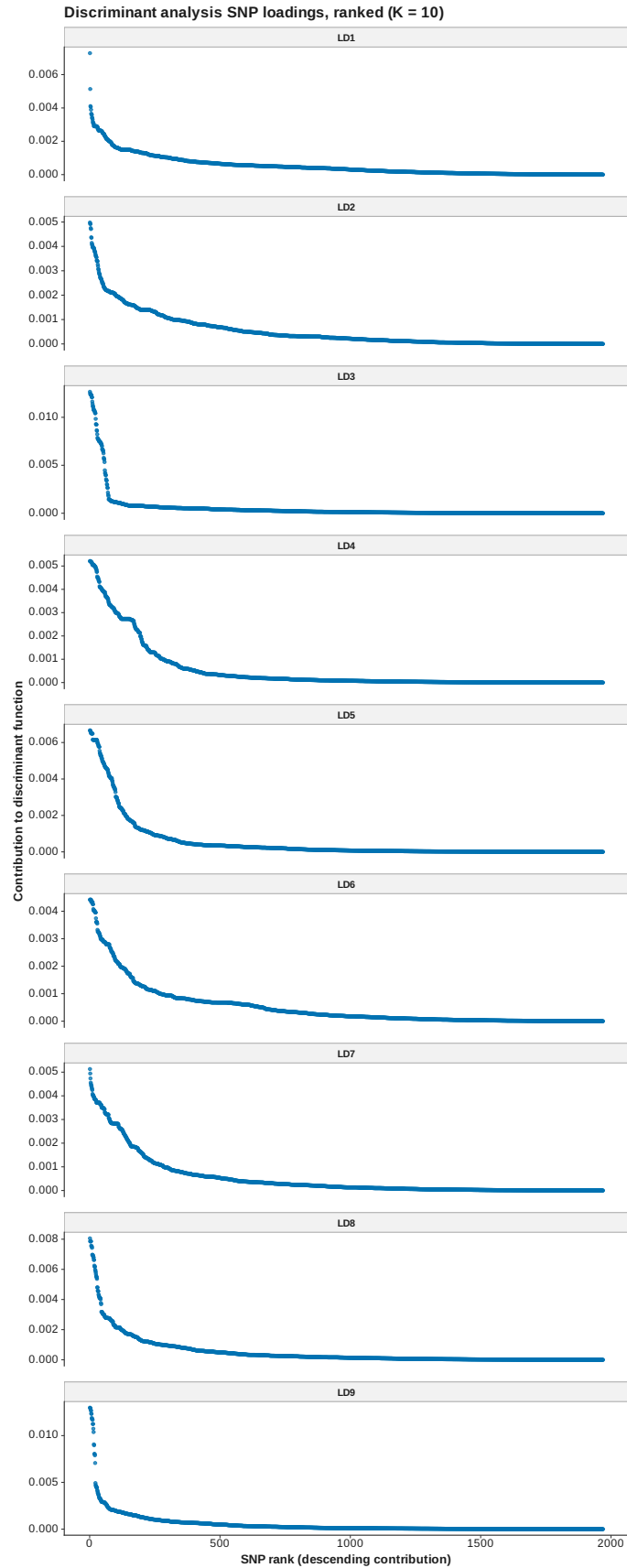


Figure 76: DAPC loadings ranked K 10

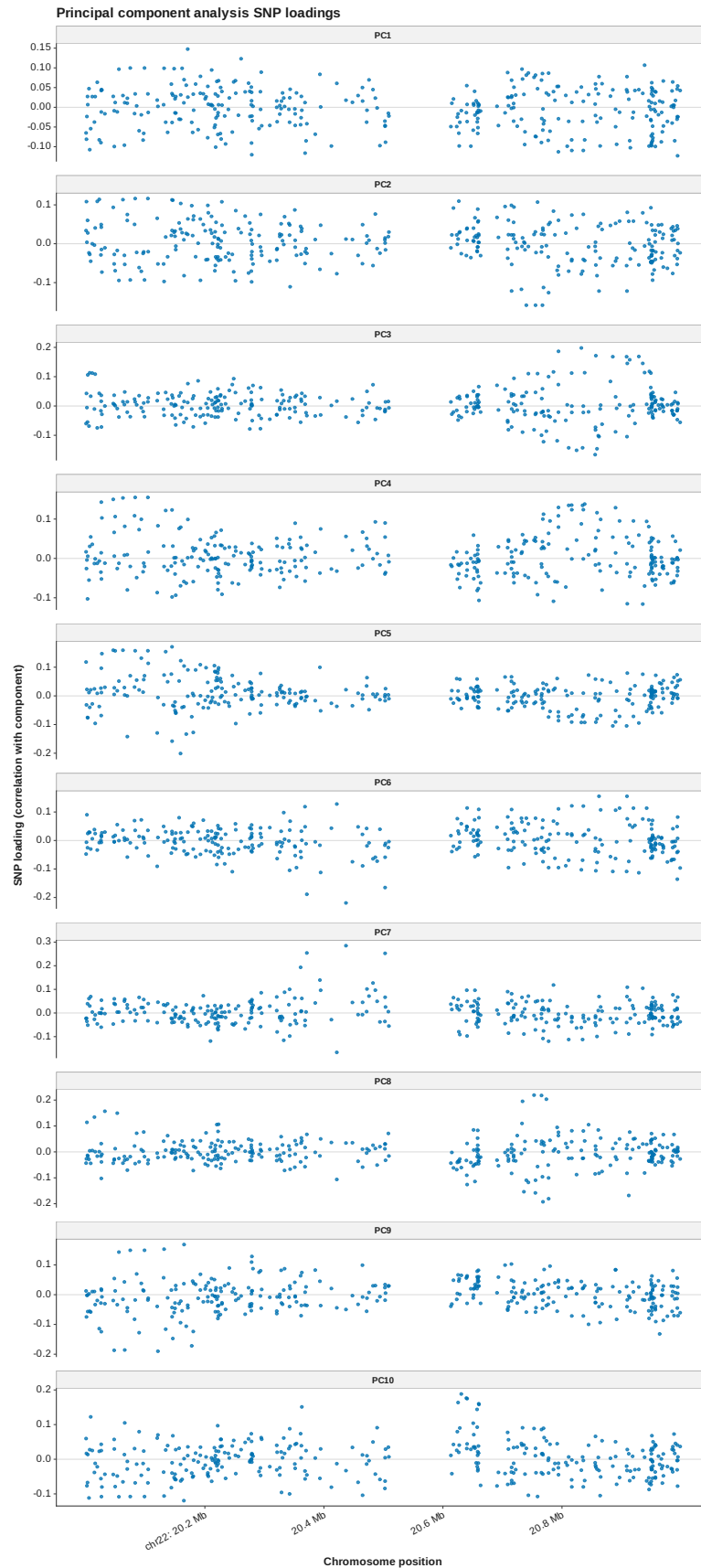


Figure 77: PCA loadings manhattan



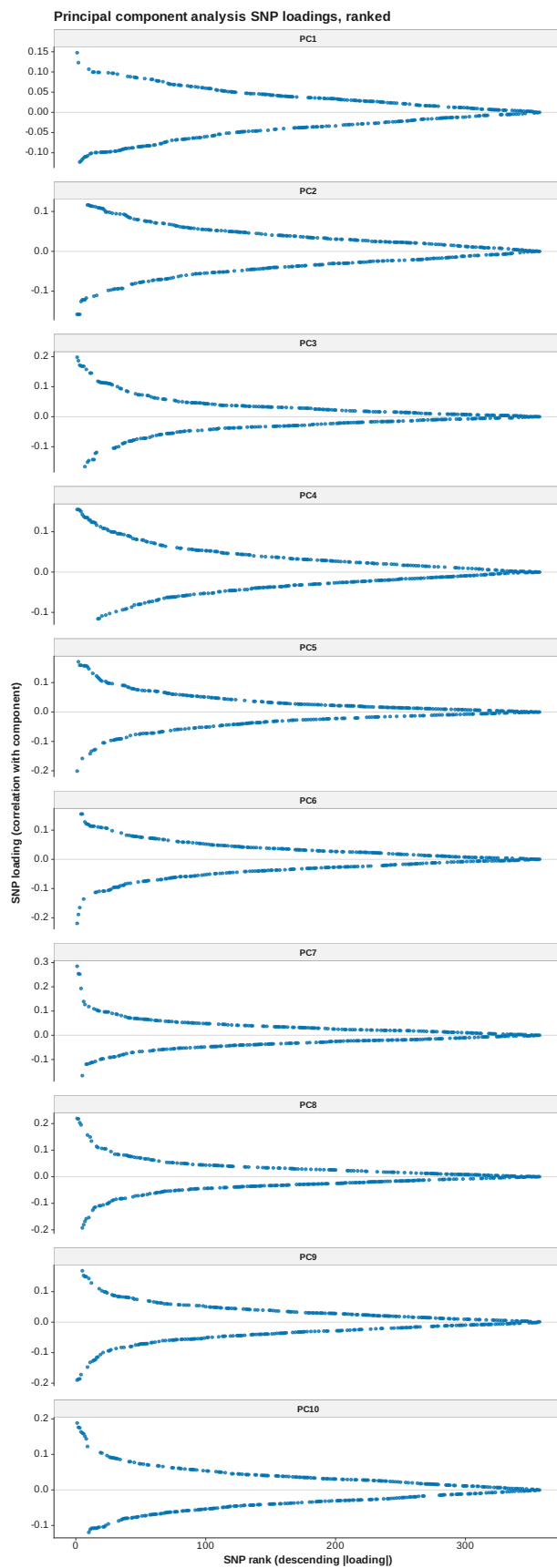


Figure 78: PCA loadings ranked

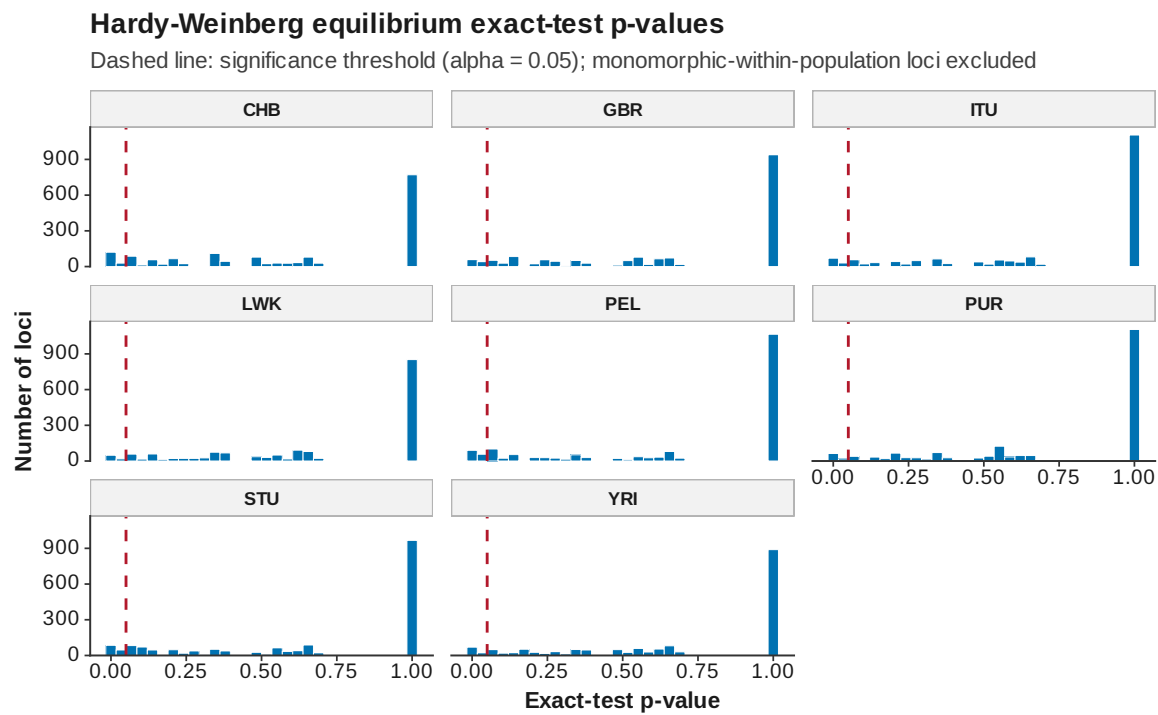


Figure 79: HWE p-values

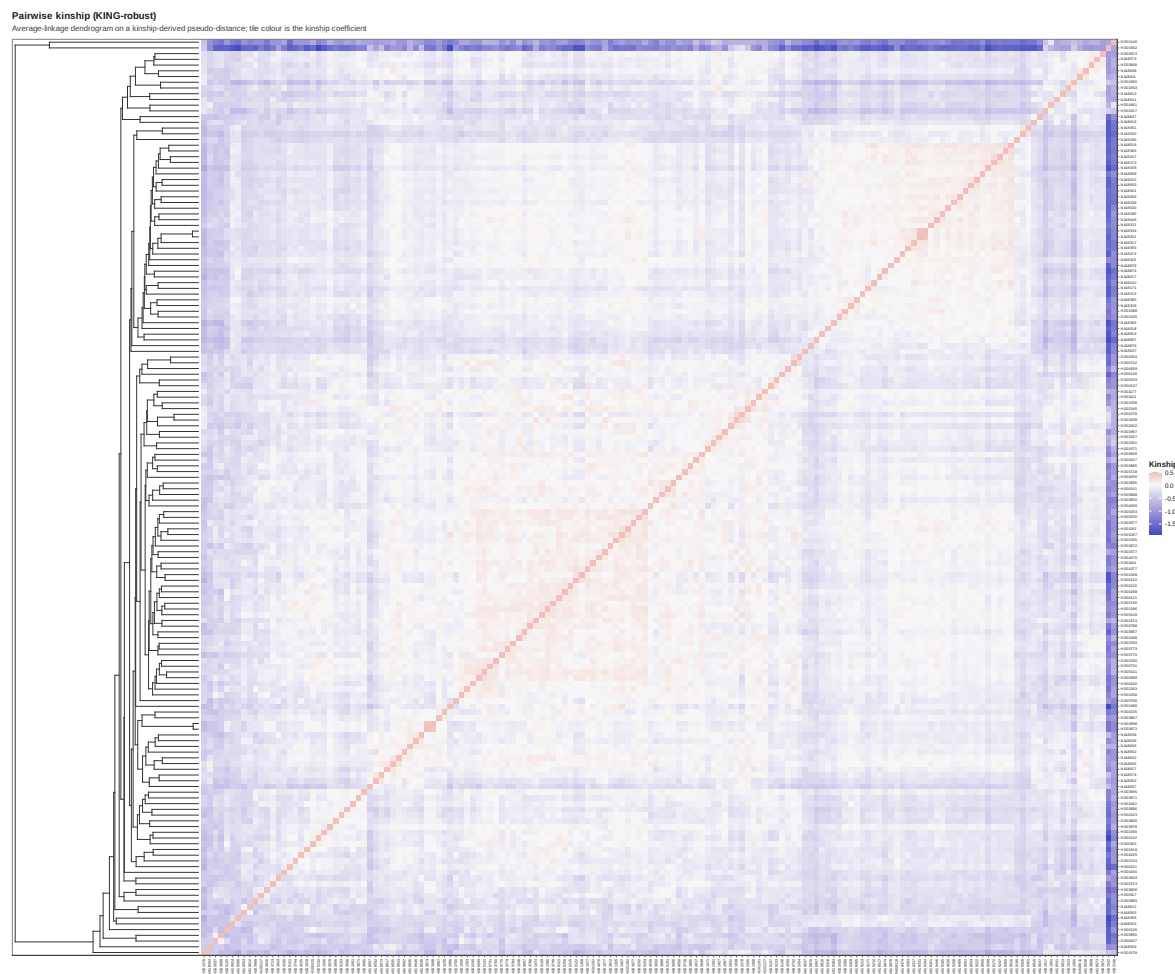


Figure 80: Kinship heatmap

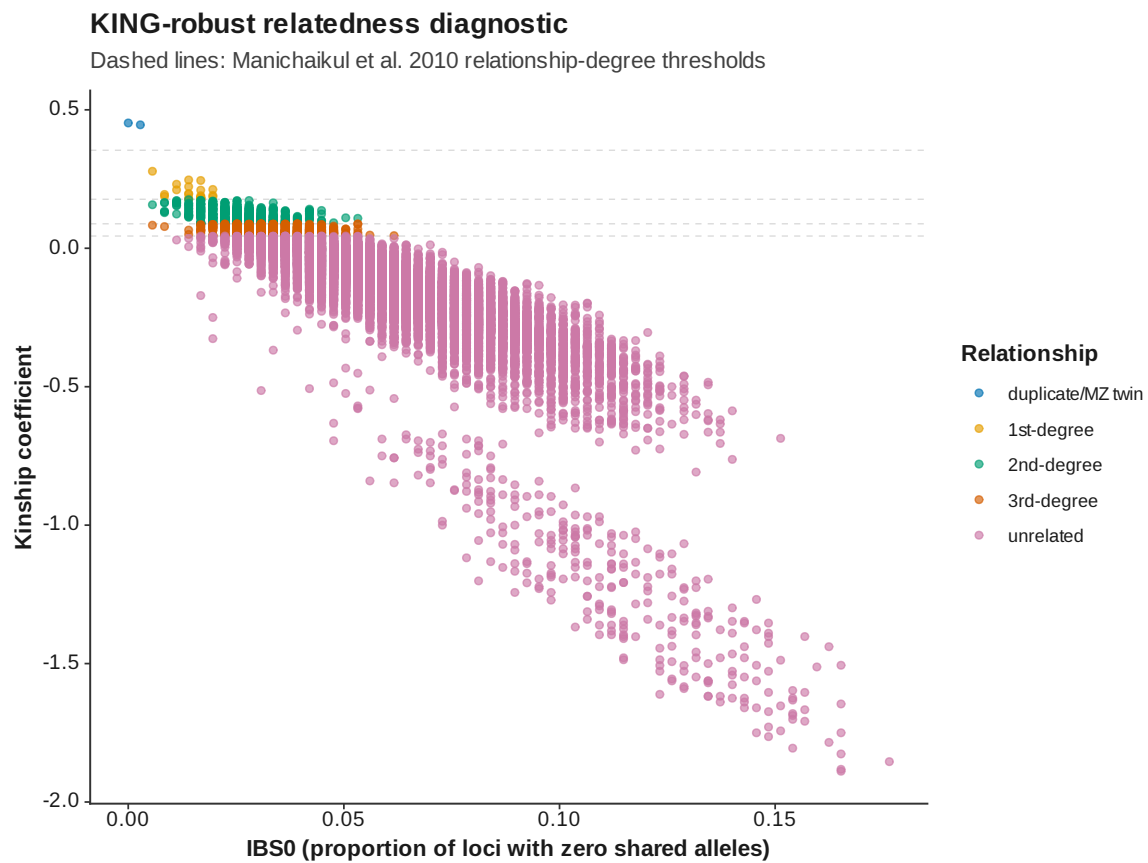


Figure 81: Kinship IBS 0 vs kinship

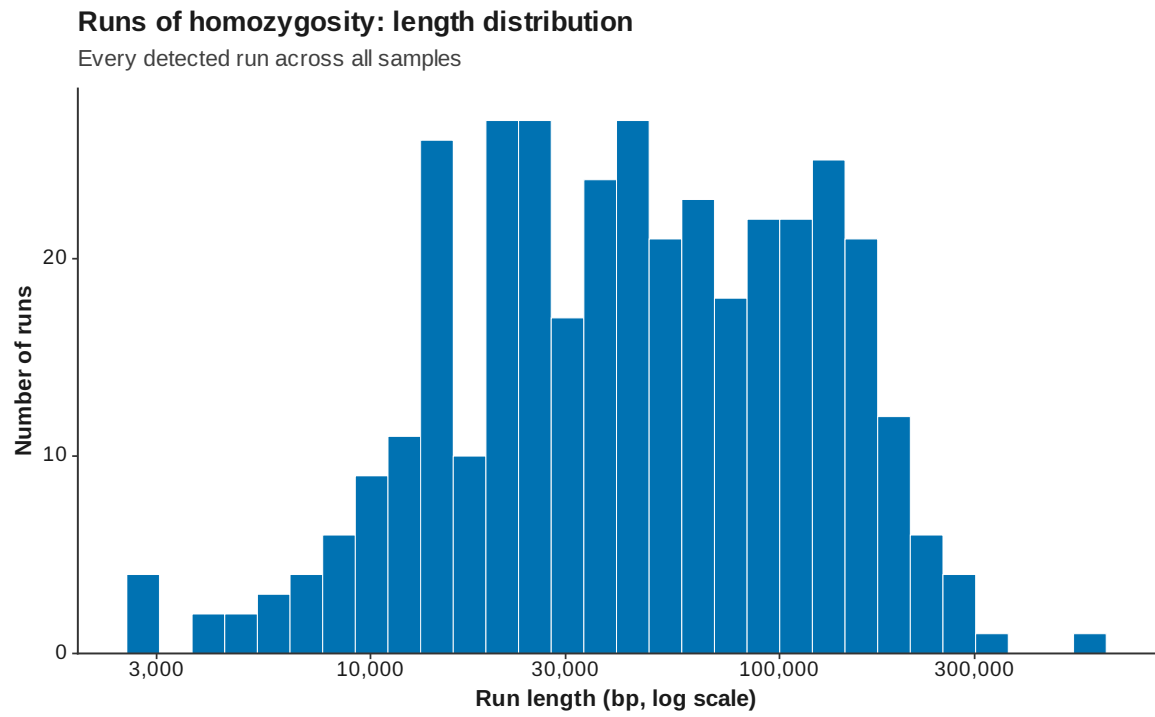


Figure 82: ROH length distribution

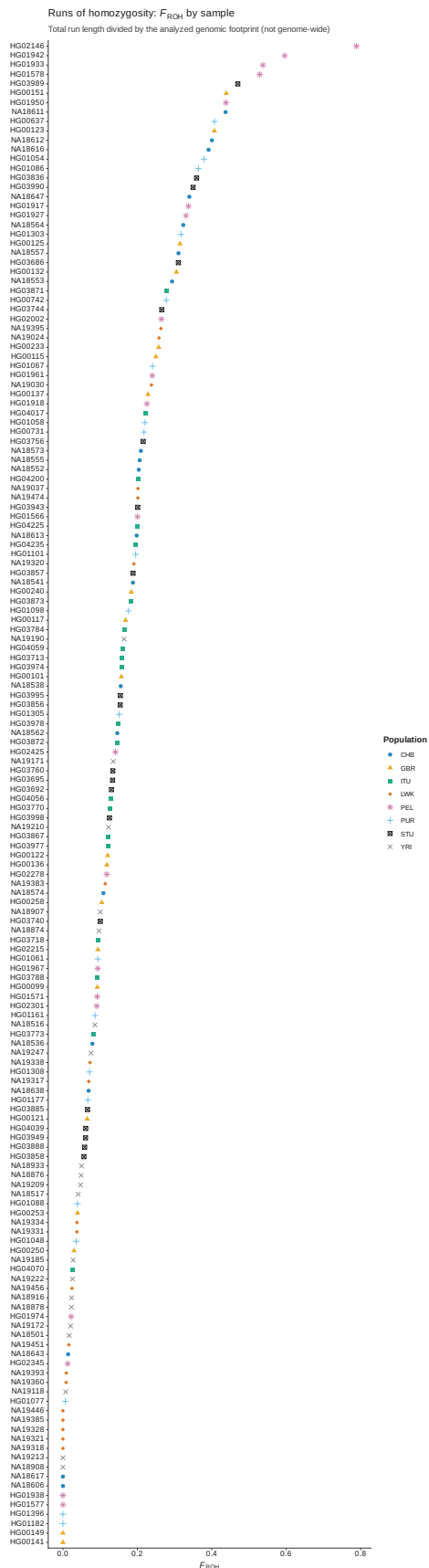


Figure 83: ROH FROH by sample

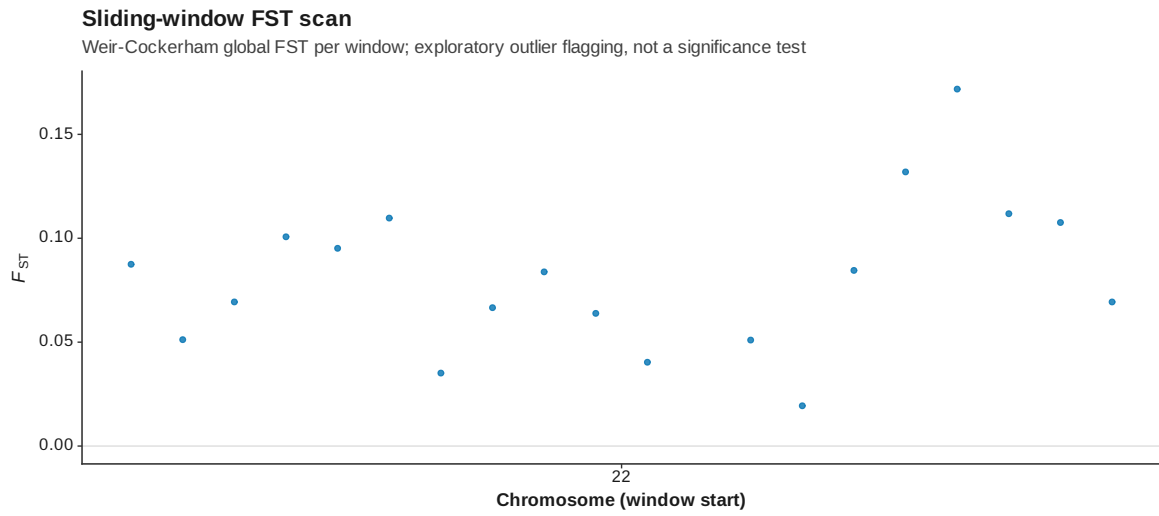


Figure 84: Genome scan FST manhattan

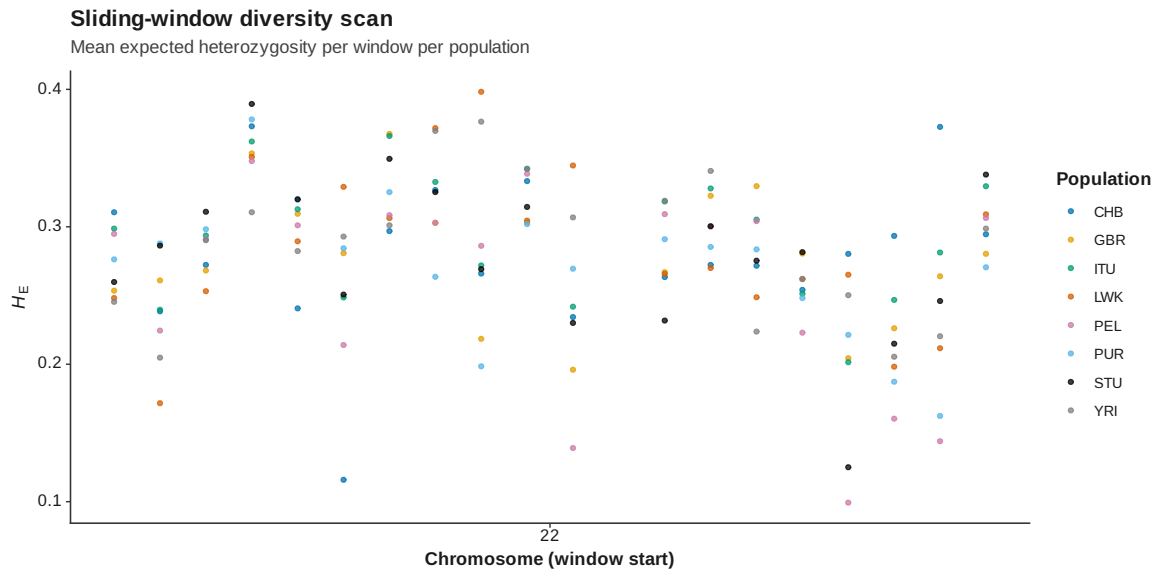


Figure 85: Genome scan diversity manhattan



### Principal component analysis

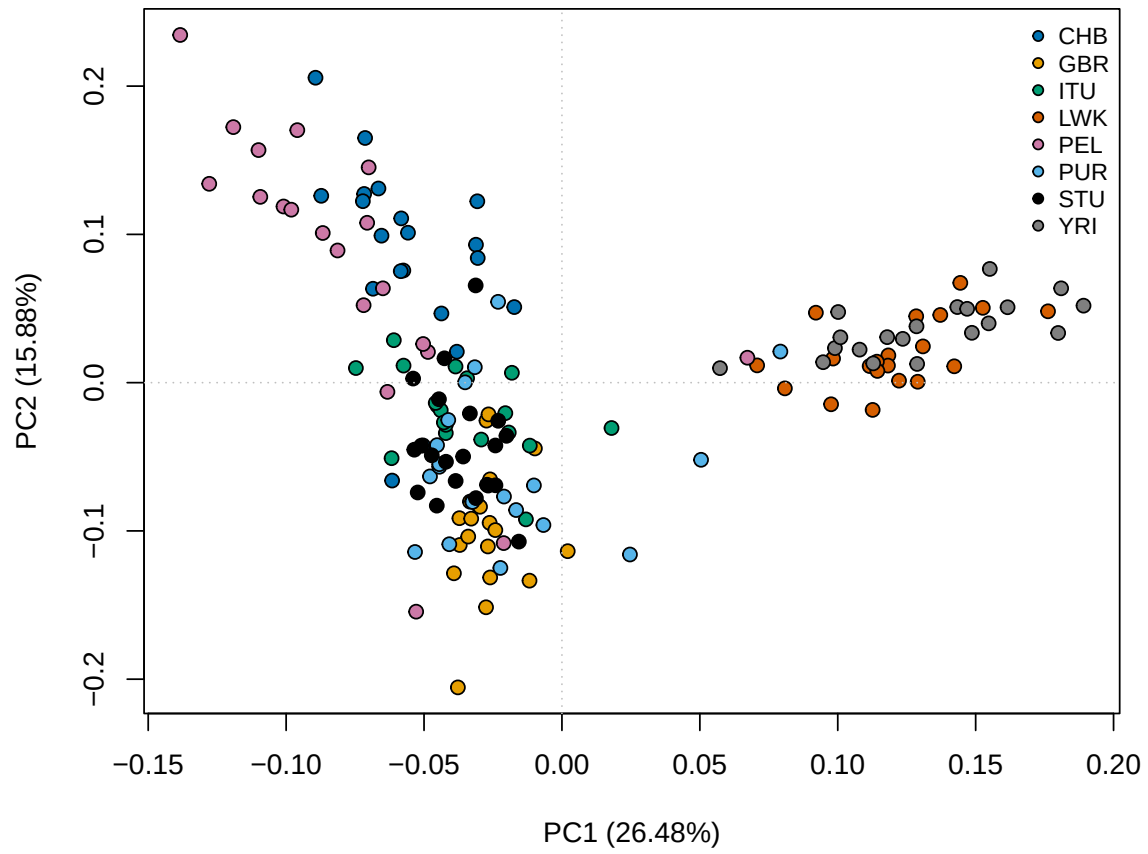


Figure 86: PCA PC 1 PC 2

## 17 Reproducibility

This report was generated from the serialized pipeline result object. Exact input hashes, settings, package versions, and stage runtimes are stored alongside the report.

### 17.1 Population-structure reproducibility

| K  | assignment_accuracy | replicate_max_rmse |
|----|---------------------|--------------------|
| 2  | 0.2500              | 0.0971             |
| 3  | 0.2938              | 0.0000             |
| 4  | 0.3562              | 0.3841             |
| 5  | 0.4062              | 0.3512             |
| 6  | 0.3938              | 0.2944             |
| 7  | 0.4812              | 0.0727             |
| 8  | 0.4625              | 0.1614             |
| 9  | 0.4688              | 0.1830             |
| 10 | 0.3812              | 0.2373             |